L03 — study notes

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Classes 3+4 — Introduction to Bayesian learning · Map of the notes

> **One claim. Bayesian learning is one rule applied over and over: posterior ∝ likelihood × prior, normalised by the predictive.** It turns Diego's 70% prior into 85% after one positive test and 99% after two, and the same arithmetic, with conjugate priors, adds data to pseudo-data (Beta, Gamma, Pareto, normal). **A point estimate is a decision**: quadratic loss reports the posterior mean, absolute loss the median, 0–1 loss the mode. **A model is judged by its marginal likelihood**, which charges for every parameter the data did not need (Greenberg's Table 3.2). **When integrals have no closed form, simulate**: Monte Carlo with independent draws, and Gibbs or Metropolis–Hastings chains whose invariant distribution is the posterior, at the price of autocorrelation. **Priors are also a modelling tool**: hierarchies let units borrow strength, and sparsity priors shrink noise while leaving signals alone.

Three layers per topic:

Layer	What it means here
U — intuition	forward vs inverse probability; the prior as earlier data; the physicists' joint density; "accept or stay"; borrowing strength; a horseshoe of shrinkage
C — course (examinable)	Bayes' rule and the predictive; Beta–Bernoulli, Gamma–Poisson/Exponential, Pareto–Uniform, normal–normal; sequential updating; Bayes estimators under four losses, Bayes risk, credible/HPD intervals; marginal likelihood, posterior odds, Bayes factors, Jeffreys' scale, the large-sample penalty, improper priors; Monte Carlo, inverse cdf, composition, accept–reject, importance sampling, SIR; Markov chains, Gibbs (with the switch point), Metropolis–Hastings, mixing and n.s.e.; hierarchical Beta–Binomial and Gaussian models; ridge/lasso as MAP, scale mixtures, normal–gamma, horseshoe
G — graduate	coherence; minimax; Chib's marginal likelihood; Rao–Blackwellisation; HMC/NUTS; Pólya–Gamma logit; SV estimation details; spike-and-slab and reversible jump. **Named, not examinable**

Reading order

Part I — the rule and the conjugate families

1. L03-01-bayes-rule-discrete — ingredients slides 1–25; `question0.pdf`; Stone (2013); discrete-prior page; G §2.1–2.2 (pp. 7–18), §3.1.3. **Diego 0.70 → 0.847 → 0.993 (Y alone 0.983); three heads leave the fair coin 0.0674; chickenpox beats smallpox 90 to 1 on base rates; sequential = batch.**
2. L03-02-conjugate-sequential-updating — sequential-Bayes page; G §2.2, §3.1.3–3.1.4, §3.2.3; B Ex. 4.4.1–4.4.2, 5.5.2; T Results 12.3–12.6. **Beta**$(\alpha_0 + s, \beta_0 + n - s)$ **and its weighted-average mean (G Fig. 2.2: 0.357, 0.315); predictive = posterior mean; uniform prior: mode = MLE; Pareto**$(\max\{x_{(n)}, \theta_0\}, \alpha_0 + n)$**.**
3. L03-03-normal-measurement-physicists — ingredients slides 26–52; `fisicos-ABCD.html`; G §3.2.3, §4.2–4.3; B Ex. 4.4.3; T Result 12.1. **Predictive** $N(\mu_0, \sigma^2 + \tau_0^2)$**; precisions add; A:** $N(890, 17.9^2)$**, B:** $N(840, 35.8^2)$**; the** t **prior C (891.1) and mixture D (876.1) by quadrature; a** t **prior's pull is bounded (1022 vs 940 at** $x = 1100$**).**

Part II — decisions and models

4. L03-04-bayes-estimators-loss — Bayes-estimator deck (16 slides); G §3.2.1–3.2.2; B §4.3–4.4, §5.5. **Mean, median, mode, quantile from four losses (Beta(2, 10): 0.167, 0.148, 0.100); the Exp–Gamma example; both estimators biased; Bayes risk** $1/72 < 1/60$**; HPD vs equal-tail.**
5. L03-05-model-comparison-bayes-factors — G §3.2.4, §4.2. **Marginal likelihood, posterior odds, Jeffreys; Table 3.2 reproduced (**-1.200 **at 0.2/0.8,** $+0.756$ **at 0.5/0.5 with** $m = 100$**); the** $\frac{1}{2}\log n$ **penalty; improper priors make** B_{12} **arbitrary.**

Part III — computing posteriors

6. L03-06-monte-carlo-simulation — ingredients slides 53–56; G Ch. 5; MS App. B; correlation and ZIP pages. **n.s.e. (**$\hat{\pi}$**: 0.052 at** $M = 1000$**); inverse cdf, composition, accept–reject (1.875, 0.533; normal 0.760, not 0.658), importance sampling with infinite variance, SIR; ZIP MLE (4.223, 0.526) with** $n = 45$**.**
7. L03-07-markov-chains-gibbs — G Ch. 6, §7.1, §7.3; T §12.1–12.3; MS §C.5. **Invariant distributions, Gibbs invariance proof, the normal mean–precision sampler, the coal switch point (1891,** $P = 0.240$**; rates 3.106 and 0.930, Gibbs vs exact),** $\rho_h = \rho^{2h}$ **and the inefficiency factor.**
8. L03-08-metropolis-hastings-hmc — G §7.2, §8.2.3; T §12.4, §12.7; logit, SV and Stan pages. **Reversibility and** $\alpha = \min\{1, \cdot\}$**; random-walk and independence kernels; acceptance** $\frac{2}{\pi}\arctan(2/s)$ **(0.445 at 2.38); Gibbs = MH with** $\alpha = 1$**; the Bayesian logit.**

Part IV — priors as models

9. L03-09-hierarchical-models-shrinkage — G §4.5–4.6; hierarchical pages. **Exchangeability; partial pooling with weight** $\eta/(\eta + n_i)$**; 13 hospitals on an exact grid vs Stan (**μ **median 0.0714 vs 0.0715); the 68-women random-coefficient Gibbs sampler (**$\sigma = 1.10$ **vs pooled 3.61).**
10. L03-10-sparsity-priors — sparse-priors deck; sparse-regression simulation; AME §3.6; I §6.2. **Ridge and lasso as posterior modes; scale mixtures; normal–gamma kurtosis** $3/\lambda$**; horseshoe**

$\kappa \sim$ **Beta(1/2, 1/2); at** $y = 6$**: 3.00, 4.59, 5.66; horseshoe nulls 8 times tighter than ridge.**

Support

11. L03-11-formulas — the site's topics and the class-plan goals mapped to notes; every result; every number.
12. L03-12-pitfalls — fifteen confusions with counterexamples and an original 25-item True/False self-test.
13. L03-13-python — the reference implementation, section by section, with its output. Runnable script: `L03_python_checks.py`.

PDF: `Notas/L03/L03-notes.pdf`, rendered from these notes (rerun `python -I ~/.claude/commands/templates/n` `Notas/L03` after any edit).

The one result to hold

$$p(\theta \mid y) = \frac{f(y \mid \theta)\,\pi(\theta)}{\int f(y \mid \theta)\pi(\theta)\,d\theta}$$

numerator: learning; denominator: the predictive, and the evidence for the mod

Everything in the unit is this formula. Its numerator gives the posterior and, through a loss, the estimate (L03-01 – L03-04). Its denominator is the predictive of the data and the marginal likelihood that ranks models (L03-05). When neither can be computed, simulation evaluates them (L03-06 – L03-08). And the prior π itself can be a model of how units or coefficients relate (L03-09 – L03-10).

Numbers and cases to have memorized

Quantity	Value	Meaning
Diego: $P(\text{ill} \mid X = 1)$, then $\mid X = 1, Y = 1$	**0.847**, **0.993**	prior 0.70; the second test does the work
Three coins after HHH	**0.0674, 0.3932, 0.5394**	predictive $P(HHH) = 0.618$
Beta–Bernoulli mean (G Fig. 2.2)	$5/14 = 0.357$; $17/54 = 0.315$	prior weight $0.29 \to 0.07$
Physicists A, B at $x = 850$	$N(890, 17.9^2)$, $N(840, 35.8^2)$	$w = 0.8$, 0.2
Normal vs t prior at $x = 1100$	940 vs **1022**	bounded influence
Beta(2, 10): mode / median / mean	0.100 / 0.148 / 0.167	three losses, three answers
Table 3.2 ($m = 10$, 0.2/0.8; $m = 100$, 0.5/0.5)	$\log_{10} B_{12} = -1.200$; $+0.756$	evidence both ways
Accept–reject Beta(3, 3)	$c = 1.875$, acceptance 0.533	$1/c$
Coal switch point	**1891**, $P = 0.240$; $\lambda_1 = 3.106$, $\lambda_2 = 0.930$	Gibbs = exact
Gibbs on $N_2(0, \rho)$	$\rho_h = \rho^{2h}$; IF 999.5 at $\rho = 0.999$	dependence costs draws
RW-MH acceptance on $N(0, 1)$	$\frac{2}{\pi}\arctan(2/s)$; 0.445 at $s = 2.38$	scale trade-off
Hospitals	separate 0.029–0.130 $\to$ hierarchical 0.057–0.089	weight $\eta/(\eta + n_i)$

Quantity	Value	Meaning
Shrinkage at $y = 6$	normal 3.00, Laplace 4.59, horseshoe 5.66	signals survive the horseshoe

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Figures

Built by `figures/make_figures.py`. Run: `python Notas/L03/figures/make_figures.py`. Every figure asserts at least one plotted number.

Figure	Shows	Lives in
L03-sequential-learning	coin posteriors vs heads in a row; G Fig. 2.2 Beta update; Pareto–Uniform posterior band vs n	L03-01-bayes-rule-discrete §5, L03-02-conjugate-sequential-updating §2
L03-normal-four-priors	priors A–D and likelihood; four posteriors; posterior mean vs x for normal and t priors	L03-03-normal-measurement-physicists §5
L03-loss-functions	Beta(2, 10) with mode/median/mean; expected-loss curves and their minima	L03-04-bayes-estimators-loss §7
L03-bayes-factor-sample-size	$\log_{10} B_{12}$ vs proportions for $m = 10, 100$ (Table 3.2); growth with m vs the BIC line	L03-05-model-comparison-bayes-factors §4
L03-monte-carlo-accept-reject	accept–reject for Beta(3, 3); running $\hat{\pi}$ in its ± 2 se funnel	L03-06-monte-carlo-simulation §2
L03-gibbs-switch-point	exact $P(k \mid y)$ vs Gibbs frequencies; rate marginals	L03-07-markov-chains-gibbs §6
L03-mcmc-mixing	Gibbs ACF ρ^{2h}; RW-MH acceptance curve	L03-07-markov-chains-gibbs §8, L03-08-metropolis-hastings-hmc §4
L03-shrinkage	hospital rates shrunk toward $E(\mu \mid y)$; shrinkage profiles of normal, Laplace, horseshoe priors	L03-09-hierarchical-models-shrinkage §4, L03-10-sparsity-priors §5

Interactive companions

- The Prior-Conflict Dial (`companions/companion-normal-prior-conflict.html`). Move the observation x, the instrument's σ, the prior mean and sd, and the t degrees of freedom; watch the normal and Student-t posteriors, the posterior-mean-vs-x curve and the **two posterior means** side by side, with the normal weight w and the t pull. Defaults reproduce slide 40 (890.0, 17.9) and prior C (891.1). Linked from L03-03-normal-measurement-physicists §6.
- The Bayes-Factor Dial (`companions/companion-bayes-factor-coins.html`). Move m, Michaela's and Lila's heads, the Beta(α, α) prior and the prior odds; watch $\log_{10} B_{12}$ with its Jeffreys label against Lila's count, the posteriors under each model, $P(M_1 \mid y)$ and

the large-sample approximation. Buttons load the ten rows of Table 3.2. Linked from L03-05-model-comparison-bayes-factors §4.

Both pages are verified by `companions/check_companions.py` against their own JavaScript run under `node` at both ends of every range, the default and landmark vectors (Table 3.2 reproduced to three decimals).

Scope and ceiling

- **Examinable (C):** the class-plan goals 1–9 as mapped in L03-11-formulas §0: Bayes' rule, predictive, sequential and conjugate updating, the normal model, Bayes estimators and losses, credible intervals, Bayes factors and posterior model probabilities, Monte Carlo (inverse cdf, composition, accept–reject, importance sampling, SIR), Markov chains, Gibbs, Metropolis–Hastings, convergence and n.s.e., hierarchical Beta–Binomial and Gaussian models, ridge/lasso as MAP, scale mixtures, normal–gamma and horseshoe.
- **Named only (G):** coherence, minimax, Chib's marginal likelihood, Rao–Blackwellisation, HMC/NUTS, Pólya–Gamma, spike-and-slab and reversible jump, SV estimation details.
- **Out of scope here:** sequential Monte Carlo (not on the site), Bayesian VARs (Class 9), the SV model as a volatility model (Class 8), probit/Tobit as models (Class 7).
- **Final exam:** no code. Expect derivations (posteriors, predictives, full conditionals), Bayes-factor arithmetic and the reading of MCMC output.

Errata found

- **Ingredients deck**, slide 40 prints physicist B's prior as $N(800, 40^2)$ and slide 44 as $N(800, 1600)$; the posterior $N(840, 35.7^2)$ and `fisicos-ABCD.html` require $N(800, 80^2)$. Slide 19 prints 0.8373093 for $0.3 \times 2.790698 = 0.8372093$ → L03-03-normal-measurement-physicists §1, L03-01-bayes-rule-discrete §3.
- `sequential-bayes-bernoulli-uniform-trials.html`: the Pareto posterior mean is typed $\theta_1\alpha_1/(\alpha_1 + 1)$ (the code uses $\alpha_1 - 1$); the prior density has a stray a → L03-02-conjugate-sequential-updating §6.
- `fisicos-ABCD.html` writes prior D with $N(\mu_A, \sigma_A^2)$ twice (the code uses A, B, C) → L03-03-normal-measurement-physicists §5.
- `correlationcoefficient.html` states the arc-sine prior but codes `1/sqrt(1+rho^2)` → L03-06-monte-carlo-simulation §7.1.
- `ZIP-fishing.html` says $n = 42$; its table and estimates need $n = 45$ → L03-06-monte-carlo-simulation §7.2.
- `hierarchical-gaussian-linear-model.html`: V_i printed without the inverse, $X_i'y_i$ without $1/\sigma^2$, and β_i's prior with τ_α (the code is right) → L03-09-hierarchical-models-shrinkage §5.
- **Bayes-estimator deck**, slide 13: "the MLE is unbiased by construction" fails for the exponential ($E = n\theta/(n-1)$) → L03-04-bayes-estimators-loss §8.
- **Greenberg**: Algorithm 5.4's acceptance probability is 0.7602, not "$1/c = 0.6577$"; Algorithm 5.5's importance sampler has infinite variance, so its reported n.s.e. is not meaningful → L03-06-monte-carlo-simulation §5–§6.
- **Stone Fig. 1.4** prints $p(\theta_s) = 0.0011$ but computes with 0.001 → L03-01-bayes-rule-discrete §6.

Course-wide position

Class	Relation to Classes 3+4
1 — review	the Exp–Gamma coal posterior (L01-04) becomes conjugate analysis; the integrated change point becomes the Gibbs switch point (L03-07 §6); p-value vs Bayes factor (L01-08) becomes model comparison; MSE becomes Bayes risk → L01 §4, §3, §8
2 — statistical learning	the logit by IRLS becomes the Bayesian logit by MH; bias–variance becomes shrinkage; cross-validated λ becomes a prior → L02 §4, §1
3+4	**prior, posterior, predictive; conjugacy; Bayes factors; Monte Carlo and MCMC; hierarchical and sparsity priors**
5+6 — causal inference	Bayesian IV and treatment models (G Ch. 11) reuse Gibbs with latent data
7 — more econometric models	Student-t errors as scale mixtures, hierarchical models, Tobit and probit by data augmentation (G §8.1–8.2)
8 — univariate time series	stochastic volatility by MCMC (T §12.7; the Petrobras page)
9 — multivariate time series	Bayesian VARs with shrinkage priors; factor SV

Rules packs: rules/07 (prior, posterior, predictive, conjugacy) · rules/08 (Monte Carlo, MCMC) · scope rules/09.

Bibliography actually used in these notes

Book / source	Where
The Classes 3+4 decks and site pages (`Aula/class-03-04/materials/`) — the spine	everywhere
Greenberg 1e §2.1–2.2, Ch. 3, Ch. 4, Ch. 5, Ch. 6, Ch. 7, §8.1–8.2	all notes
Tsay AFTS 3e §12.1–12.4, §12.7	notes 02, 03, 07, 08
Bolfarine & Sandoval (2010) §4.2–4.4, §5.5	notes 02, 03, 04
Morettin & Singer (2021) App. B §B.2, §B.6; App. C §C.5	notes 06, 07
Izbicki & Mendonça (2020) §3.6; **ISLP** §6.2	note 10
Carvalho, Polson & Scott (2010); **Griffin & Brown (2010)**; **Stone (2013)** as reproduced in the decks	notes 01, 10

Related project material: Aula/COURSE.md (the site's Classes 3+4 list) · Map/class-plan.md (goals, readings, practice) · Map/exercises-index.md (exercise IDs). The book-exercise list, quiz, audio prompt and narration for this unit are produced separately (see `.agent/plans/all-units-study-set.md`).

Practice (number and printed page only; statements in the books): G 2.2 (p. 19), 2.4 (p. 19), 3.3 (p. 39), 3.4 (p. 39), 3.2 (p. 39), 3.8 (p. 40), 4.5 (p. 58); G 5.4 (p. 75), 6.2 (p. 87), 7.1 (p. 105), 7.3 (p. 106), 7.5 (p. 106), 7.8 (p. 107); G 8.1 (p. 132), 8.8 (p. 132); T 12.1–12.3 (p. 670); B 4.3 (p. 71), 4.5 (p. 71), 4.9 (p. 72), 4.10 (p. 72), 5.11 (p. 89); MS B.16 (p. 478), B.19 (p. 478), C.5 (p. 489).

Ingredients deck slides 1–25, three coins, Stone · Bayes' rule on a finite parameter space: posterior = prior × likelihood / predictive

Claim. When the unknown takes finitely many values, Bayes' rule is arithmetic: multiply each prior probability by the likelihood of what was seen, add the products to get the **predictive** probability of the data, and divide. Diego's positive test moves $P(\text{ill})$ from **0.70** to **0.847**, and a second, better test moves it to **0.993**. Three heads in a row leave coin 1 (the fair one) with **0.0674**. The likelihood alone answers a different question: smallpox explains the spots better than chickenpox ($0.9 > 0.8$), yet chickenpox is **90 times** more probable because it is 100 times more common. Updating one datum at a time gives exactly the posterior of updating on all data at once. Prev: — · Next: L03-02-conjugate-sequential-updating · Map: L03-00-map

Notation.

Symbol	Meaning	Range / value
θ	state of nature (the unknown)	finite set, e.g. $\{0,1\}$, {coin 1, 2, 3}
x, y	observed data (test results, tosses)	discrete
$p(\theta)$	prior probability	sums to 1 over θ
$p(x \mid \theta)$	likelihood: probability of the data under θ, read as a function of θ	$[0,1]$
$p(x)$	predictive (marginal) probability of the data	$\sum_\theta p(x \mid \theta)p(\theta)$
$p(\theta \mid x)$	posterior probability	sums to 1 over θ
H	the information held at a given moment	—

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1 · U — two questions that look alike

Diego feels unwell. His doctor wants $P(\text{ill} \mid \text{test positive})$. The number that comes out of the laboratory's validation study is $P(\text{test positive} \mid \text{ill}) = 0.95$. The deck's slide 14 calls confusing the two "the common (and wrong!) answer". Bayes' rule is the bridge between **forward probability** (from the state of nature to the data: what the laboratory measures) and **inverse probability** (from the data back to the state: what the doctor needs). Stone's Fig. 1.12 draws the two directions as two loops: $\theta_{true} \to$ physics $\to x$ on top, and $x \to$ model $\to$ likelihood $\to$ Bayes' rule $\to \theta_{est}$ on the bottom, with the prior entering only on the way back.

The likelihood is a forward quantity read backwards. It is a probability over data, but in Bayes' rule it is used as a function of θ with the data fixed. It does not sum to one over θ (Greenberg, p. 14). The prior is what turns it into a probability over θ.

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2 · C — Bayes' rule (Greenberg (2.1)–(2.3))

Start from the definition of conditional probability, Greenberg's Axiom 4: $P(A \mid B) = P(AB)/P(B)$. Since $P(AB) = P(BA)$,

$$P(A \mid B)P(B) = P(B \mid A)P(A). \tag{1}$$

Divide both sides of (1) by $P(B) > 0$:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}. \tag{2}$$

Set $A = \theta$ and $B = x$ in (2). This is Greenberg's (2.1) for discrete data:

$$p(\theta \mid x) = \frac{p(x \mid \theta)\, p(\theta)}{p(x)}. \tag{3}$$

The denominator comes from the **law of total probability** (question0.pdf slides 12–14): the events "$\theta = \theta_k$" partition the sample space, so

$$p(x) = \sum_k p(x \mid \theta_k)\, p(\theta_k). \tag{4}$$

(4) is the **predictive probability of the data**: what the model and the prior together predict before x is seen. The ingredients deck writes (3) in words on slide 18:

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{predictive}}. \tag{5}$$

Because $p(x)$ does not depend on θ, (3) is often written as Greenberg's (2.3):

$$p(\theta \mid x) \;\propto\; p(x \mid \theta)\, p(\theta), \tag{6}$$

and the constant is recovered at the end by making the posterior sum to one.

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3 · Worked — Diego, one test (slides 4–19)

Model. $\theta = 1$ if Diego is ill, $\theta = 0$ if not. The test X has

	$\theta = 0$ (healthy)	$\theta = 1$ (ill)
$P(X = 1 \mid \theta)$	0.40 (false positive)	0.95
$P(X = 0 \mid \theta)$	0.60	0.05 (false negative)
prior $p(\theta)$	0.30	0.70

Maximum likelihood (slides 8–9). $X = 1$ is more likely under $\theta = 1$:

$$\frac{P(X=1 \mid \theta=1)}{P(X=1 \mid \theta=0)} = \frac{0.95}{0.40} = 2.375, \qquad \text{so } \hat{\theta}_{MLE} = 1.$$

Predictive (slide 12). Substitute the table into (4):

$$P(X=1) = (0.40)(0.30) + (0.95)(0.70) = 0.120 + 0.665 = 0.785. \tag{7}$$

Posterior (slide 16). Substitute (7) and the table into (3):

$$P(\theta=1 \mid X=1) = 0.70 \times \frac{0.95}{0.785} = 0.70 \times 1.210191 = \mathbf{0.8471338}. \tag{8}$$

The factor $0.95/0.785 = 1.210$ is the **likelihood relative to the predictive**: the datum raises the probability of every state under which it was more likely than average.

If instead $X = 0$ (slide 19). The likelihood ratio is $0.60/0.05 = 12$, so $\hat{\theta}_{MLE} = 0$, and

$$P(X=0) = (0.60)(0.30) + (0.05)(0.70) = 0.180 + 0.035 = 0.215,$$

$$P(\theta=0 \mid X=0) = 0.30 \times \frac{0.60}{0.215} = 0.30 \times 2.790698 = \mathbf{0.8372093}.$$

(The slide prints 0.8373093. The product of its own two factors is 0.8372093, so the printed value has a typo in the fourth decimal.)

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4 · C — sequential learning: yesterday's posterior is today's prior (slides 20–25; Greenberg §3.1.3)

The doctor orders a second test Y, more reliable than X: $P(Y=0 \mid \theta=1) = 0.01$ and $P(Y=1 \mid \theta=0) = 0.04$. The tests are conditionally independent given θ.

General statement (Greenberg (3.1)). For two data sets y_1, y_2:

$$p(\theta \mid y_1, y_2) \propto f(y_1, y_2 \mid \theta) p(\theta) = f(y_2 \mid y_1, \theta) f(y_1 \mid \theta) p(\theta) = f(y_2 \mid y_1, \theta)\, p(\theta \mid y_1) \cdot f(y_1), \tag{9}$$

where the last step uses (6) for $p(\theta \mid y_1)$, and $f(y_1)$ is a constant. With conditional independence, $f(y_2 \mid y_1, \theta) = f(y_2 \mid \theta)$, so (9) is Bayes' rule with $p(\theta \mid y_1)$ **in the place of the prior**.

Predictive for Y given $X = 1$ (slide 21). Apply (4) with the posterior (8) as the weights:

$$P(Y=1 \mid X=1) = (0.04)(0.1528662) + (0.99)(0.8471338) = 0.0061146 + 0.8386625 = 0.8447771. \tag{10}$$

Posterior after both (slide 22). Apply (3) with prior (8) and predictive (10):

$$P(\theta=1 \mid X=1, Y=1) = 0.8471338 \times \frac{0.99}{0.8447771} = 0.8471338 \times 1.171907 = \mathbf{0.992762}. \tag{11}$$

Batch check. Using both tests at once, the likelihood of $(X = 1, Y = 1)$ is $0.95 \times 0.99 = 0.9405$ under $\theta = 1$ and $0.40 \times 0.04 = 0.016$ under $\theta = 0$:

$$P(\theta = 1 \mid X = 1, Y = 1) = \frac{0.70 \times 0.9405}{0.70 \times 0.9405 + 0.30 \times 0.016} = \frac{0.65835}{0.65835 + 0.0048} = \frac{0.65835}{0.66315} = 0.992762.$$

The sequential and batch answers agree to every digit, as (9) says they must.

The deck's summary (slide 23).

Information H	$P(\theta = 1 \mid H)$
before X and Y	0.70
after $X = 1$	0.847
after $X = 1$, $Y = 1$	0.993
after $Y = 1$ alone	$\dfrac{0.99 \times 0.70}{0.99 \times 0.70 + 0.04 \times 0.30} = \dfrac{0.693}{0.705} = 0.982979$

> **"Don't consider test X, unless it is 'cost' free" (slide 25).** Y alone gets to 98.3%; adding X buys one more point. The value of a datum is how far it moves the posterior, and that depends on what is already known.

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5 · Worked — three coins (question0.pdf, question0-R.txt)

Coins with $P(\text{head}) = 0.5,\ 0.9,\ 1.0$, picked at random: $p(\text{coin } k) = 1/3$. Three tosses give three heads.

Likelihoods (slide 6). $(0.5)^3 = 0.125$, $(0.9)^3 = 0.729$, $(1.0)^3 = 1.000$.

Predictive (slide 14). By (4):

$$P(HHH) = (0.125)\tfrac{1}{3} + (0.729)\tfrac{1}{3} + (1.000)\tfrac{1}{3} = \frac{1.854}{3} = 0.618. \tag{12}$$

Posteriors (slides 15–17). By (3):

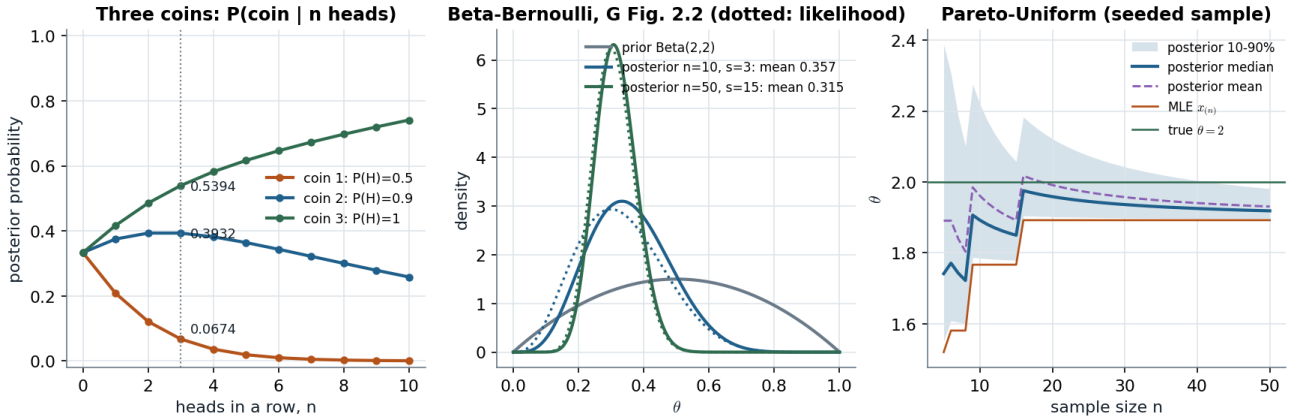
$$P(\text{coin } 1 \mid HHH) = \frac{(0.125)(1/3)}{0.618} = \frac{0.0416667}{0.618} = \mathbf{0.0674},$$

$$P(\text{coin } 2 \mid HHH) = \frac{0.243}{0.618} = \mathbf{0.3932}, \qquad P(\text{coin } 3 \mid HHH) = \frac{0.333333}{0.618} = \mathbf{0.5394}.$$

"What if a fourth flip turned out head as well?" (slide 18). Treat the three-head posterior as the prior, by (9):

$$P(\text{coin } k \mid H^4) \propto p_k \times P(\text{coin } k \mid HHH) : \quad 0.5(0.0674) = 0.0337,\ 0.9(0.3932) = 0.3539,\ 1.0(0.5394) = 0.5394.$$

The sum is 0.9270, so the posteriors are 0.0364, 0.3818, 0.5819. The batch route gives the same: $p_k^4 / \sum_j p_j^4 = 0.0625/1.7186,\ 0.6561/1.7186,\ 1/1.7186$.



Left panel: (3) with $p(\text{coin } k) = 1/3$ and likelihood p_k^n, the R script `question0-R.txt` *evaluated exactly. Middle and right: the conjugate updates of L03-02-conjugate-sequential-updating.*

Read the left panel from left to right. Every head multiplies coin 1's weight by 0.5, coin 2's by 0.9 and coin 3's by 1, and renormalises. Coin 2 first gains (it explains heads better than coin 1) and then loses (coin 3 explains them better still). Coin 3 can never be ruled in with certainty by heads alone, because coin 2 also produces them: after ten heads it holds 0.741. One tail would settle it, because its likelihood under coin 3 is 0.

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6 · Worked — base rates (Stone 2013, Figs 1.2–1.7)

Spots (Fig. 1.4). Symptoms x; chickenpox θ_c with $p(x \mid \theta_c) = 0.8$ and $p(\theta_c) = 0.1$; smallpox θ_s with $p(x \mid \theta_s) = 0.9$ and $p(\theta_s) = 0.001$. Stone takes $p(x) = 0.081$:

$$p(\theta_c \mid x) = \frac{0.8 \times 0.1}{0.081} = 0.988, \qquad p(\theta_s \mid x) = \frac{0.9 \times 0.001}{0.081} = 0.011. \tag{13}$$

The likelihood favours smallpox; the posterior favours chickenpox by a factor $0.988/0.011 \approx 90$. The ratio of posteriors is the ratio of likelihoods times the ratio of priors: $(0.8/0.9) \times (0.1/0.001) = 0.889 \times 100 = 88.9$. (The figure's prior box prints $p(\theta_s) = 0.0011$; its arithmetic uses 0.001. $p(x) = 0.081$ is rounded: the two diseases alone give $0.08 + 0.0009 = 0.0809$, which yields 0.989 and 0.011.)

"Four candles" (Figs 1.5–1.7). The acoustic signal fits "fork handles" a little better (0.7 vs 0.6), but "four candles" is nine times more frequent (0.9 vs 0.1): $p(x) = 0.6(0.9) + 0.7(0.1) = 0.54 + 0.07 = 0.61$, so $p(\text{candles} \mid x) = 0.54/0.61 = 0.885$ and $p(\text{handles} \mid x) = 0.07/0.61 = 0.115$.

> **The posterior ratio is the likelihood ratio times the prior ratio.** A datum that is only slightly more likely under a rare state cannot overcome a large difference in base rates.

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7 · C — a discrete prior for a Bernoulli probability (bernoulli-trials-discrete-prior.html)

The same arithmetic works when θ is a probability restricted to a few values. The site page uses $\theta \in \{0.3, 0.5, 0.8\}$ with prior $\{0.4, 0.5, 0.1\}$ and n Bernoulli trials with s_n successes. The likelihood of the sequence is $\theta^{s_n}(1-\theta)^{n-s_n}$, so by (3)

$$P(\theta = \theta_j \mid x_{1:n}) = \frac{p_j\,\theta_j^{s_n}(1-\theta_j)^{n-s_n}}{\sum_{l=1}^{3} p_l\,\theta_l^{s_n}(1-\theta_l)^{n-s_n}}, \tag{14}$$

which is the line `prob[i,] = ptheta*theta^s[i]*(1-theta)^(i-s[i])` of the R code, renormalised.

Two facts are visible in (14). First, it depends on the data only through (s_n, n): the order of the successes is irrelevant (sufficiency). Second, the prior only shifts the starting point. With 40 successes in 50 trials, (14) gives $P(\theta = 0.8 \mid x) = 0.99967$ even though the prior gave $\theta = 0.8$ only 10%: the likelihood ratio $(0.8/0.5)^{40}(0.2/0.5)^{10} = 1.5 \times 10^4$ swamps the prior odds of 0.1/0.5. (The page simulates with `set.seed(12345)` in R; Python cannot reproduce that stream, so the notes quote the formula and this deterministic case, not the page's path.)

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8 · G — subjective probability and coherence (Greenberg §2.1.2, pp. 8–11)

Greenberg grounds the prior in **subjective probability**: a probability is the price at which you would buy or sell a bet that pays 1 if the event occurs. **Coherence** (no set of bets that loses for sure) forces these prices to obey the axioms of probability, so Bayes' rule applies to beliefs as it does to frequencies. The frequentist reading (§2.1.1) cannot give θ a distribution, because a parameter is not the outcome of a repeated experiment. Named here for orientation; the coherence argument is not examinable.

Links

- Next: the same rule with a continuous θ and conjugate families, L03-02-conjugate-sequential-updating. Comparing coins as **models**: L03-05-model-comparison-bayes-factors §5. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §1.
- Earlier: the first Bayesian example (Exp prior, Poisson data), L01 §4; a p-value is not $P(H_0 \mid x)$, L01 §8.
- Rules: rules/07 (prior, posterior, predictive, sequential Bayes).
- Reading: G §2.1–2.2 (pp. 7–18), §3.1.3 (pp. 24–25).
- Practice (statements in the books): **G 2.1** (p. 19, coherence) is background; the class list starts at **G 2.2** (p. 19), see L03-02-conjugate-sequential-updating.

Site page "Sequential Bayesian Learning", G §2.2, §3.1.3–3.1.4, §3.2.3 · Conjugate updating: the posterior adds the data to the prior's pseudo-data

Claim. For a conjugate pair the posterior is the prior's family with **updated parameters**, and the update is a count: the Beta prior adds successes and failures, the Gamma prior adds events and exposure, the Pareto prior raises its lower bound to the sample maximum and adds n to its shape. The posterior mean is a **weighted average** of the prior mean and the MLE, with the prior's weight falling like $1/n$. Because the update is a count, updating in batches or one observation at a time gives the same posterior. The **predictive** probability that the next Bernoulli trial succeeds equals the posterior mean of θ. With a uniform prior the posterior **mode is the MLE** s/n; the mean $(1+s)/(2+n)$ is not. Prev: L03-01-bayes-rule-discrete · Next: L03-03-normal-measurement-physicists · Map: L03-00-map

Notation.

Symbol	Meaning	Range / value
$x_1, \dots, x_n$	data (Bernoulli, Poisson, exponential or uniform)	—
$s_n = \sum x_i$	successes (Bernoulli) or total count (Poisson)	$0, \dots, n$ / $0, 1, \dots$
α_0, β_0	Beta prior parameters (Greenberg's α, β; Bolfarine's a, b)	> 0
α_1, β_1	posterior parameters	> 0
$x_{(n)}$	sample maximum	> 0
θ_0, α_0 (Pareto)	Pareto lower bound and shape	> 0
$B(a, b)$	Beta function $\Gamma(a)\Gamma(b)/\Gamma(a+b)$	—

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1 · U — the prior as data seen earlier

Greenberg's reading (p. 17): in a Beta(α_0, β_0) prior, α_0 is "the number of heads obtained in the experiment on which the prior is based" and β_0 the number of tails. After the data, the posterior is what you would have believed had you seen both experiments. This is why conjugate updating is sequential by construction, and why a prior can be judged by asking **how many observations it is worth** ($\alpha_0 + \beta_0$ for the Beta, β_0 periods of exposure for the Gamma, as in L01-04 where the Exp(1/2) prior was worth half a year).

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2 · C — Beta–Bernoulli (G §2.2; site page §1; B Ex. 4.4.1)

Likelihood (Greenberg (2.4)). For n independent tosses,

$$p(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{s_n}(1-\theta)^{n-s_n}. \tag{1}$$

Prior. $\theta \sim \text{Beta}(\alpha_0, \beta_0)$:

$$p(\theta) = \frac{\theta^{\alpha_0-1}(1-\theta)^{\beta_0-1}}{B(\alpha_0,\beta_0)}, \quad E(\theta) = \frac{\alpha_0}{\alpha_0+\beta_0}, \quad V(\theta) = \frac{\alpha_0\beta_0}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)}, \quad \text{Mode} = \frac{\alpha_0-1}{\alpha_0+\beta_0-2}. \tag{2}$$

Posterior. Substitute (1) and (2) into $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$ (L03-01-bayes-rule-discrete (6)), dropping $B(\alpha_0, \beta_0)$:

$$p(\theta \mid x) \;\propto\; (\theta^{s_n}(1-\theta)^{n-s_n})(\theta^{\alpha_0-1}(1-\theta)^{\beta_0-1}) = \theta^{(\alpha_0+s_n)-1}(1-\theta)^{(\beta_0+n-s_n)-1}. \tag{3}$$

(3) is a Beta kernel:

$$\theta \mid x \;\sim\; \text{Beta}(\alpha_1, \beta_1), \qquad \alpha_1 = \alpha_0 + s_n, \quad \beta_1 = \beta_0 + n - s_n. \tag{4}$$

Posterior mean as a weighted average (Greenberg (2.5)). By (2) applied to (4),

$$E(\theta \mid x) = \frac{\alpha_0 + s_n}{\alpha_0 + \beta_0 + n}. \tag{5}$$

Split the numerator into α_0 and $s_n = n\bar{x}$:

$$E(\theta \mid x) = \underbrace{\frac{\alpha_0+\beta_0}{\alpha_0+\beta_0+n}}_{w} \cdot \frac{\alpha_0}{\alpha_0+\beta_0} + \underbrace{\frac{n}{\alpha_0+\beta_0+n}}_{1-w} \cdot \bar{x}. \tag{6}$$

Check: the first product is $\alpha_0/(\alpha_0+\beta_0+n)$, the second $s_n/(\alpha_0+\beta_0+n)$; they add to (5).

Greenberg's Fig. 2.2 in numbers (prior Beta(2,2), mean 0.5):

Data	Posterior (4)	w in (6)	$E(\theta \mid x)$ by (6)	MLE
$n = 10$, $s = 3$	Beta(5, 9)	$4/14 = 0.2857$	$0.2857(0.5) + 0.7143(0.3) = 0.1429 + 0.2143 = 0.3571$	0.3
$n = 50$, $s = 15$	Beta(17, 37)	$4/54 = 0.0741$	$0.0741(0.5) + 0.9259(0.3) = 0.0370 + 0.2778 = 0.3148$	0.3

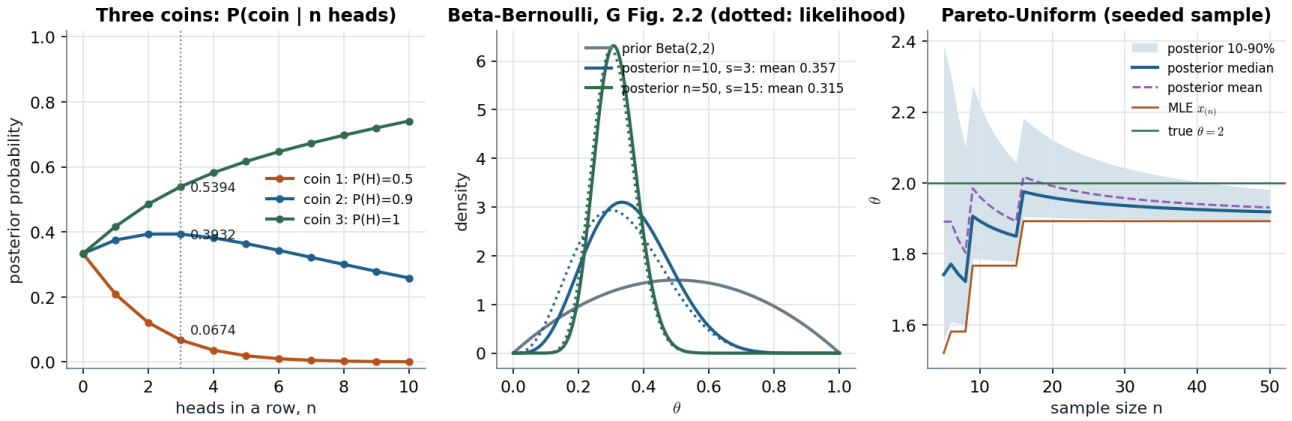
The same proportion of heads, five times the data: the prior's weight falls from 29% to 7%.

Uniform prior (site page; B (4.4.10)–(4.4.11)). $\alpha_0 = \beta_0 = 1$ in (4)–(5) and (2):

$$\theta \mid x \sim \text{Beta}(1+s_n, 1+n-s_n), \quad E(\theta \mid x) = \frac{1+s_n}{2+n}, \quad \text{Mode}(\theta \mid x) = \frac{(1+s_n)-1}{(2+n)-2} = \frac{s_n}{n} = \hat{\theta}_{MLE}, \tag{7}$$

$$V(\theta \mid x) = \frac{(1+s_n)(1+n-s_n)}{(2+n)^2(3+n)}. \tag{8}$$

With $s = 1$ in $n = 10$ (the sample of L01-01): mode 0.100, mean $2/12 = 0.167$, posterior sd $\sqrt{2 \times 10/(144 \times 13)} = 0.103$.



Middle: Greenberg Fig. 2.2 recomputed from (4). Right: Pareto–Uniform update (13) on a seeded $U(0, 2)$ sample, prior Pareto(1.5, 0.1) as on the site page. Left panel: L03-01-bayes-rule-discrete §5.

In the middle panel each dotted likelihood (normalised to integrate to one) sits almost on top of its posterior; the prior only nudges the posterior toward 0.5, and less so for $n = 50$. That is (6) seen as densities: posterior $\approx$ likelihood when w is small.

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3 · C — sequential updating is exact (G §3.1.3, p. 25)

First experiment: n_1 trials, s_1 successes. By (4) the posterior is Beta$(\alpha_0 + s_1,\ \beta_0 + n_1 - s_1)$. Use it as the prior for the second experiment $(n_2,\ s_2)$ and apply (4) again:

$$\theta \mid s_1, s_2 \sim \text{Beta}((\alpha_0{+}s_1){+}s_2,\ (\beta_0{+}n_1{-}s_1){+}(n_2{-}s_2)) = \text{Beta}(\alpha_0{+}(s_1{+}s_2),\ \beta_0{+}(n_1{+}n_2){-}(s_1{+}s_2)), \tag{9}$$

which is the posterior from Beta(α_0, β_0) and all $n_1 + n_2$ trials at once. The site page's recursion `alphan = cumsum(x)+1; betan = 2:(n+1)-cumsum(x)` is (9) applied one trial at a time.

> **The order and grouping of the data do not matter; only the sufficient statistic does.** This is L03-01-bayes-rule-discrete (9) for a continuous θ.

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4 · C — the predictive probability of the next trial (G §3.2.3, pp. 32–33)

Introduce θ and integrate it out:

$$P(x_{n+1} = 1 \mid x) = \int_0^1 P(x_{n+1} = 1 \mid \theta)\, p(\theta \mid x)\, d\theta = \int_0^1 \theta\, \frac{\theta^{\alpha_1 - 1}(1-\theta)^{\beta_1 - 1}}{B(\alpha_1, \beta_1)}\, d\theta. \tag{10}$$

The integrand is the Beta$(\alpha_1 + 1, \beta_1)$ kernel, whose integral is $B(\alpha_1 + 1, \beta_1)$:

$$P(x_{n+1}=1 \mid x)=\frac{B(\alpha_1+1,\beta_1)}{B(\alpha_1,\beta_1)}=\frac{\Gamma(\alpha_1+1)\Gamma(\alpha_1+\beta_1)}{\Gamma(\alpha_1)\Gamma(\alpha_1+\beta_1+1)}=\frac{\alpha_1}{\alpha_1+\beta_1}=\frac{\alpha_0+s_n}{\alpha_0+\beta_0+n}, \tag{11}$$

using $\Gamma(a+1)=a\Gamma(a)$ twice. (11) equals the posterior mean (5): for a Bernoulli variable $E(x \mid \theta)=P(x=1 \mid \theta)=\theta$. With $\alpha_0=\beta_0=1$, (11) is Laplace's rule of succession $(s_n+1)/(n+2)$. For Fig. 2.2's first case, $5/14=0.357$.

> **The predictive averages the model over the posterior; it does not plug in an estimate.** Plugging in the MLE would give s_n/n, which is 0 after n failures, a forecast of certainty that no finite sample justifies.

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5 · C — Gamma–Poisson and Gamma–Exponential (T Results 12.4–12.5; B Ex. 4.4.2; deck slides 4–5)

With a Gamma(α_0,β_0) prior (shape, rate), $p(\lambda)\propto\lambda^{\alpha_0-1}e^{-\beta_0\lambda}$:

Data	Likelihood kernel	Posterior	Posterior mean
Poisson(λ)	$\lambda^{s_n}e^{-n\lambda}$	Gamma$(\alpha_0+s_n,\ \beta_0+n)$	$\dfrac{\alpha_0+s_n}{\beta_0+n}$
Exponential(θ), density $\theta e^{-\theta x}$	$\theta^n e^{-\theta\sum x_i}$	Gamma$(\alpha_0+n,\ \beta_0+\sum x_i)$	$\dfrac{\alpha_0+n}{\beta_0+\sum x_i}$

Each row is (3) with a Gamma kernel: multiply the kernels and add exponents. The Poisson row is Tsay's Result 12.4 and Bolfarine's Ex. 4.4.2; the exponential row is Result 12.5 and the "Modelo Exponencial-Gama" of the Bayes-estimator deck (slides 4–5), used in L03-04-bayes-estimators-loss. L01-04's coal example is the Poisson row with $\alpha_0=1$, $\beta_0=1/2$, $n=41$, $s_n=127$: Gamma(128, 41.5), mean $128/41.5=3.084$.

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6 · C — Pareto–Uniform (site page §2; B Ex. 5.5.2)

Likelihood. For $x_i\sim U(0,\theta)$ iid,

$$p(x_1,\dots,x_n \mid \theta)=\prod_{i=1}^{n}\frac{1}{\theta}I(x_i\le\theta)=\frac{1}{\theta^n}I(x_{(n)}\le\theta). \tag{12}$$

The likelihood is zero below the sample maximum and decreasing above it, so $\hat\theta_{MLE}=x_{(n)}$ (the boundary MLE of L01-05).

Prior. Pareto(θ_0,α_0): $p(\theta)=\alpha_0\theta_0^{\alpha_0}\theta^{-(\alpha_0+1)}$ for $\theta\ge\theta_0$, zero below. (The site page types the numerator as $\alpha_0\theta_0^{\alpha}$ and adds a stray a in the denominator.)

Posterior. Multiply (12) by the prior and keep the factors in θ:

$$p(\theta \mid x)\ \propto\ \Big(\frac{1}{\theta^n}I(\theta\ge x_{(n)})\Big)\Big(\frac{1}{\theta^{\alpha_0+1}}I(\theta\ge\theta_0)\Big)=\frac{1}{\theta^{(\alpha_0+n)+1}}\,I(\theta\ge\max\{x_{(n)},\theta_0\}). \tag{13}$$

(13) is a Pareto kernel:

$$\theta \mid x \sim \text{Pareto}(\theta_1, \alpha_1), \qquad \theta_1 = \max\{x_{(n)}, \theta_0\}, \quad \alpha_1 = \alpha_0 + n. \tag{14}$$

Summaries. For Pareto(θ_1, α_1):

$$E(\theta \mid x) = \frac{\alpha_1 \theta_1}{\alpha_1 - 1} \ (\alpha_1 > 1), \qquad \text{Median} = \theta_1 2^{1/\alpha_1}, \qquad V(\theta \mid x) = \frac{\theta_1^2 \alpha_1}{(\alpha_1 - 1)^2(\alpha_1 - 2)} \ (\alpha_1 > 2). \tag{15}$$

The mean follows from $\int_{\theta_1}^{\infty} \theta\, \alpha_1 \theta_1^{\alpha_1} \theta^{-\alpha_1 - 1} d\theta = \alpha_1 \theta_1^{\alpha_1} \theta_1^{1-\alpha_1}/(\alpha_1 - 1)$. The site page's text writes $E(\theta \mid x) = \theta_1 \alpha_1/(\alpha_1 + 1)$, which is smaller than θ_1 and so outside the support; its code uses (15).

Worked (seeded sample). The site page draws $n = 50$ from $U(0, 2)$ with prior Pareto(1.5, 0.1). Python cannot replay R's seed, so the notes use a seeded NumPy sample with the same design: $x_{(50)} = 1.8923$, so by (14) $\theta_1 = \max\{1.8923, 1.5\} = 1.8923$ and $\alpha_1 = 0.1 + 50 = 50.1$. By (15):

$$E(\theta \mid x) = \frac{50.1 \times 1.8923}{49.1} = 1.9308, \qquad \text{Median} = 1.8923 \times 2^{1/50.1} = 1.9186.$$

The 5% and 95% posterior quantiles are $\theta_1(0.95)^{-1/\alpha_1} = 1.8942$ and $\theta_1(0.05)^{-1/\alpha_1} = 2.0089$; the true $\theta = 2$ lies inside.

> **The posterior never goes below the data's maximum, and the MLE never goes above it.** Every posterior summary sits at or above $x_{(n)}$, which is the bias correction the frequentist $\frac{n+1}{n} x_{(n)}$ of L01-05 makes by hand. The right panel of the figure shows the band hugging $x_{(n)}$ from above and narrowing like $1/n$.

Bayesian interval (B Ex. 5.5.2, (5.5.3)). For coverage $1 - \epsilon$ (Bolfarine's $\gamma = 1 - \alpha$), solving $P(\theta \le t_1 \mid x) = \epsilon/2$ and $P(\theta \ge t_2 \mid x) = \epsilon/2$ with the Pareto cdf $1 - (\theta_1/t)^{\alpha_1}$ gives the equal-tail interval $\left[\theta_1(1 - \epsilon/2)^{-1/\alpha_1},\ \theta_1(\epsilon/2)^{-1/\alpha_1}\right]$ (with $\epsilon = 0.10$, the 1.8942–2.0089 above). It is not the HPD interval, which here starts at θ_1 because the density is decreasing.

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7 · C — large samples (G §3.1.4, pp. 25–27)

Write the posterior as $p(\theta \mid y) \propto p(\theta) \exp[n\bar{l}(\theta \mid y)]$, where $\bar{l}$ is the mean log-likelihood contribution. A second-order expansion of $\bar{l}$ around the MLE $\hat{\theta}$, where the first derivative vanishes, gives

$$p(\theta \mid y) \ \approx \ p(\theta) \exp\Big[-\frac{n}{2v}(\theta - \hat{\theta})^2 \Big], \qquad v = [-\bar{l}''(\hat{\theta} \mid y)]^{-1}, \tag{16}$$

so for large n the posterior is approximately $N(\hat{\theta}, v/n)$, provided $p(\hat{\theta}) \neq 0$. Three consequences (Greenberg p. 27): the prior's role shrinks; the posterior concentrates at the true value (the Bayesian counterpart of consistency); and it is approximately normal around the MLE. For the Beta–Bernoulli, (8) gives $V(\theta \mid x) \approx \bar{x}(1 - \bar{x})/n$, the inverse Fisher information of L01-01. The proviso matters for the Pareto prior: a lower bound θ_0 above the true θ gives the true value zero posterior probability for every n.

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8 · G — other conjugate pairs (named)

Dirichlet–multinomial (G §3.1.2, pp. 23–24): $D(\alpha)$ prior, posterior $D(\alpha + y)$, marginal of θ_1 a Beta. Beta–negative binomial (T Result 12.6, p. 621). Normal–normal and normal–inverse-gamma: L03-03-normal-measurement-physicists. Inverse chi-squared for a variance (T Result 12.8): used inside the Gibbs samplers of L03-07-markov-chains-gibbs.

Links

- Prev: L03-01-bayes-rule-discrete; next: the normal model, where the weight w becomes a ratio of variances, L03-03-normal-measurement-physicists. Point estimates from these posteriors: L03-04-bayes-estimators-loss. Formulas: L03-11-formulas; code: L03-13-python §2.
- Earlier: Exp–Gamma on the coal data, L01 §4; the $U(0, \theta)$ estimators, L01 §5.
- Rules: rules/07 (conjugate families, sequential updating).
- Practice (statements in the books): **G 2.2** (p. 19, gamma–Poisson), **G 2.4** (p. 19, Pareto–uniform), **G 3.2** (p. 39, Poisson predictive), **G 3.4** (p. 39, sequential Poisson updating), **B 4.10** (p. 72, Pareto prior for $U(0, \theta)$), **T 12.1** (p. 670).

Ingredients deck slides 26–52, fisicos-ABCD · The normal measurement model: predictive, posterior, and what a heavy-tailed prior changes

Claim. With $x \mid \theta \sim N(\theta, \sigma^2)$ and a normal prior $N(\mu_0, \tau_0^2)$, the predictive is $N(\mu_0, \sigma^2 + \tau_0^2)$ and the posterior is normal with mean $w\mu_0 + (1-w)x$ and variance $w\tau_0^2$, where $w = \sigma^2/(\sigma^2 + \tau_0^2)$: **precisions add, and the mean is the precision-weighted average.** For $x = 850$, $\sigma = 40$, physicist A (prior $N(900, 20^2)$) ends at $N(890,\ 17.9^2)$ and physicist B ($N(800, 80^2)$) at $N(840,\ 35.8^2)$. Physicist C holds a Student-t_5 prior with A's mean and variance; there is no closed form, and quadrature gives a posterior mean of **891.1**. At $x = 850$ A and C barely differ, but the normal prior pulls by the fixed share w however far x lands, while the t prior's pull **dies away** when data and prior conflict: at $x = 1100$ the posterior means are 940 (normal) and 1022 (t). Prev: L03-02-conjugate-sequential-updating · Next: L03-04-bayes-estimators-loss · Map: L03-00-map

Notation.

Symbol	Meaning	Range / value
θ	the physical quantity	$\mathbb{R}$
x	one measurement	850 (slide 39)
σ	measurement sd	40 (slide 28)
μ_0, τ_0	prior mean and sd (normal prior)	A: 900, 20; B: 800, 80
w	weight on the prior mean, $\sigma^2/(\sigma^2 + \tau_0^2)$	$(0, 1)$
μ_1, τ_1^2	posterior mean and variance	—
h, h_0	precisions $1/\sigma^2$, $1/\tau_0^2$	> 0
$t_\nu(\mu, s^2)$	Student-t with ν df, location μ, scale s; variance $\nu s^2/(\nu-2)$	C: $t_5(900, 240)$

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1 · U — two physicists, one instrument (slides 26–33)

The deck's second example learns a physical constant from one noisy reading. The instrument's error is known: $x \mid \theta \sim N(\theta, 40^2)$ (slide 28). Two physicists bring different experience:

- **A (large experience):** $\theta \sim N(900, 20^2)$;
- **B (not so experienced):** $\theta \sim N(800, 80^2)$.

Slide 33 draws each physicist's **joint** density $p(x, \theta) = p(x \mid \theta)p(\theta)$: a tilted ellipse for A, a broad one for B. Everything below is a slice or a projection of that joint density: the predictive is its projection on the x axis; the posterior is its slice at the observed x.

Erratum. Slide 40 prints B's prior as $N(800, (40)^2)$ and slide 44 as $N(800, 1600)$. The posterior on slide 40, $N(840, 35.7^2)$, and `fisicos-ABCD.html` (`sig.B = 80`) both require $\tau_0 = 80$, which is the value used here.

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2 · C — the predictive distribution (slides 34–36; G Ex. 3.3 setting)

Write the measurement as the quantity plus an independent error:

$$x = \theta + e, \qquad \theta \sim N(\mu_0, \tau_0^2), \quad e \sim N(0, \sigma^2), \quad \theta \perp e. \tag{1}$$

A sum of independent normals is normal, with means and variances added:

$$x \sim N(\mu_0 + 0,\ \tau_0^2 + \sigma^2). \tag{2}$$

This is slide 35's result $x \sim N(\theta_0, \sigma^2 + \tau_0^2)$, obtained there by computing $p(x) = \int p(x \mid \theta)p(\theta)\, d\theta$ and completing the square in θ. Numbers:

	Predictive	Predictive sd
A	$N(900,\ 1600 + 400) = N(900, 2000)$	$\sqrt{2000} = 44.72$
B	$N(800,\ 1600 + 6400) = N(800, 8000)$	$\sqrt{8000} = 89.44$

The predictive is wider than the instrument. Before measuring, a physicist is uncertain both about θ and about the error; (2) adds the two. The observed $x = 850$ is $(850 - 900)/44.72 = -1.12$ predictive sd from A's centre and $(850 - 800)/89.44 = 0.56$ from B's: neither physicist is surprised.

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3 · C — the posterior (slides 37–38; B Ex. 4.4.3; T Result 12.1)

Keep only the factors of $p(x \mid \theta)p(\theta)$ that involve θ:

$$p(\theta \mid x) \propto \exp\Big\{ -\frac{(x-\theta)^2}{2\sigma^2} \Big\} \exp\Big\{ -\frac{(\theta-\mu_0)^2}{2\tau_0^2} \Big\} = \exp\Big\{ -\frac{1}{2}\Big[\frac{\theta^2 - 2\theta x}{\sigma^2} + \frac{\theta^2 - 2\theta\mu_0}{\tau_0^2} \Big] + c \Big\}, \tag{3}$$

where c collects x^2/σ^2 and μ_0^2/τ_0^2. Collect θ^2 and θ:

$$p(\theta \mid x) \propto \exp\Big\{ -\frac{1}{2}\Big[\theta^2 \Big(\frac{1}{\sigma^2} + \frac{1}{\tau_0^2} \Big) - 2\theta \Big(\frac{x}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \Big) \Big] \Big\}. \tag{4}$$

Define the posterior precision and mean

$$\frac{1}{\tau_1^2} = \frac{1}{\sigma^2} + \frac{1}{\tau_0^2}, \qquad \mu_1 = \tau_1^2 \Big(\frac{x}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \Big). \tag{5}$$

Substitute (5) into (4) and complete the square: the bracket is $(\theta^2 - 2\theta\mu_1)/\tau_1^2 = (\theta - \mu_1)^2/\tau_1^2 - \mu_1^2/\tau_1^2$, and the last term is a constant, so

$$\theta \mid x \sim N(\mu_1, \tau_1^2). \tag{6}$$

Slide 37's form. From (5), $\tau_1^2 = \sigma^2\tau_0^2/(\sigma^2 + \tau_0^2)$. Then

$$\mu_1 = \frac{\sigma^2\tau_0^2}{\sigma^2+\tau_0^2}\Big(\frac{x}{\sigma^2}+\frac{\mu_0}{\tau_0^2}\Big) = \frac{\sigma^2}{\sigma^2+\tau_0^2}\mu_0 + \frac{\tau_0^2}{\sigma^2+\tau_0^2}x. \tag{7}$$

With $w = \sigma^2/(\sigma^2+\tau_0^2) \in (0,1)$ (slide 38):

$$E(\theta \mid x) = w\,E(\theta) + (1-w)\,x, \qquad V(\theta \mid x) = w\,V(\theta). \tag{8}$$

> **Precisions add; means are precision-weighted (Tsay, p. 619).** (5) says the posterior precision is the prior precision plus the data precision, and (7) weights each source by its share of the total precision. The weight on the prior, w, is large when the instrument is noisy relative to the prior ($\sigma^2 \gg \tau_0^2$).

Limits (slide 38). As $\tau_0^2 \to \infty$ (a vague prior), $w \to 0$ and $\theta \mid x \to N(x, \sigma^2)$: "the posterior collapses at the observed value x". As $\tau_0^2 \to 0$, $w \to 1$: the data are ignored.

n observations (B Ex. 4.4.3; T Result 12.1). With $x_1, \ldots, x_n$ iid $N(\theta, \sigma^2)$, the sample mean $\bar{x} \sim N(\theta, \sigma^2/n)$ is sufficient, so (5)–(8) hold with σ^2/n in place of σ^2: $w_n = \sigma^2/(\sigma^2 + n\tau_0^2) \to 0$ like $1/n$.

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4 · Worked — physicists A and B at $x = 850$ (slides 39–41)

A: $w = 1600/(1600+400) = 0.8$. Substitute into (8):

$$E(\theta \mid x) = 0.8{\times}900{+}0.2{\times}850 = 720{+}170 = \mathbf{890}, \qquad V(\theta \mid x) = 0.8{\times}400 = 320, \quad \text{sd} = \sqrt{320} = \mathbf{17.89}.$$

B: $w = 1600/(1600+6400) = 0.2$:

$$E(\theta \mid x) = 0.2{\times}800{+}0.8{\times}850 = 160{+}680 = \mathbf{840}, \qquad V(\theta \mid x) = 0.2{\times}6400 = 1280, \quad \text{sd} = \sqrt{1280} = \mathbf{35.78}.$$

Slide 40 prints $N(890, (17.9)^2)$ and $N(840, (35.7)^2)$; 35.78 rounds to 35.8.

> **The experienced physicist barely moves; the inexperienced one mostly follows the data.** A moves 10 units toward x (20% of the 50-unit gap), B moves 40 units (80% of the gap). Both posteriors are narrower than both their prior and the instrument: $17.9 < \min(20, 40)$, $35.8 < \min(80, 40)$.

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5 · C — physicist C: a Student-t prior (slides 44–52)

Prior C (slide 45). $\theta \sim t_5(900, 240)$, whose variance is

$$V(\theta) = \frac{\nu}{\nu-2}s^2 = \frac{5}{3} \times 240 = 400 = V(\text{prior A}). \tag{9}$$

Same centre, same variance as A, but heavier tails: slide 47's log densities show C's log prior falling like $-(\nu+1)\log|\theta-900| = -6\log|\theta-900|$ instead of $-(\theta-900)^2/800$.

No closed form (slide 48). The predictive and the posterior are

$$p(x) = \int \frac{e^{-(x-\theta)^2/2\sigma^2}}{\sqrt{2\pi\sigma^2}} \frac{\Gamma(3)}{\Gamma(5/2)\sqrt{5\pi\,240}} \Big[1+\frac{(\theta-900)^2}{5\times 240}\Big]^{-3} d\theta, \qquad p(\theta \mid x) \propto e^{-(x-\theta)^2/2\sigma^2} \Big[1+\frac{(\theta-900)^2}{1200}\Big]^{-3}, \tag{10}$$

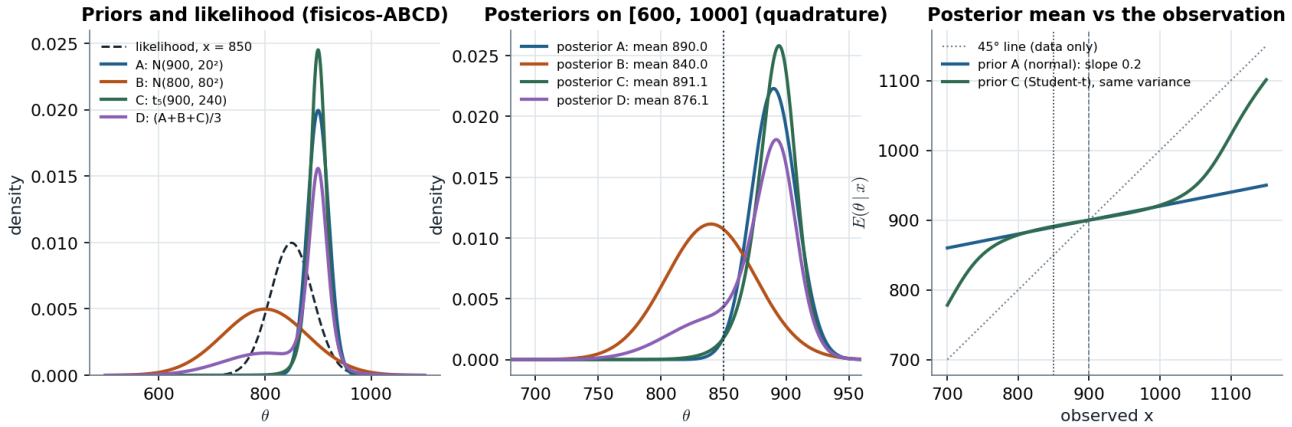
neither of which is a standard form. The deck postpones them to Monte Carlo (footnotes "yet to be learned!", slides 49 and 51); `fisicos-ABCD.html` uses sampling importance resampling (L03-06-monte-carlo-simulation §7). A one-dimensional posterior can also be computed **exactly** by quadrature: evaluate the kernel (10) on a fine grid and normalise.

Prior D (`fisicos-ABCD.html`). The equal mixture $\frac{1}{3}N(900, 20^2) + \frac{1}{3}N(800, 80^2) + \frac{1}{3}t_5(900, 240)$ of the three physicists. (The page's text writes $N(\mu_A, \sigma_A^2)$ twice; its code uses A, B and C.)

The page's table against quadrature. The SIR draws come from $U(600, 1000)$ proposals, so the targets are the posteriors restricted to $[600, 1000]$ (the restriction changes nothing at two decimals).

Posterior	$E(\theta \mid x)$ SIR / exact	Median SIR / exact	$P(\theta < x \mid x)$ SIR / exact	5% quantile SIR / exact
A	890.06 / 890.00	890.10 / 890.00	0.01 / 0.013	860.52 / 860.58
B	840.02 / 840.00	840.09 / 840.00	0.61 / 0.610	781.18 / 781.15
C	891.16 / 891.12	892.45 / 892.41	0.02 / 0.022	860.30 / 860.31
D	876.15 / 876.08	884.35 / 884.30	0.19 / 0.187	806.53 / 806.40

The SIR column agrees with quadrature to Monte Carlo error (a million draws). C's posterior is slightly right of A's and slightly **narrower** (sd 17.70 vs 17.89): near the centre a t prior with A's variance is more peaked than A. D is bimodal-looking with a long left shoulder: B's component keeps values near 800 alive, so its median (884.3) exceeds its mean (876.1).



Left and middle: (6) for A and B, quadrature of (10) for C and D, at $x = 850$, $\sigma = 40$ (slides 39–52; `fisicos-ABCD.html`). Right: $E(\theta \mid x)$ against x for prior A, by (7), and prior C, by quadrature.

The right panel is the comparative static that matters. Under the normal prior the posterior mean is the straight line $0.8 \times 900 + 0.2x$: the pull toward 900 is $0.8(x - 900)$, proportional to the conflict, without limit. Under the t prior the curve hugs the normal line for $|x - 900| \lesssim 100$ and then turns back toward the 45° line: the prior gives way when the data contradict it strongly.

6 · C — why the t prior gives way: bounded influence

Take logs of the posterior kernel and differentiate. The posterior mode $\tilde{\theta}$ solves

$$\frac{x-\tilde{\theta}}{\sigma^2} = \psi(\tilde{\theta}), \qquad \psi(\theta) = -\frac{d}{d\theta}\log p(\theta). \tag{11}$$

The left side is the data's pull toward x; ψ is the prior's pull back toward μ_0.

- **Normal prior:** $\log p(\theta) = -(\theta-\mu_0)^2/2\tau_0^2 + c$, so $\psi(\theta) = (\theta-\mu_0)/\tau_0^2$. It grows linearly without bound. Substituting into (11) gives $\tilde{\theta} = \mu_1$ of (5) for every x.
- **Student-t prior:** $\log p(\theta) = -\frac{\nu+1}{2}\log\left(1+\frac{(\theta-\mu_0)^2}{\nu s^2}\right) + c$, so

$$\psi(\theta) = \frac{(\nu+1)(\theta-\mu_0)}{\nu s^2 + (\theta-\mu_0)^2}. \tag{12}$$

(12) rises to a maximum at $|\theta-\mu_0| = s\sqrt{\nu}$ and then falls to 0. The prior's pull is **bounded** and vanishes in the tails. For C: $s\sqrt{\nu} = \sqrt{240\times 5} = 34.6$, and the largest pull is $\psi = 6/(2\times 34.6) = 0.087$, against a data pull of $(x-\theta)/1600$. When x is far from 900, the data pull wins and $\tilde{\theta} \to x$.

Numbers (quadrature). At $x = 1100$, 4.47 predictive sd from A's centre: normal posterior mean $0.8\times 900 + 0.2\times 1100 = 940.0$; t posterior mean **1022.0**, posterior sd 53.6 (against 17.9): the t posterior is wide because it is torn between the prior and the data.

> **A normal prior is never contradicted; it is averaged in.** A heavy-tailed prior encodes "I am fairly sure, unless the data say otherwise". Greenberg makes the same point about likelihoods in §3.1.1 (pp. 20–21): a Student-t puts more probability in the tails than a normal with the same variance, and the choice can be compared by posterior odds. The Petrobras page applies it to returns (L03-08-metropolis-hastings-hmc §7).
>
> **Interactive companion.** The Prior-Conflict Dial (`companions/companion-normal-prior-conflict.html`) — move the observation x, the instrument's σ, the prior mean and sd, and the t degrees of freedom; watch both posteriors and the posterior-mean-vs-x curve, with the normal pull w and the t pull side by side. Defaults reproduce 890.0 / 17.9 and 891.1; "Set x = μ_0" shows the no-conflict case.

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7 · G — improper priors and the regression version (G §4.2–4.3)

- **Flat prior.** $p(\theta) \propto c$ on $\mathbb{R}$ is improper (§4.2, pp. 43–44); with the normal likelihood it still gives the proper posterior $N(x, \sigma^2)$, the $\tau_0 \to \infty$ limit of §3. It is unusable for Bayes factors (L03-05-model-comparison-bayes-factors §6).
- **Normal linear regression (G (4.2)–(4.7), pp. 45–46).** With $\beta \mid \sigma^2 \sim N_K(\beta_0, \sigma^2 B_0)$ and $\sigma^2 \sim IG(\alpha_0/2, \delta_0/2)$, the posterior is $\beta \mid \sigma^2, y \sim N_K(\bar{\beta}, \sigma^2 B_1)$ with $B_1 = (X'X + B_0^{-1})^{-1}$, $\bar{\beta} = B_1(X'y + B_0^{-1}\beta_0)$, and $\sigma^2 \mid y \sim IG(\alpha_1/2, \delta_1/2)$, $\alpha_1 = \alpha_0 + n$; marginally $\beta \mid y$ is multivariate t (4.7). $\bar{\beta}$ is the matrix version of (7): a precision-weighted average of β_0 and the OLS estimate. Dropping the proportionality to σ^2 (§4.9) loses the closed form and leads to the Gibbs sampler of L03-07-markov-chains-gibbs.

Links

- Prev: L03-02-conjugate-sequential-updating; next: which summary of this posterior to report, L03-04-bayes-estimators-loss. Monte Carlo for physicist C: L03-06-monte-carlo-simulation §7. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §3.
- Earlier: OLS as a projection and the Gaussian likelihood, L01 §6. Later: shrinkage estimators built on (7), L03-09-hierarchical-models-shrinkage, L03-10-sparsity-priors.
- Rules: rules/07 (normal–normal conjugacy, predictive).
- Practice (statements in the books): **G 3.3** (p. 39, normal predictive), **G 4.5** (p. 58, marginal likelihood of the normal–inverse-gamma model), **T 12.1** (p. 670, normal–normal posterior).

Bayes-estimator deck (16 slides), G §3.2.1–3.2.2, B §4.3–4.4, §5.5 · A point estimate is the answer to a loss function

> **Claim.** The posterior is the whole answer; a single number requires a **loss function**. The **Bayes estimator** minimises the posterior expected loss: quadratic loss gives the **posterior mean**, absolute loss the **median**, 0–1 loss the **mode** (MAP), and Greenberg's bilinear loss the quantile of order $a/(a+b)$. With a uniform prior and $s = 1$ success in $n = 10$ trials the posterior is Beta(2, 10): mode **0.100** (= MLE), median **0.148**, mean **0.167**, so the three losses give three different answers. Bayes estimators are usually **biased**, yet they minimise the **Bayes risk** and become the MLE as $n \to \infty$ or as the prior becomes vague. Interval summaries are **credible intervals**: probability statements about θ given the data, with the HPD interval the shortest. Prev: L03-03-normal-measurement-physicists · Next: L03-05-model-comparison-bayes-factors · Map: L03-00-map

Notation (the deck writes $\delta(x)$ and $\pi(\theta \mid x)$; Greenberg $\hat{\theta}$ and $\pi(\theta \mid y)$).

Symbol	Meaning	Range
$L(\theta, a)$	loss when the truth is θ and the estimate is a	≥ 0
$\rho(a \mid x)$	posterior expected loss $E[L(\theta, a) \mid x]$	≥ 0
$\delta(x)$	Bayes estimator, $\arg\min_a \rho(a \mid x)$	—
$F(\cdot \mid x)$	posterior cdf	$[0, 1]$
$R(\theta, d)$	frequentist risk of the rule d, $E_\theta[L(\theta, d(X))]$	—
$r(\pi, d)$	Bayes risk $\int R(\theta, d)\pi(\theta)d\theta$	—

---

1 · U — why the posterior is not yet an answer

A posterior is a distribution. A forecaster who must publish one number, a regulator who must set one threshold, an estimator plugged into the next model: each needs a single a. Which number depends on **what an error costs**. Overestimating a bridge's load capacity costs more than underestimating it; missing a fraud costs more than a false alarm. The loss function writes those costs down, and the Bayes estimator is the number that does best **on average over what is still uncertain about** θ, given everything observed (deck slides 6–8).

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2 · C — loss, expected loss, Bayes estimator (deck slides 6–8; G §3.2.1; B Def. 4.3.1–4.3.2)

For each true value θ and estimate a, $L(\theta, a) \geq 0$ measures the loss. Before the data, the **prior expected loss** of a is $\int L(\theta, a)\pi(\theta)d\theta$ (slide 6). After observing x:

$$\rho(a \mid x) = E[L(\theta, a) \mid x] = \int_\Theta L(\theta, a)\, \pi(\theta \mid x)\, d\theta, \tag{1}$$

$$\delta(x) = \arg\min_a \ \rho(a \mid x). \tag{2}$$

(2) is the deck's definition of the Bayes estimator (slide 8). "The optimal estimator depends on the choice of the loss function." Greenberg (p. 31) stresses that (2) is mechanical: given a loss, there is nothing to invent, only a minimisation.

---

3 · C — quadratic loss gives the posterior mean (slide 9; G p. 30; B Thm 4.4.1)

$L(\theta, a) = (\theta - a)^2$. Write $\theta - a = (\theta - \bar{\theta}) + (\bar{\theta} - a)$ with $\bar{\theta} = E(\theta \mid x)$ and expand inside (1):

$$\rho(a \mid x) = E[(\theta - \bar{\theta})^2 \mid x] + 2(\bar{\theta} - a)E[\theta - \bar{\theta} \mid x] + (\bar{\theta} - a)^2. \tag{3}$$

The middle expectation is $\bar{\theta} - \bar{\theta} = 0$, so

$$\rho(a \mid x) = V(\theta \mid x) + (\bar{\theta} - a)^2. \tag{4}$$

(4) is minimised at $a = \bar{\theta}$:

$$\delta(x) = E(\theta \mid x), \qquad \min_a \rho(a \mid x) = V(\theta \mid x). \tag{5}$$

Greenberg reaches the same by differentiating under the integral: $-2\int(\theta - \hat{\theta})\pi(\theta \mid y)d\theta = 0$. (4) is the posterior twin of MSE = variance + bias² (L01-05): the expectation is over θ given x, not over x given θ.

---

4 · C — absolute loss gives the posterior median (slide 10; G Ex. 3.5 setting)

$L(\theta, a) = |\theta - a|$. Split (1) at a:

$$\rho(a \mid x) = \int_{-\infty}^{a} (a - \theta)\pi(\theta \mid x)d\theta + \int_{a}^{\infty} (\theta - a)\pi(\theta \mid x)d\theta. \tag{6}$$

Differentiate with respect to a (Leibniz rule; the boundary terms are zero because the integrands vanish at $\theta = a$):

$$\frac{d\rho}{da} = \int_{-\infty}^{a} \pi(\theta \mid x)d\theta - \int_{a}^{\infty} \pi(\theta \mid x)d\theta = F(a \mid x) - [1 - F(a \mid x)] = 2F(a \mid x) - 1. \tag{7}$$

(7) is negative below the median and positive above it, so

$$\delta(x) = \text{Median}(\theta \mid x), \quad F(\delta \mid x) = \tfrac{1}{2}. \tag{8}$$

The deck: "more robust than quadratic loss: it penalises errors linearly and is less sensitive to extreme values."

---

5 · C — 0–1 loss gives the posterior mode (slide 11; G Ex. 3.6 setting)

$L(\theta, a) = 1$ if $|\theta - a| \geq \varepsilon$ and 0 otherwise, for a small $\varepsilon > 0$. Then

$$\rho(a \mid x) = P(|\theta - a| \geq \varepsilon \mid x) = 1 - \int_{a-\varepsilon}^{a+\varepsilon} \pi(\theta \mid x) d\theta \approx 1 - 2\varepsilon\, \pi(a \mid x). \tag{9}$$

Minimising (9) means maximising the posterior density at a; as $\varepsilon \to 0$,

$$\delta(x) = \text{Mode}(\theta \mid x) \quad \text{(the maximum a posteriori, MAP, estimate).} \tag{10}$$

---

6 · C — bilinear loss gives a quantile (G p. 29, L_3)

Greenberg's bilinear loss charges a per unit when the estimate is too low ($\theta > \hat{\theta}$) and b per unit when it is too high. Repeating (6)–(7) with the two weights:

$$\frac{d\rho}{d\hat{\theta}} = -a\,[1 - F(\hat{\theta} \mid x)] + b\, F(\hat{\theta} \mid x) = 0 \implies F(\hat{\theta} \mid x) = \frac{a}{a+b}. \tag{11}$$

The Bayes estimator is the posterior quantile of order $a/(a+b)$; $a = b$ gives back the median. If over-estimation is three times as costly as under-estimation ($a = 1$, $b = 3$), report the 25% quantile.

Summary (deck slide 12).

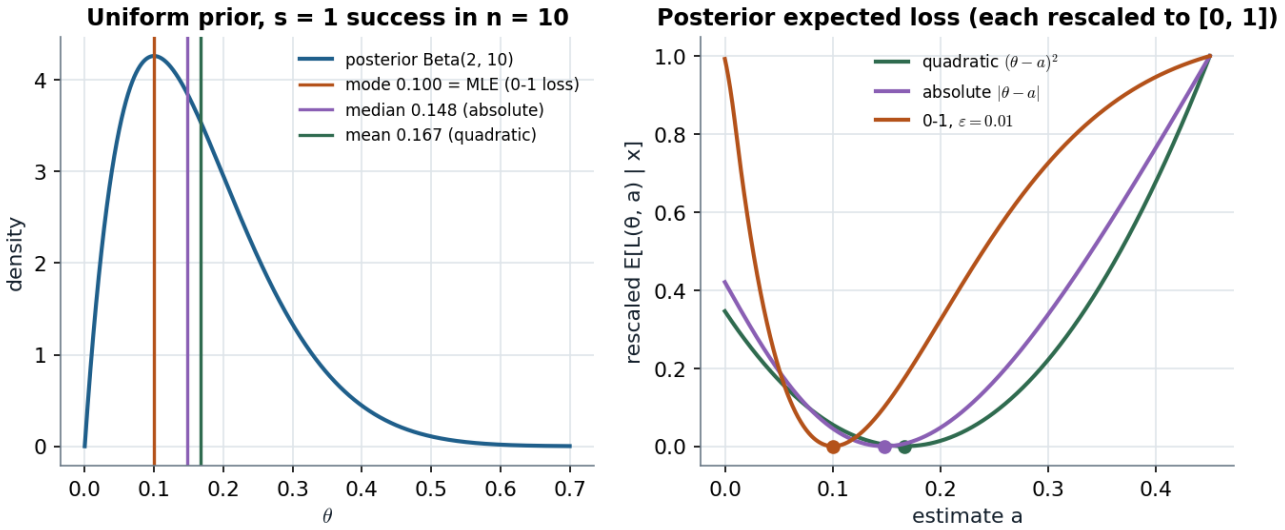
Loss $L(\theta, a)$	Bayes estimator $\delta(x)$
quadratic $(\theta - a)^2$	posterior mean
absolute $\lvert\theta - a\rvert$	posterior median
0–1, $\mathbf{1}[\lvert\theta - a\rvert \geq \varepsilon]$	posterior mode (MAP)
bilinear a/b	posterior quantile $a/(a+b)$

---

7 · Worked — Beta(2, 10): one posterior, four answers

Uniform prior, $s = 1$ success in $n = 10$ trials (L03-02-conjugate-sequential-updating (7)): $\theta \mid x \sim \text{Beta}(2, 10)$.

Estimator	Formula	Value
mode (0–1)	$(\alpha_1 - 1)/(\alpha_1 + \beta_1 - 2) = 1/10$	**0.1000** = MLE s/n
median (absolute)	$F^{-1}(0.5)$	**0.1480**
mean (quadratic)	$\alpha_1/(\alpha_1 + \beta_1) = 2/12$	**0.1667**
25% quantile (bilinear, $a = 1$, $b = 3$)	$F^{-1}(0.25)$	0.0876
minimum expected quadratic loss	$V(\theta \mid x) = \dfrac{2 \times 10}{12^2 \times 13}$	0.010684



Left: the posterior Beta$(2, 10)$ *with its three classical summaries. Right:* $\rho(a \mid x)$ *of (1) for quadratic, absolute and 0–1 (*$\varepsilon = 0.01$*) loss, computed by quadrature and rescaled; dots mark the minima (5), (8), (10).*

For a right-skewed posterior the ordering is **mode < median < mean**: the mean is pulled by the long right tail, the mode ignores it. The 0–1 curve is the steepest and the narrowest: it only rewards landing near the peak. The quadratic curve is the flattest near its minimum, which is why small departures from the mean cost little. The same ordering holds for L01-04's coal posterior Gamma(128, 41.5): mode 3.0602 < median 3.0763 < mean 3.0843, all three close because 41 years of data make the posterior nearly symmetric.

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8 · C — the Exponential–Gamma example (deck slides 4–5, 13–15)

$x_1, \ldots, x_n \mid \theta \sim \text{Exp}(\theta)$ (density $\theta e^{-\theta x}$) and $\theta \sim \text{Gama}(\alpha_0, \beta_0)$. The kernels (slide 4) are $\pi(\theta) \propto \theta^{\alpha_0 - 1} e^{-\beta_0 \theta}$ and $L(\theta; x) \propto \theta^n e^{-\theta \sum x_i}$, so

$$\pi(\theta \mid x) \propto \theta^{(\alpha_0 + n) - 1} e^{-(\beta_0 + \sum x_i)\theta}, \qquad \theta \mid x \sim \text{Gama}\Big(\alpha_0 + n,\ \beta_0 + \sum x_i\Big). \tag{12}$$

Under quadratic loss, by (5),

$$\delta(x) = E(\theta \mid x) = \frac{\alpha_0 + n}{\beta_0 + \sum x_i}, \qquad \hat{\theta}_{EMV} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}. \tag{13}$$

Observation 1 (slide 13): the Bayes estimator is biased. True, and the reason is the prior's pull. But the slide's contrast "while the MLE is unbiased by construction" fails here: with $S = \sum x_i \sim \text{Gama}(n, \theta)$,

$$E\Big[\frac{1}{S}\Big] = \int_0^\infty \frac{1}{s} \frac{\theta^n s^{n-1} e^{-\theta s}}{\Gamma(n)} ds = \frac{\theta^n \Gamma(n-1)}{\Gamma(n) \theta^{n-1}} = \frac{\theta}{n-1}, \quad \text{so} \quad E(\hat{\theta}_{EMV}) = \frac{n}{n-1}\theta \neq \theta. \tag{14}$$

Both estimators are biased; neither is unbiased "by construction". (Maximum likelihood guarantees invariance and large-sample efficiency, not unbiasedness; L01-01 showed the same for the negative binomial.)

Observation 2 (slide 14): asymptotic efficiency. As $n \to \infty$, the prior's contribution (α_0, β_0) in (13) is swamped by $(n, \sum x_i)$, so $E(\theta \mid x) - \hat{\theta}_{EMV} \to 0$ and the Bayes estimator inherits the MLE's efficiency (L03-02-conjugate-sequential-updating §7).

Vague prior (slide 15). $\alpha_0 \to 0$, $\beta_0 \to 0$ in (13) gives $E(\theta \mid x) \to n/\sum x_i$. A seeded illustration: $n = 30$ draws with $\theta = 2$, prior Gama(2, 1): $\delta = 2.1636$ against $\hat{\theta} = 2.1755$.

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9 · C — Bayes risk: why the posterior mean is optimal before the data too (B §4.3–4.4)

The frequentist risk of a rule d is $R(\theta, d) = E_\theta[L(\theta, d(X))]$, an average over data for fixed θ. Averaging it over the prior gives the **Bayes risk** (B Def. 4.3.1):

$$r(\pi, d) = \int R(\theta, d)\pi(\theta)d\theta = \iint L(\theta, d(x)) f(x \mid \theta)\pi(\theta)\, dx\, d\theta. \tag{15}$$

Write $f(x \mid \theta)\pi(\theta) = \pi(\theta \mid x)g(x)$ (B (4.4.3)) and swap the integrals:

$$r(\pi, d) = \int \Big[\int L(\theta, d(x))\pi(\theta \mid x)d\theta \Big] g(x)\, dx = \int \rho(d(x) \mid x)\, g(x)\, dx. \tag{16}$$

(16) is minimised by minimising the bracket for every x, which is (2): **the Bayes estimator minimises the Bayes risk** (B Def. 4.3.2, Thm 4.4.1 for quadratic loss).

Worked (B Ex. 4.4.1). Bernoulli data, uniform prior, quadratic loss: $\delta = (S+1)/(n+2)$ (B (4.4.11)). Its risk is variance plus squared bias:

$$R(\theta, \delta) = \frac{n\theta(1-\theta)}{(n+2)^2} + \Big(\frac{n\theta + 1}{n+2} - \theta\Big)^2 = \frac{n\theta(1-\theta) + (1-2\theta)^2}{(n+2)^2}. \tag{17}$$

Integrating (17) over $\theta \sim U(0,1)$ ($E\theta = \frac{1}{2}$, $E\theta^2 = \frac{1}{3}$):

$$r(\pi, \delta) = \frac{n/6 + 1/3}{(n+2)^2} = \frac{1}{6(n+2)}, \qquad r(\pi, \bar{X}) = \int_0^1 \frac{\theta(1-\theta)}{n} d\theta = \frac{1}{6n}. \tag{18}$$

For $n = 10$: $1/72 = 0.0139$ against $1/60 = 0.0167$. The biased estimator has the smaller average risk.

Two more facts (B pp. 64–70). The Bayes estimator depends on the data only through a sufficient statistic (Thm 4.4.2). And it is **not invariant**: if $\hat{\theta}$ is the Bayes estimator of θ, $g(\hat{\theta})$ need not be the Bayes estimator of $g(\theta)$, because $E[g(\theta) \mid x] \neq g(E[\theta \mid x])$ for nonlinear g (the MLE is invariant; L01-01).

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10 · C — interval estimates: credible and HPD (G §3.2.2; B §5.5)

A $100(1-\epsilon)\%$ **credible interval** $[t_1, t_2]$ satisfies $\int_{t_1}^{t_2} \pi(\theta \mid x)d\theta = 1 - \epsilon$ (B Def. 5.5.1). It is a probability statement about θ given the observed data. A confidence interval is a statement about the procedure over repeated samples (G p. 31). Among all intervals with the same content the shortest is the **highest posterior density** (HPD) interval, whose endpoints have equal density.

Beta(2, 10), 95%. Equal tails: $(F^{-1}(0.025), F^{-1}(0.975)) = (0.0228,\ 0.4128)$, length 0.390. HPD: $(0.0063,\ 0.3675)$, length 0.361, with $\pi(0.0063 \mid x) = \pi(0.3675 \mid x)$. The HPD moves toward the mode because the posterior is skewed. For a symmetric posterior (normal, B Ex. 5.5.1) the two coincide: $\bar{\theta} \pm 1.96\,\text{sd}$.

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11 · G — minimax (named)

Bolfarine §4.2 (pp. 60–61) ranks rules by their worst-case risk $\max_\theta R(\theta, d)$ instead of the average (15); Ex. 4.2.2 finds $d(X) = X$ minimax for a Poisson mean under $(\theta - a)^2/\theta$, and Ex. 4.3.2 the Bayes rule $X/2$ within $\{cX\}$ under an Exp(1) prior. Not examinable here.

Links

- Prev: L03-03-normal-measurement-physicists (a symmetric posterior: mean = median = mode); next: choosing between models instead of values, L03-05-model-comparison-bayes-factors. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §4.
- Earlier: MSE = variance + bias², L01 §5; posterior mean vs mode on the coal data, L01 §4.
- Rules: rules/07.
- Practice (statements in the books): **G 3.5** (p. 39, median under absolute loss), **G 3.6** (p. 39, 0–1 loss and the mode), **B 4.3** (p. 71), **B 4.5** (p. 71), **B 4.9** (p. 72), **B 4.10** (p. 72), **B 5.11** (p. 89, symmetric Bayesian interval).

G §3.2.4 · Bayes factors: the marginal likelihood rewards fit and charges for every free parameter

Claim. A model is a likelihood plus a prior. Its **marginal likelihood** $m(y) = \int f(y \mid \theta)\pi(\theta)d\theta$ is the likelihood **averaged over the prior**, and posterior odds = prior odds × **Bayes factor** $B_{12} = m_1(y)/m_2(y)$. Greenberg's Michaela–Lila example compares "one coin probability" with "two": with $m = 10$ tosses each and proportions 0.2 / 0.8, $\log_{10} B_{12} = -1.200$ (strong evidence for two probabilities); with 0.5 / 0.5, $+0.297$ for the simpler model, and $+0.756$ at $m = 100$. The simpler model gains when the data agree with it because the bigger model spreads its prior over values the data rule out: in large samples $\log B_{12} \approx \log LR - \frac{d_1 - d_2}{2}\log n$, a BIC-type penalty. Bayes factors need **proper** priors; an improper prior makes B_{12} arbitrary. Prev: L03-04-bayes-estimators-loss · Next: L03-06-monte-carlo-simulation · Map: L03-00-map

Notation.

Symbol	Meaning	Range
M_i	model i: prior $p_i = P(M_i)$, parameters θ_i (d_i of them), prior $\pi_i(\theta_i)$, likelihood $f_i(y \mid \theta_i)$	$i = 1, 2, ...$
$m_i(y)$	marginal likelihood of M_i	> 0
B_{12}	Bayes factor $m_1(y)/m_2(y)$	> 0
R_{12}	posterior odds $P(M_1 \mid y)/P(M_2 \mid y)$	> 0
m	tosses by each girl (Michaela–Lila)	integer
s_1, s_2	heads by Michaela, by Lila	$0, \dots, m$
n	total sample size	here $2m$

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1 · U — comparing explanations, not testing a point

A frequentist test asks whether the data would be unusual if H_0 were true, and refers a statistic to its sampling distribution (G p. 36). The Bayesian question is different and symmetric: **which model, given the data, is more probable?** It applies to non-nested models (a normal vs a Student-t likelihood, y vs $\log y$) as easily as to nested ones, and it can end in "both are still plausible; wait for more data" (G p. 35). The three-coin problem of L03-01-bayes-rule-discrete was already a model comparison: each coin is a model with no free parameter, and its marginal likelihood is just its likelihood.

---

2 · C — posterior model probabilities (G (3.3)–(3.4))

Apply Bayes' rule to the model index and introduce the parameters, as for the predictive (L03-02-conjugate-sequential-updating (10)):

$$P(M_1 \mid y) = \frac{p_1 f_1(y \mid M_1)}{f(y)} = \frac{p_1 \int f_1(y \mid \theta_1, M_1)\pi_1(\theta_1 \mid M_1)d\theta_1}{f(y)}, \tag{1}$$

$$f(y) = p_1 \int f_1(y \mid \theta_1, M_1)\pi_1(\theta_1 \mid M_1)d\theta_1 + p_2 \int f_2(y \mid \theta_2, M_2)\pi_2(\theta_2 \mid M_2)d\theta_2. \tag{2}$$

Each integral in (2) is a **marginal likelihood** (G (3.4)):

$$m_i(y) = \int f_i(y \mid \theta_i, M_i)\, \pi_i(\theta_i \mid M_i)\, d\theta_i, \tag{3}$$

"the expected value of the likelihood function with respect to the prior distribution" (p. 34). It is also the normalising constant of model i's posterior, so the constants of f and π must be kept.

---

3 · C — posterior odds and the Bayes factor (G pp. 34–35)

Divide (1) for M_1 by (1) for M_2; $f(y)$ cancels:

$$R_{12} = \frac{P(M_1 \mid y)}{P(M_2 \mid y)} = \underbrace{\frac{p_1}{p_2}}_{\text{prior odds}} \times \underbrace{\frac{m_1(y)}{m_2(y)}}_{B_{12}}. \tag{4}$$

With two models, $P(M_1 \mid y) = R_{12}/(1 + R_{12}) = 1/(1 + (p_2/p_1)/B_{12})$.

Jeffreys' guidelines (G Table 3.1, p. 35), for evidence in favour of M_1 (mirror the signs for M_2):

$\log_{10} R_{12}$	Interpretation
> 2	decisive support for M_1
3/2 to 2	very strong evidence
1 to 3/2	strong evidence
1/2 to 1	substantial evidence
0 to 1/2	weak evidence

Greenberg: "these values are arbitrary, as is the 5% level of significance". If $\log_{10} B_{12} = 2$, the prior odds would have to be below 1/100 for the posterior odds to favour M_2.

---

4 · Worked — Michaela and Lila (G pp. 37–38, Table 3.2)

Michaela tosses a coin m times and gets s_1 heads; Lila tosses it m times and gets s_2. M_1: $\theta_1 = \theta_2 = \theta$ (the way they toss does not matter). M_2: $\theta_1 \neq \theta_2$. Priors: Beta(1, 1), i.e. uniform, on θ under M_1 and on each θ_j under M_2.

M_1. The likelihood of the $2m$ tosses is $\theta^{s_1+s_2}(1-\theta)^{2m-s_1-s_2}$; integrate against the uniform prior with the Beta integral $\int_0^1 \theta^{a-1}(1-\theta)^{b-1}d\theta = B(a,b)$:

$$m_1(y) = \int_0^1 \theta^{s_1+s_2}(1-\theta)^{2m-s_1-s_2}d\theta = B(s_1+s_2+1,\ 2m-s_1-s_2+1) = \frac{\Gamma(2m\bar{y}+1)\Gamma(2m-2m\bar{y}+1)}{\Gamma(2m+2)}, \tag{5}$$

Greenberg's expression with $\bar{y} = (s_1 + s_2)/2m$.

M_2**.** The two probabilities are independent a priori and each appears in its own factor of the likelihood, so the double integral factors (G (3.5)):

$$m_2(y) = B(s_1{+}1, m{-}s_1{+}1)\, B(s_2{+}1, m{-}s_2{+}1) = \frac{\Gamma(m\bar{y}_1+1)\Gamma(m-m\bar{y}_1+1)}{\Gamma(m+2)} \cdot \frac{\Gamma(m\bar{y}_2+1)\Gamma(m-m\bar{y}_2+1)}{\Gamma(m+2)}. \tag{6}$$

One row in full ($m = 10$, $s_1 = 2$, $s_2 = 8$). By (5), $m_1 = B(11, 11)$, $\log_{10} m_1 = -6.5888$. By (6), $m_2 = B(3, 9)B(9, 3)$, $\log_{10} m_2 = -5.3892$. Then

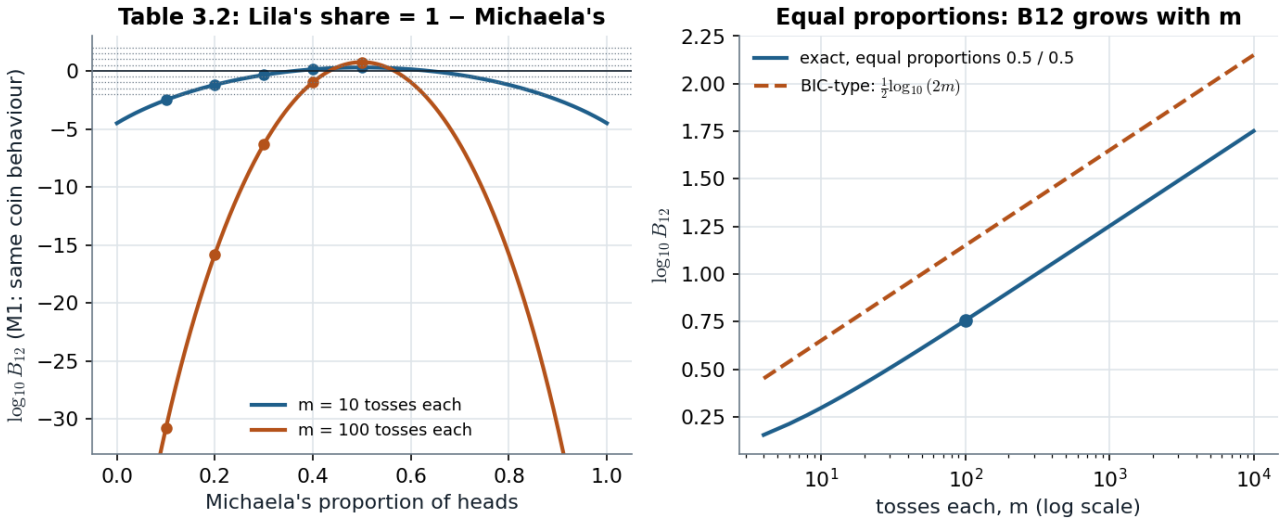
$$\log_{10} B_{12} = -6.5888 - (-5.3892) = -1.1996 \approx \mathbf{-1.200}.$$

With even prior odds, $P(M_1 \mid y) = 1/(1 + 10^{1.1996}) = 0.059$: strong evidence that the girls toss differently.

Table 3.2 reproduced (Lila's proportion = 1− Michaela's):

Michaela	Lila	$\log_{10} B_{12}$, $m = 10$	$\log_{10} B_{12}$, $m = 100$
0.1	0.9	−2.506	−30.775
0.2	0.8	−1.200	−15.793
0.3	0.7	−0.348	−6.316
0.4	0.6	0.138	−0.975
0.5	0.5	0.297	0.756

Every entry is recomputed from (5)–(6) in L03-13-python §5 and agrees to the printed three decimals.



Left: (5)–(6) evaluated on a continuum of proportions (the Gamma functions accept non-integers), Table 3.2 as dots. Right: $\log_{10} B_{12}$ *at* $s_1 = s_2 = m/2$ *against* m*, with the large-sample approximation (8).*

Two comparative statics. Across proportions, B_{12} is largest when both girls agree (0.5 / 0.5) and collapses as they diverge; **more data sharpen the verdict in both directions** (the $m = 100$ arch is taller and steeper). Across sample sizes at perfect agreement, the evidence for the simpler model keeps

growing, like $\frac{1}{2}\log_{10} m$: two free probabilities are never refuted by agreeing data, only penalised for being unnecessary.

> **Interactive companion.** The Bayes-Factor Dial (`companions/companion-bayes-factor-coins.html`) — move m, the two head counts, the prior Beta(α,α) and the prior odds; watch $\log_{10} B_{12}$ against Lila's count with the Jeffreys bands, the Beta posteriors under each model, and the large-sample approximation. The buttons load the ten rows of Table 3.2.

---

5 · Worked — coins and tests as models

Three coins (L03-01-bayes-rule-discrete §5). With no free parameters, $m_k(HHH) = p_k^3$, so $B_{31} = 1/0.125 = 8$ and $B_{21} = 0.729/0.125 = 5.832$; with equal prior odds these give the posteriors 0.5394, 0.3932 and 0.0674. Here the Bayes factor is the likelihood ratio, because there is nothing to integrate.

Diego (L03-01-bayes-rule-discrete §3). "Ill" vs "healthy" are two models; the Bayes factor of $X = 1$ is $0.95/0.40 = 2.375$ ($\log_{10} = 0.38$, weak evidence) and prior odds $0.7/0.3$ turn it into posterior odds 5.54, i.e. $P = 0.847$.

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6 · C — what the marginal likelihood penalises (G p. 37)

Exponentiate the second-order expansion of the log-likelihood around the MLE $\hat\theta_i$ (L03-02-conjugate-sequential-updating (16), G (3.2)) and substitute into (3). For large n the exponential dominates the prior, so $\pi_i(\theta_i) \approx \pi_i(\hat\theta_i)$ on the region that matters:

$$m_i(y) \approx L_i(\hat\theta_i \mid y)\,\pi_i(\hat\theta_i) \int \exp\Big[-\frac{n}{2}(\theta_i-\hat\theta_i)' V_i^{-1}(\theta_i-\hat\theta_i)\Big] d\theta_i = L_i(\hat\theta_i \mid y)\,\pi_i(\hat\theta_i)\,(2\pi)^{d_i/2} n^{-d_i/2} |V_i|^{1/2}. \tag{7}$$

Take the log of the ratio of (7) for $i = 1, 2$:

$$\log B_{12} \approx \underbrace{\log \frac{L_1(\hat\theta_1 \mid y)}{L_2(\hat\theta_2 \mid y)} - \frac{d_1 - d_2}{2}\log n}_{\text{grows with } n} + \underbrace{\log \frac{\pi_1(\hat\theta_1)}{\pi_2(\hat\theta_2)} + \frac{1}{2}\log\frac{|V_1|}{|V_2|} + \frac{d_1-d_2}{2}\log(2\pi)}_{\text{bounded}}. \tag{8}$$

The first bracket is the log-likelihood ratio minus **half the log sample size per extra parameter**: the Schwarz/BIC penalty of L01-06, derived rather than imposed.

Check on Michaela–Lila. $d_1 - d_2 = 1 - 2 = -1$ and $n = 2m$, so the first bracket is $\log LR + \frac{1}{2}\log(2m)$. For (2, 8) at $m = 10$: $\log_{10} LR = -1.674$ and the approximation gives $-1.674 + \frac{1}{2}\log_{10} 20 = -1.024$, against the exact -1.200. At equal proportions $LR = 1$, and with uniform priors ($\pi = 1$, $V = \theta(1-\theta) = 1/4$) the bounded bracket evaluates to $-\frac{1}{2}\log(2\pi)$; the exact value tends to $\frac{1}{2}\log_{10}(m/\pi)$, which stays $\frac{1}{2}\log_{10}(2\pi) = 0.399$ below the BIC line, the constant gap in the right panel.

> **Occam's razor is built in.** M_2 can fit anything M_1 fits, so its maximised likelihood is always at least as high. Its marginal likelihood averages over a prior spread across (θ_1, θ_2) pairs the data contradict, and that average is lower whenever the extra freedom is not needed.

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7 · C — Bayes factors need proper priors (G §4.2, p. 44)

An improper prior is defined only up to a constant: $c\,\pi(\theta)$ is "the same" prior for every $c > 0$. Substituting into (3) and (4):

$$B_{12} = \frac{c_1 \int f_1(y \mid \theta_1)\pi_1(\theta_1)d\theta_1}{c_2 \int f_2(y \mid \theta_2)\pi_2(\theta_2)d\theta_2}, \tag{9}$$

and the arbitrary ratio c_1/c_2 sets B_{12} to any value, "even if only one of the prior distributions is improper". A posterior for θ within a model survives an improper prior (the constant cancels in L03-01-bayes-rule-discrete (3)); a comparison between models does not. A related warning: a proper but very diffuse prior makes m_i small, so (8)'s $\log \pi_i(\widehat{\theta}_i)$ term favours the smaller model however good the bigger one's fit (the Jeffreys–Lindley effect of L01-08).

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8 · C — model averaging and invariance (G pp. 35–36)

Prediction under model uncertainty. Weight each model's predictive by its posterior probability:

$$f(y_f \mid y) = \sum_{i=1}^{m} P(M_i \mid y) \int f_i(y_f \mid \theta_i, y, M_i)\, \pi_i(\theta_i \mid y, M_i)\, d\theta_i. \tag{10}$$

Models with small $P(M_i \mid y)$ contribute little. The Vanguard table in L03-10-sparsity-priors uses Bayesian model averaging (BMA) as the benchmark.

Invariance to the scale of y. If M_1 models y and M_2 models $z = g(y)$ with g monotone, transforming M_2 back to the y scale multiplies its density by the Jacobian $|dz/dy|$, and the same factor appears in every comparison on the y scale; comparing y with $\log y$ is legitimate once both densities are on the same scale (G p. 36).

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9 · G — computing marginal likelihoods (named)

When the posterior is not of a known form, $m(y)$ is not available in closed form. Chib's method (G §7.1.2, pp. 95–96) uses the identity $f(y) = f(y \mid \theta^*)\pi(\theta^*)/\pi(\theta^* \mid y)$ at a point θ^* and estimates the denominator from Gibbs output; Chib–Jeliazkov (§7.2.2) does the same for MH output. Greenberg's union-wage example (§8.1.1, p. 113) reports $\log_{10} B = 5.390$ for including union membership. Not examinable here.

Links

- Prev: L03-04-bayes-estimators-loss; next: computing the integrals that have no closed form, L03-06-monte-carlo-simulation. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §5.
- Earlier: p-value vs Bayes factor at $X = 1.8$, L01 §8; BIC as a selection criterion, L01 §6.
- Rules: rules/07.
- Practice (statements in the books): **G 3.7** (p. 40, verify (3.5)), **G 3.8** (p. 40, Bayes factors for two Poisson rates; listed in the class plan), **G 4.5** (p. 58, analytic marginal likelihood).

Ingredients deck slides 53–56, G Ch. 5, MS App. B · Monte Carlo: replace an integral by an average of draws, and know its error

Claim. Every Bayesian summary is an integral (a posterior mean, a probability, a predictive, a marginal likelihood). Monte Carlo writes it as an expectation, draws M independent values and averages; the error falls like $1/\sqrt{M}$ and is measured by the **numerical standard error**. Hit-or-miss gives $\hat{\pi}$ with standard error 0.052 at $M = 1000$, so the deck's 3.196 is one error away from π. When direct draws are impossible: the **inverse cdf** for univariate targets, **composition** for mixtures (a Student-t is a normal with a gamma precision), **accept–reject** with acceptance probability $1/c$ (Beta(3,3) from uniforms: $c = 1.875$, 53%), and **importance sampling / SIR**, which reweight draws from another density. Importance sampling fails silently when the proposal's tails are too thin: Greenberg's own Beta(2,3) example has **infinite variance**. Prev: L03-05-model-comparison-bayes-factors · Next: L03-07-markov-chains-gibbs · Map: L03-00-map

Notation.

Symbol	Meaning	Range
f	target density (often a posterior known up to a constant)	—
g or h	proposal / importance density we can sample	—
M (deck), G (Greenberg), n (MS)	number of draws	integer
$\bar{g}_M$	Monte Carlo average of $g(X^{(i)})$	—
n.s.e.	numerical standard error $\sqrt{\widehat{V}(g(X))/M}$	> 0
c	accept–reject constant with $f \le c\, g$	≥ 1 for normalised f, g
w_i	importance weight $f(x_i)/g(x_i)$	≥ 0

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1 · U — integrals that cannot be done by hand (slides 42–43, 48–52)

The physicist C posterior of L03-03-normal-measurement-physicists needs $\int \theta\, p(x \mid \theta) p(\theta) d\theta$ with a Student-t prior: no closed form. The deck: "Bayesian computation was done on limited, unrealistic models until the Monte Carlo revolution (and the computing revolution) of the late 1980's and early 1990's" (slide 43). The idea is the law of large numbers: if we can draw from a distribution, sample averages approximate its expectations.

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2 · C — Monte Carlo integration and its error (slides 53–56; G p. 63, §5.6; MS §B.2)

For $X \sim f$ and a function g with $E|g(X)| < \infty$,

$$E[g(X)] = \int g(x) f(x) dx \;\approx\; \bar{g}_M = \frac{1}{M} \sum_{i=1}^{M} g(X^{(i)}), \qquad X^{(i)} \overset{iid}{\sim} f. \tag{1}$$

$\bar{g}_M$ is unbiased, converges by the law of large numbers, and has variance $V(g(X))/M$; Greenberg's **numerical standard error** (p. 73) is

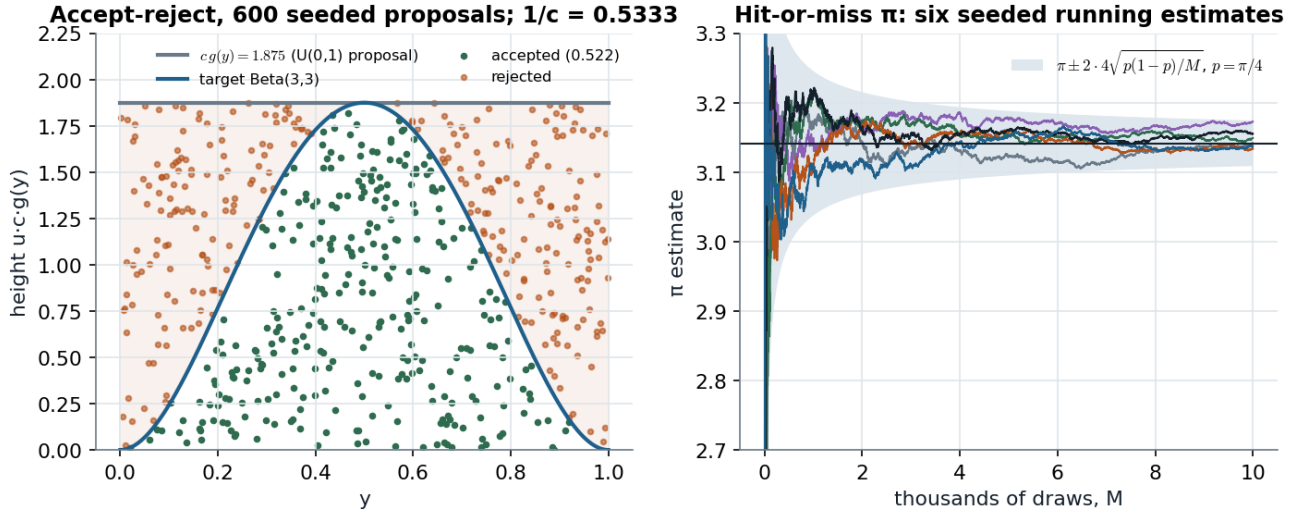
$$\text{n.s.e.} = \sqrt{\widehat{V}(g(X))/M}. \tag{2}$$

Probabilities are the case $g = \mathbf{1}_A$: $P(A) \approx$ the fraction of draws in A (G p. 63).

Worked — hit-or-miss π (slides 53–56). Draw (x_i, y_i) uniform on the unit square; $P(x^2 + y^2 < 1) = \pi/4$ (the area of a quarter disc), so $\hat{\pi} = 4\times$ (fraction inside). The indicator is Bernoulli with $p = \pi/4 = 0.7854$; by (2)

$$\text{se}(\hat{\pi}) = 4\sqrt{\frac{p(1-p)}{M}} = 4\sqrt{\frac{0.7854 \times 0.2146}{M}} = \frac{1.6422}{\sqrt{M}}. \tag{3}$$

At $M = 1000$: $1.6422/31.623 = 0.0519$. The deck's $\hat{\pi} = 3.196$ (slide 54) is $(3.196 - 3.1416)/0.0519 = 1.05$ standard errors high, an ordinary draw. At $M = 10{,}000$: 0.0164; a seeded Python run gives 3.1452.



Left: accept–reject for Beta(3, 3) with a $U(0,1)$ proposal, Algorithm 5.3 (G p. 67). Right: running hit-or-miss estimates of π (slides 55–56) inside $\pi \pm 2$ se of (3). Both panels seeded.

The right panel is slide 56 with its envelope drawn: every path wanders, none escapes the $\pm 2/\sqrt{M}$ funnel for long, and the funnel narrows slowly: **four times the draws for half the error**.

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3 · C — the inverse-cdf (probability integral transform) method (G §5.1; MS §B.4)

If $U \sim U(0,1)$ and F is a continuous cdf, $Z = F^{-1}(U)$ has cdf F:

$$P(Z \le z) = P(F^{-1}(U) \le z) = P(U \le F(z)) = F(z). \tag{4}$$

Example (G pp. 64–65). $f(y) = \frac{3}{8}y^2$ on $[0,2]$: $F(y) = y^3/8$, so $Y = 2U^{1/3}$. *Truncation (G (5.1)).* To draw X from F restricted to $[c_1, c_2]$, set $X = F^{-1}(F(c_1) + U[F(c_2) - F(c_1)])$; this is how the truncated normals of the Tobit and probit samplers are drawn (G §8.2). The method needs the full cdf and is univariate only.

4 · C — composition and scale mixtures (G §5.2, §4.5)

If $f(x) = \int g(x \mid y)h(y)dy$, draw $y \sim h$ then $x \sim g(\cdot \mid y)$; the x's are draws from f.

The Student-t as a normal scale mixture (G p. 52). Let $u \mid \lambda \sim N(0, \sigma^2/\lambda)$ and $\lambda \sim G(\nu/2, \nu/2)$. The joint density is

$$f(u, \lambda) \propto \lambda^{1/2} e^{-\lambda u^2/2\sigma^2} \cdot \lambda^{\nu/2-1} e^{-\nu\lambda/2} = \lambda^{(\nu+1)/2-1} \exp\Big[-\lambda \frac{u^2 + \nu\sigma^2}{2\sigma^2}\Big]. \tag{5}$$

(5) is a $G(\frac{\nu+1}{2}, \frac{u^2+\nu\sigma^2}{2\sigma^2})$ kernel in λ; integrating it out leaves the reciprocal of its rate raised to the shape:

$$f(u) \propto \Big(\frac{u^2 + \nu\sigma^2}{2\sigma^2}\Big)^{-(\nu+1)/2} \propto \Big(1 + \frac{u^2}{\nu\sigma^2}\Big)^{-(\nu+1)/2}, \tag{6}$$

the $t_\nu(0, \sigma^2)$ kernel. So a t draw is a gamma draw followed by a normal draw. Seeded check with $\nu = 5$, 400,000 draws: variance 1.660 (theory 5/3), $P(|u| > 2.571) = 0.0502$ (theory 0.05). This representation is why Student-t errors and priors fit inside Gibbs samplers (L03-07-markov-chains-gibbs, G §8.1.2) and why the horseshoe is a normal with a random scale (L03-10-sparsity-priors).

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5 · C — accept–reject (G §5.3; MS §B.6.1)

Algorithm 5.2. Find a density g we can sample and a constant c with $f(y) \le c\, g(y)$ for all y. (1) Draw $y \sim g$. (2) Draw $u \sim U(0, 1)$. (3) Accept y if $u \le f(y)/[c\, g(y)]$; otherwise return to (1).

Why it works (G p. 67). By Bayes' rule, the density of an accepted value is

$$h(y \mid \text{accept}) = \frac{P(\text{accept} \mid y)g(y)}{\int P(\text{accept} \mid y)g(y)dy} = \frac{[f(y)/cg(y)]\, g(y)}{\int [f(y)/cg(y)]\, g(y)dy} = \frac{f(y)/c}{1/c} = f(y), \tag{7}$$

and the acceptance probability is the denominator, $1/c$. Choose c as small as possible. If f is known only up to a constant, $f = kr$ with $r \le cg$, accepting with probability r/cg still yields draws from f; then c need not exceed 1.

Worked — Beta(3, 3) from $U(0, 1)$ (Algorithm 5.3). $f(y) = 30y^2(1-y)^2$ peaks at $y = \frac{1}{2}$: $c = 30 \times \frac{1}{16} = 1.875$, acceptance $1/1.875 = 0.5333$. A proposal $y = 0.15$ has $f(0.15) = 30(0.0225)(0.7225) = 0.4877$ and is accepted with probability $0.4877/1.875 = 0.2601$. Seeded run of 200,000: acceptance 0.5339; accepted draws have mean 0.500 and variance 0.0357 ($= 1/28$).

Worked — $N(0, 1)$ from a Laplace proposal (Algorithm 5.4). By symmetry, sample $|y|$ from the half-normal $f(x) = \sqrt{2/\pi}\, e^{-x^2/2}$ with proposal $g(x) = e^{-x}$, $x > 0$, and attach a random sign. The ratio $f/g = \sqrt{2/\pi}\, e^{x - x^2/2}$ is maximised at $x = 1$:

$$c = \sqrt{2/\pi}\, e^{1/2} = \sqrt{2e/\pi} = 1.3155, \qquad \text{acceptance } 1/c = 0.7602. \tag{8}$$

Greenberg's test (5.2), accept if $u \le e^{x - x^2/2 - 1/2}$, is exactly $u \le f(x)/[c\, g(x)]$ with (8). His text gives "the probability of acceptance is $1/c = 0.6577$": $0.6577 = \sqrt{e/2\pi}$ is the constant that dominates the

unnormalised kernel $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, whose integral over $x > 0$ is $\frac{1}{2}$, so the acceptance rate is $\frac{1}{2}/0.6577 = 0.7602$ either way. A seeded run of 200,000 accepts 0.7605.

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6 · C — importance sampling (G §5.4; MS §B.6.2)

Multiply and divide by a density h we can sample:

$$E_f[g(X)] = \int g(x)\frac{f(x)}{h(x)}h(x)dx \;\approx\; \frac{1}{G}\sum_{i=1}^{G} g(X^{(i)})\, w_i, \qquad w_i = \frac{f(X^{(i)})}{h(X^{(i)})},\; X^{(i)} \sim h. \tag{9}$$

When f is known only up to a constant, use the self-normalised form (MS (B.17)):

$$\tilde{\theta} = \frac{\sum_i w_i g(X^{(i)})}{\sum_i w_i}, \tag{10}$$

consistent if the support of h covers that of f (Geweke 1989, as cited by MS).

The variance condition (G p. 70). The estimator's second moment is $E_h[(gf/h)^2] = \int g^2 f^2/h\, dx$, large "when the tail values of $h(\cdot)$ are very small compared to the tail values of $f(\cdot)$".

Worked — Greenberg's Algorithm 5.5, and its hidden flaw. Target $E[(1+x^2)^{-1}]$ for $x \sim \text{Exp}(1)$ truncated to $[0,1]$, $f(x) = e^{-x}/(1-e^{-1})$; importance density $h = \text{Beta}(2,3)$, $h(x) = 12x(1-x)^2$. The exact value is 0.8302 (G p. 71; quadrature 0.830217). Greenberg reports 0.8268 with n.s.e. 0.0030 from $G = 10{,}000$; a seeded Python run gives 0.8003 with a computed n.s.e. of 0.0089, three "standard errors" away. The reason is near $x = 0$: $f(0)g(0) = 1.582$ while $h(x) \approx 12x$, so

$$\int_0^{\varepsilon} \frac{g^2 f^2}{h}dx \approx \int_0^{\varepsilon} \frac{1.582^2}{12x}dx = \infty. \tag{11}$$

The estimator has infinite variance. It is still consistent, but its sample standard error means nothing and convergence is erratic. (Numerically, each factor of 100 closer to zero adds $1.582^2 \log(100)/12 = 0.960$ to the truncated second moment.) A proposal with a positive density at 0, such as $U(0,1)$ or the truncated exponential itself, removes the problem.

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7 · C — sampling importance resampling (SIR) (MS §B.6.2; fisicos-ABCD, correlation, ZIP pages)

(a) Draw $x_1, \ldots, x_n$ from g. (b) Compute normalised weights $w_i = \frac{f(x_i)/g(x_i)}{\sum_j f(x_j)/g(x_j)}$.

(b) Resample from $\{x_i\}$ with probabilities w_i. The resampled values are approximately draws from f; MS's argument: the resampled cdf $\sum_{i:x_i \le a} w_i$ converges to $\int_{-\infty}^{a} f$ by the strong law (p. 470).

With a uniform g on a box, $w_i \propto$ likelihood $\times$ prior: the lecturer's pages use exactly this.

Worked — physicist C (`fisicos-ABCD.html`). 10^6 proposals from $U(600, 1000)$, weights $N(850 \mid \theta, 40^2)\, t_5(\theta \mid 900, 240)$, 10^6 resampled draws. Seeded Python: mean 891.13 against the quadrature value 891.12. The **effective sample size** $(\sum w_i)^2/\sum w_i^2$ is 144,275: only one draw in seven carries real weight, because most uniform proposals land where the posterior is negligible.

7.1 · Worked — the correlation coefficient (`correlationcoefficient.html`)

Standardised pairs $x_i = (x_{i1}, x_{i2})$, $n = 14$, modelled as bivariate normal with unit variances and correlation ρ:

$$p(x_{1:n} \mid \rho) = \Big(\frac{1}{2\pi\sqrt{1-\rho^2}}\Big)^n \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\sum_i (x_{i1}^2 + x_{i2}^2) - 2\rho\sum_i x_{i1}x_{i2}\Big]\Big\}, \tag{12}$$

which "has no known form, regardless of which prior one uses". Data: heights and weights of 14 students (the 15th, the lecturer, removed as an outlier), sample correlation 0.615. Posterior quantiles (2.5%, 50%, 97.5%):

Prior	Page (SIR)	Exact (quadrature on ρ)
uniform $p(\rho) = \frac{1}{2}$	0.102, 0.569, 0.784	0.105, 0.567, 0.784
what the page codes, $\propto 1/\sqrt{1+\rho^2}$	0.081, 0.558, 0.783	0.089, 0.556, 0.781
arc-sine $\frac{1}{\pi}(1-\rho^2)^{-1/2}$, as stated	—	0.130, 0.589, 0.794

The page states the arc-sine prior but codes `prior = 1/sqrt(1+rho^2)`; its second column is the posterior under that other prior. The true arc-sine prior puts more mass near ± 1 and moves the posterior up. With 14 observations the prior visibly matters in the lower tail.

7.2 · Worked — zero-inflated Poisson (`ZIP-fishing.html`)

Fish caught by groups that brought a camper and children. Some groups never fish, so zeros come from two sources: with probability π a structural zero, otherwise a Poisson(λ) count:

$$P(y_i \mid \lambda, \pi) = \begin{cases} \pi + (1-\pi)e^{-\lambda}, & y_i = 0, \\ (1-\pi)\dfrac{\lambda^{y_i}e^{-\lambda}}{y_i!}, & y_i = 1, 2, \ldots \end{cases} \qquad E(y_i) = (1-\pi)\lambda, \quad V(y_i) = (1-\pi)\lambda(1+\pi\lambda). \tag{13}$$

Data. The page's frequency table: 24 zeros, then counts 1 (6), 2 (2), 3 (4), 4 (2), 5 (3), 7, 8, 14, 16 (one each). That is $n = 45$ groups (the text says 42; the printed estimates below need 45), $n_0 = 24$, $\sum y_i = 90$, $\bar{y} = 2$, $s^2 = 12.318$.

MLE. Set the two score equations of $\log L = n_0 \log D + (n - n_0)[\log(1-\pi) - \lambda] + \sum y_i \log\lambda$, $D = \pi + (1-\pi)e^{-\lambda}$, to zero. The π-equation gives $D = n_0(1-\pi)(1-e^{-\lambda})/(n-n_0)$; substituting it into the λ-equation leaves

$$\frac{\lambda}{1-e^{-\lambda}} = \bar{z} = \frac{\sum y_i}{n - n_0} = \frac{90}{21} = 4.2857, \qquad \hat{\pi} = 1 - \frac{\bar{y}}{\hat{\lambda}}, \tag{14}$$

whose root is $\hat{\lambda} = \bar{z} + W_0(-\bar{z}e^{-\bar{z}}) = \mathbf{4.223}$ (W_0 the principal Lambert branch), and $\hat{\pi} = 1 - 2/4.223 = \mathbf{0.526}$. (14) says the fitted probability of a positive count, $(1-\hat{\pi})(1-e^{-\hat{\lambda}})$, equals the observed share 21/45.

Method of moments. Equate $\bar{y}$ and s^2 to (13)'s moments: $\lambda_{mm} = (s^2+\bar{y}^2)/\bar{y} - 1 = (12.318+4)/2 - 1 = 7.159$ and $\pi_{mm} = (s^2 - \bar{y})/(s^2 + \bar{y}^2 - \bar{y}) = 10.318/14.318 = 0.721$, the page's values. The two large

counts (14, 16) inflate s^2 and drive the moment estimates; the page's fitted-frequency table shows the MLE fitting the zeros exactly and MM overpredicting them (32 vs 24).

Posterior. Uniform priors on $\lambda \in (1, 8)$ and $\pi \in (0.1, 0.9)$ (the page's SIR box). On a fine grid the posterior means are $E(\lambda \mid y) = 4.266$ and $E(\pi \mid y) = 0.524$, close to the MLE because the priors are flat over the region the likelihood occupies. The page's posterior predictive intervals for each count come from pushing the SIR draws through (13).

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8 · C — using simulated output (G §5.6)

Draws $\theta^{(1)}, \ldots, \theta^{(G)}$ from a joint posterior give, column by column, draws from each **marginal** without computing any integral (G p. 74); functions of the draws (ratios, XY, predicted probabilities) are draws of the function. Quantiles of the sorted draws give credible intervals; kernel-smoothed histograms give densities. Greenberg's illustration $Z = XY$ with $X \sim \text{Beta}(3, 3)$, $Y \sim \text{Beta}(5, 3)$: mean 0.3146, n.s.e. 0.0021, 95% interval (0.0719, 0.6279) from $G = 5000$.

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9 · G — multivariate draws (G §5.5, named)

$X = \mu + C'Z$ with $Z \sim N_p(0, I)$ and $\Sigma = C'C$ (Cholesky) gives $X \sim N_p(\mu, \Sigma)$, used in every Gibbs step that draws a coefficient vector. Accept–reject and importance sampling become hard in many dimensions (a dominating density over the whole support, tails heavy enough everywhere), which is what motivates MCMC.

Links

- Prev: L03-05-model-comparison-bayes-factors; next: dependent draws from a Markov chain, L03-07-markov-chains-gibbs. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §6.
- Earlier: Monte Carlo for sampling distributions and the bootstrap, L01 §5; L01 §2.
- Rules: rules/08 (Monte Carlo integration).
- Practice (statements in the books): **G 5.4** (p. 75, accept–reject Beta(3,3)), **MS B.16** (p. 478, accept–reject for a truncated gamma), **MS B.19** (p. 478, discrete accept–reject).

G Ch. 6, §7.1, §7.3, T §12.1–12.3 · Gibbs sampling: a Markov chain whose invariant distribution is the posterior

> **Claim.** When a posterior cannot be sampled directly but each **full conditional** can, draw the blocks one at a time, each from its conditional given the latest values of the others. The draws form a Markov chain whose **invariant distribution is the joint posterior**; after a burn-in, their histograms are the marginals. On the coal-mining series with the L01-04 prior on both rates, a three-block Gibbs sampler puts the switch at **1891** (posterior probability 0.240, against the exact 0.240), with $E(\lambda_1 \mid y) = 3.106$ and $E(\lambda_2 \mid y) = 0.930$. The price is **dependence**: on a bivariate normal with correlation ρ, the chain's lag-h autocorrelation is ρ^{2h}, so $\rho = 0.999$ needs about 1000 draws for the information of one independent draw. Prev: L03-06-monte-carlo-simulation · Next: L03-08-metropolis-hastings-hmc · Map: L03-00-map

Notation.

Symbol	Meaning	Range
$P = \{p_{ij}\}$	transition matrix, $p_{ij} = P(X_{t+1} = j \mid X_t = i)$	rows sum to 1
π	invariant distribution, $\pi P = \pi$	probability vector / density
$p(x, y)$	transition kernel (continuous state)	density in y
$\theta = (\theta_1, \dots, \theta_d)$	parameter blocks	—
θ_{-j}	all blocks except θ_j	—
B, G	burn-in length, retained draws	integers
k	switch point: last year of the first regime	$1, \dots, n$
λ_1, λ_2	Poisson rates before and after the switch	> 0
$S_k = \sum_{i \le k} y_i$	cumulative count	—
ρ_h	lag-h autocorrelation of a chain	$[-1, 1]$

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1 · U — independent draws are a luxury

Chapter 5's methods (L03-06-monte-carlo-simulation) need a density we can sample whose shape is close to the target: hard to find in many dimensions. MCMC reverses the question. Instead of a distribution that is easy to sample, build a **process** that is easy to run and that settles into the target. The draws are no longer independent, but averages still converge (Theorem 6.3), and the recipe works for "a great variety of distributions without having to undertake an intensive analysis of the special features of the distribution" (G p. 76).

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2 · C — Markov chains on a finite state space (G §6.1)

A process X_t on $S = \{1, \dots, s\}$ is a homogeneous **Markov chain** if

$$p_{ij} = P(X_{t+1} = j \mid X_t = i) \quad \text{does not depend on } t \text{ or on } X_{t-1}, X_{t-2}, \dots \tag{1}$$

The n-step transition probabilities are the entries of P^n (G (6.3)). A distribution π is **invariant** if

$$\pi_j = \sum_i \pi_i p_{ij} \ \text{ for all } j, \qquad \text{i.e. } \pi P = \pi. \tag{2}$$

Worked (G (6.2), p. 79). $P = \begin{pmatrix} 0.750 & 0.250 \\ 0.125 & 0.875 \end{pmatrix}$. The first column of (2) gives $\pi_1 = 0.750\pi_1 + 0.125\pi_2$. Substitute $\pi_2 = 1 - \pi_1$:

$$\pi_1 = 0.750\pi_1 + 0.125(1 - \pi_1) \implies 0.375\pi_1 = 0.125 \implies \pi = (1/3,\ 2/3). \tag{3}$$

Powers of P (G p. 80): $P^{10} = \begin{pmatrix} 0.339 & 0.661 \\ 0.330 & 0.670 \end{pmatrix}$ and $P^{20} = \begin{pmatrix} 0.333 & 0.667 \\ 0.333 & 0.667 \end{pmatrix}$: every row converges to π, so after 20 steps the starting state is forgotten to three decimals. The speed is set by the second eigenvalue of P, $1 - 0.250 - 0.125 = 0.625$ (G Ex. 6.5): the distance to π shrinks like 0.625^n.

When convergence fails (G pp. 78–81). A **reducible** chain (block-diagonal P) never leaves the block it starts in, and has many invariant distributions; a **periodic** chain alternates and P^n does not converge. Theorem 6.2: if P is irreducible and aperiodic on a finite space, there is a unique invariant π and $|p_{ij}^{(n)} - \pi_j| \le r^{n/\nu}$ with $r < 1$.

---

3 · C — general state spaces and the ergodic average (G §6.2–6.3)

For continuous states the transition kernel $p(x, y)$ replaces p_{ij}, and a density π is invariant if (G (6.9))

$$\pi(y) = \int \pi(x) p(x, y)\, dx. \tag{4}$$

Example (G pp. 85–86). The AR(1) $y_t = \phi y_{t-1} + u_t$, $|\phi| < 1$, $u_t \sim N(0, \sigma^2)$, has invariant density $N(0, \sigma^2/(1 - \phi^2))$, the equilibrium distribution of L01-07: if $y_{t-1} \sim N(0, v)$ with $v = \sigma^2/(1 - \phi^2)$, then $V(y_t) = \phi^2 v + \sigma^2 = v$.

Theorem 6.3 (ergodic average). For an irreducible chain whose states are positive recurrent, there is a unique invariant π and, for any starting distribution,

$$\frac{1}{n} \sum_{m=1}^{n} f(X_m) \ \longrightarrow \ E_\pi f(X) \quad \text{whenever } E_\pi |f(X)| < \infty. \tag{5}$$

(5) is the law of large numbers that MCMC needs: chain averages estimate posterior expectations. **MCMC turns Markov-chain theory around** (G p. 84): instead of finding π for a given P, it starts from the target π and constructs a kernel with that invariant distribution (Theorems 6.5–6.6 then give convergence for π-irreducible, aperiodic kernels).

---

4 · C — the Gibbs algorithm (G §7.1.1; T §12.2)

Algorithm 7.1 (two blocks). Choose $x_2^{(0)}$. At iteration g draw

$$x_1^{(g)} \sim f(x_1 \mid x_2^{(g-1)}), \qquad x_2^{(g)} \sim f(x_2 \mid x_1^{(g)}). \tag{6}$$

Discard the first B draws (burn-in); keep G draws. With d blocks (Algorithm 7.2; Tsay's three parameters in (12.1)), each block is drawn from its full conditional given the most recent values of all others.

Why the target is invariant (G pp. 91–92). The Gibbs kernel from $x = (x_1, x_2)$ to $y = (y_1, y_2)$ is $p(x, y) = f(y_1 \mid x_2) f(y_2 \mid y_1)$. Check (4):

$$\int p(x,y) f(x) dx = \iint f(y_1 \mid x_2) f(y_2 \mid y_1) f(x_1, x_2)\, dx_1 dx_2 = f(y_2 \mid y_1) \int f(y_1 \mid x_2) f(x_2)\, dx_2 = f(y_2 \mid y_1) f(y_1) = f \tag{7}$$

The second equality integrates x_1 out of $f(x_1, x_2)$; the third is the law of total probability.

How to find a full conditional (G p. 55). Write the joint posterior (likelihood × prior), keep only the factors that contain the block, and recognise the kernel. Everything else goes into the constant. Conditionally conjugate priors (G §4.9) make every kernel standard.

Using the output (T (12.3)). Point estimates and variances are sample moments of the retained draws; differences, ratios and other functions are computed draw by draw (T p. 616).

---

5 · Worked — normal mean and precision (G Algorithm 7.3, (7.1)–(7.2))

$y_i \sim N(\mu, h^{-1})$ iid, priors $\mu \sim N(\mu_0, h_0^{-1})$ and $h \sim G(\alpha_0/2, \delta_0/2)$, independent (G §4.9: not conjugate jointly). The joint posterior is

$$\pi(\mu, h \mid y) \propto h^{n/2} \exp\Big[-\frac{h}{2}\sum_i (y_i - \mu)^2\Big] \exp\Big[-\frac{h_0}{2}(\mu - \mu_0)^2\Big] h^{\alpha_0/2-1} \exp\Big[-\frac{\delta_0 h}{2}\Big]. \tag{8}$$

$h \mid \mu, y$. Keep the factors of (8) with h:

$$\pi(h \mid \mu, y) \propto h^{(\alpha_0+n)/2-1} \exp\Big[-h\,\frac{\delta_0 + \sum_i (y_i - \mu)^2}{2}\Big] \;\Rightarrow\; h \mid \mu, y \sim G\Big(\frac{\alpha_0 + n}{2},\; \frac{\delta_0 + \sum_i (y_i - \mu)^2}{2}\Big). \tag{9}$$

$\mu \mid h, y$. Keep the factors with μ; this is L03-03-normal-measurement-physicists (5) with data precision hn and prior precision h_0:

$$\mu \mid h, y \sim N\Big(\frac{h_0\mu_0 + hn\bar{y}}{h_0 + hn},\; (h_0 + hn)^{-1}\Big). \tag{10}$$

Alternate (9) and (10). On a seeded data set with the design of the lecturer's Stan `exemplo1` ($n = 100$, $y \sim N(1.6, 0.2^2)$, prior $\mu \sim N(1.7, 0.3^2)$): 5000 draws after 1000 burn-in give $E(\mu \mid y) = 1.6079$ against $\bar{y} = 1.6077$, and $1/\sqrt{E(h \mid y)} = 0.195$, the sample sd. The prior on μ is worth $h_0/(hn) = (1/0.09)/(100/0.04) = 0.4\%$ of the data.

---

6 · Worked — the coal-mining switch point (G (7.3)–(7.4), pp. 93–94)

Model. Counts $y_1, \dots, y_n$ ($n = 112$ years, 1851–1962, $\sum y_i = 191$) with

$$y_i \sim \text{Poisson}(\lambda_1)\ (i \le k), \qquad y_i \sim \text{Poisson}(\lambda_2)\ (i > k), \tag{11}$$

priors $\lambda_j \sim G(\alpha_0, \beta_0)$ and k uniform on $\{1, \dots, n\}$ ($k = n$ means no change). Here both rates get **the L01-04 prior** $G(1,\ 1/2)$, i.e. Exp(1/2), so the results connect to L01-04.

Joint posterior. Likelihood × priors, dropping $\prod y_i!$ and constants:

$$\pi(\lambda_1, \lambda_2, k \mid y) \propto \lambda_1^{\alpha_0 - 1} e^{-\beta_0 \lambda_1} \lambda_2^{\alpha_0 - 1} e^{-\beta_0 \lambda_2} \cdot \lambda_1^{S_k} e^{-k\lambda_1} \cdot \lambda_2^{S_n - S_k} e^{-(n-k)\lambda_2}. \tag{12}$$

Full conditionals. Keep the factors of (12) with each block:

$$\lambda_1 \mid k, y \sim G(\alpha_0 + S_k,\ \beta_0 + k), \qquad \lambda_2 \mid k, y \sim G(\alpha_0 + S_n - S_k,\ \beta_0 + n - k), \tag{13}$$

$$P(k \mid \lambda_1, \lambda_2, y) = \frac{e^{k(\lambda_2 - \lambda_1)} (\lambda_1/\lambda_2)^{S_k}}{\sum_{k'=1}^{n} e^{k'(\lambda_2 - \lambda_1)} (\lambda_1/\lambda_2)^{S_{k'}}}, \tag{14}$$

where (14) divides (12) by the factors free of k, $e^{-n\lambda_2} \lambda_2^{S_n}$. Each sweep: two gamma draws and one draw from a discrete distribution on 112 points.

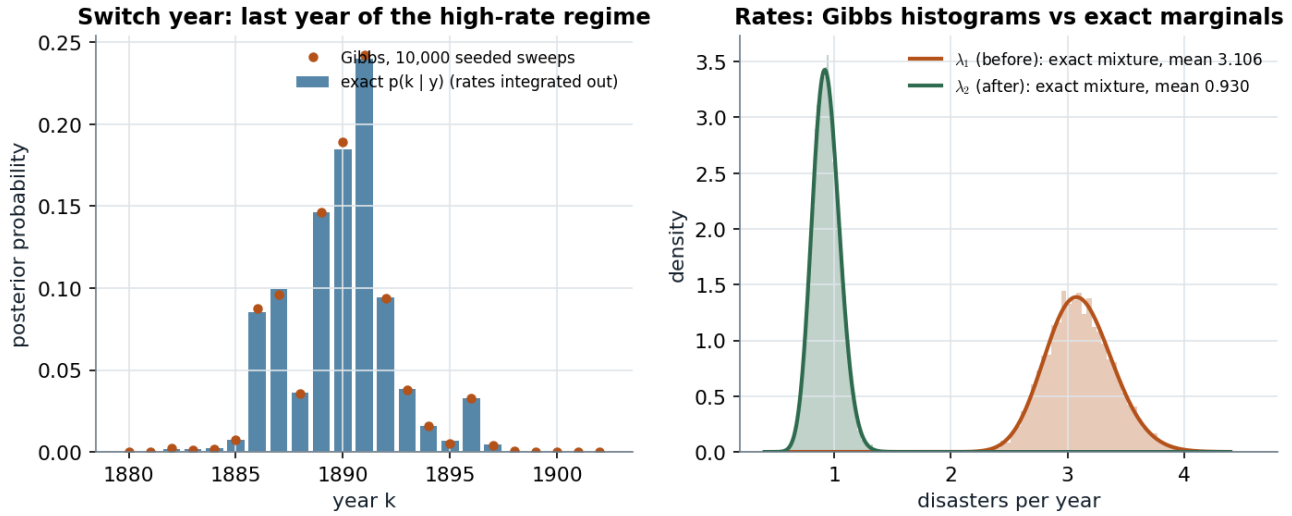
An exact check. The rates can be integrated out of (12) with the gamma integral (L01-04 (2)), giving the marginal posterior of k in closed form:

$$P(k \mid y) \propto \frac{\Gamma(\alpha_0 + S_k)}{(\beta_0 + k)^{\alpha_0 + S_k}} \cdot \frac{\Gamma(\alpha_0 + S_n - S_k)}{(\beta_0 + n - k)^{\alpha_0 + S_n - S_k}}. \tag{15}$$

This is the integrated likelihood of L01-03 §8, now with a prior. It lets the Gibbs output be checked against the truth.

Year	k	S_k	$P(k \mid y)$ exact	$E(\lambda_1 \mid k, y)$	$E(\lambda_2 \mid k, y)$
1886	36	117	0.0854	3.233	0.980
1887	37	119	0.0995	3.200	0.967
1889	39	123	0.1463	3.139	0.939
1890	40	125	0.1849	3.111	0.924
1891	**41**	**127**	**0.2402**	**3.084**	**0.909**
1892	42	128	0.0945	3.035	0.908
1896	46	134	0.0328	2.903	0.872

The years 1886–1896 hold 98.1% of the posterior mass. At $k = 41$, $E(\lambda_1 \mid k, y) = 128/41.5 = 3.084$: L01-04's posterior mean for 1851–1891, the same computation. Averaging (13) over (15): $E(\lambda_1 \mid y) = 3.106$ (sd 0.289) and $E(\lambda_2 \mid y) = 0.930$ (sd 0.117). A Gibbs run of 20,000 sweeps after 1000 burn-in gives 3.107 and 0.931, and its histogram of k is within total-variation distance 0.010 of (15).



Left: (15) as bars, Gibbs frequencies from (13)–(14) as dots (10,000 seeded sweeps). Right: Gibbs histograms of the rates against the exact marginals $\sum_k P(k \mid y)\, G(\lambda \mid \cdot)$ *from (13).*

The switch is located to within a few years but not to one year: the posterior is jagged because a single year's count (a 4 in 1886, a 3 in 1896) can make a candidate split better or worse. The rates are well determined: about 3.1 disasters a year before, 0.93 after, a ratio of about 3.3, as in L01-03's profile likelihood (3.098 vs 0.901 at 1891).

> **Profiling vs integrating, once more.** L01-03 maximised over λ_1, λ_2 for each τ; (15) integrates them out against a prior, and the Gibbs sampler does the integration by simulation. Here all three agree on 1891; with a vaguer prior or a shorter series they need not.

---

7 · C — data augmentation (T §12.1; MS §C.5)

A "parameter" can be anything unobserved: a missing value, a latent volatility, a latent utility, a regime indicator (T p. 615). Adding such variables z to the sampler (Tanner–Wong's **data augmentation**) often turns an intractable posterior into easy conditionals. MS (C.16)–(C.17) write the two identities the algorithm iterates:

$$p(\theta \mid y) = \int p(\theta \mid y, z) p(z \mid y)\, dz, \qquad p(z \mid y) = \int p(z \mid \theta, y) p(\theta \mid y)\, d\theta; \tag{16}$$

drawing $z \sim p(z \mid \theta, y)$ and then $\theta \sim p(\theta \mid y, z)$ is a two-block Gibbs sampler. The switch point k is such a variable; so are the Student-t precisions λ_i of L03-06-monte-carlo-simulation §4 (G §8.1.2), the latent utilities of the probit (G Algorithm 8.3) and the log-volatilities of the SV model (L03-08-metropolis-hastings-hmc §7).

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8 · C — mixing, burn-in and the numerical standard error (G p. 94, §7.3)

When Gibbs mixes badly (G p. 94). Sample $(X_1, X_2) \sim N_2(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix})$ by Gibbs with $X_1 \mid x_2 \sim N(\rho x_2, 1-\rho^2)$ and $X_2 \mid x_1 \sim N(\rho x_1, 1-\rho^2)$. Substitute one conditional into the other:

$$x_1^{(g)} = \rho x_2^{(g-1)} + e_1 = \rho(\rho x_1^{(g-1)} + e_2) + e_1 = \rho^2 x_1^{(g-1)} + (\rho e_2 + e_1), \tag{17}$$

so the x_1 chain is an AR(1) with coefficient ρ^2, and its lag-h autocorrelation is

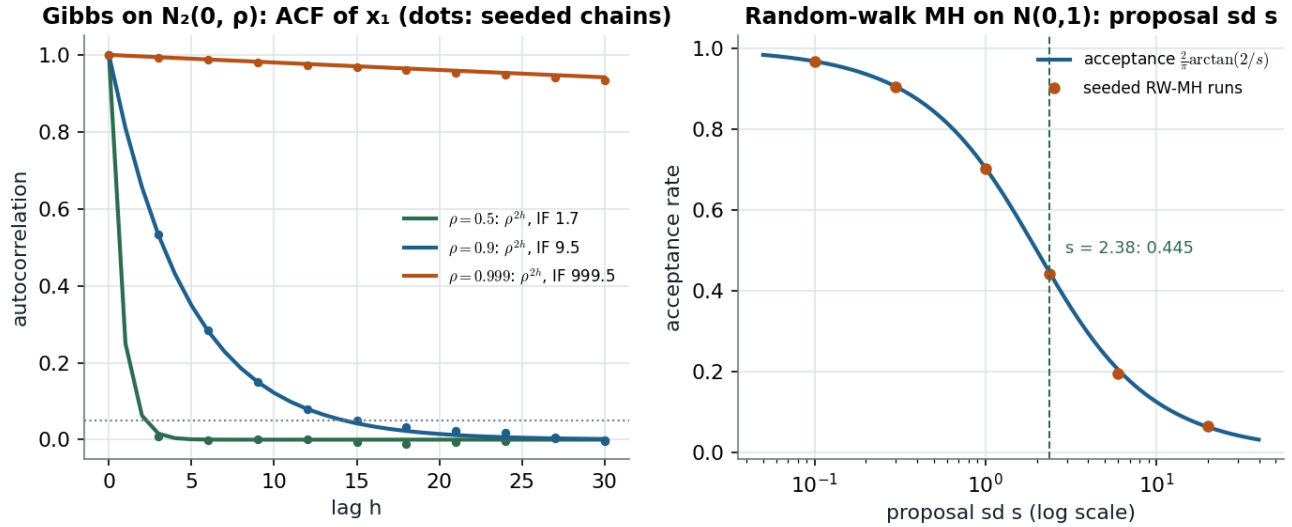
$$\rho_h = \rho^{2h}. \tag{18}$$

For $\rho = 0.5$: $\rho_1 = 0.25$, and $|\rho_h| < 0.05$ from lag 3 ($0.5^6 = 0.016$), Greenberg's report. For $\rho = 0.999$: $\rho_{20} = 0.998^{20} = 0.961$, "over 0.900 even at lag 20"; with $G = 5000$ the sample mean of x_1 was 1.084 or -0.623 depending on the start (G p. 94), against the true 0.

The cost of dependence. The variance of a chain average is $\frac{V(X)}{G}(1 + 2\sum_{h\geq 1}\rho_h)$; the bracket is the **inefficiency factor** (autocorrelation time; G p. 103, the reciprocal of the relative numerical efficiency). For an AR(1) with coefficient ϕ it is $(1+\phi)/(1-\phi)$; with $\phi = \rho^2$:

$$\mathrm{IF} = \frac{1+\rho^2}{1-\rho^2}: \quad \rho = 0.5 \Rightarrow 1.67, \qquad \rho = 0.9 \Rightarrow 9.5, \qquad \rho = 0.999 \Rightarrow 999.5. \tag{19}$$

A chain of G draws carries the information of G/IF independent ones.



Left: (18) for three correlations, with autocorrelations of seeded Gibbs chains (G p. 94, Fig. 7.2). Right: random-walk Metropolis acceptance, L03-08-metropolis-hastings-hmc (9).

Remedies. Block the correlated parameters together (T p. 617: "when parameters are highly correlated, it pays to draw them jointly"), reparameterise, run several chains from dispersed starts.

Numerical standard error by batch means (G §7.3). Find a lag j_0 with $|\hat{\rho}_{j_0}| < 0.05$, cut the G draws into $N = G/j_0$ batches, and use the variance of the batch means:

$$\text{n.s.e.} = \sqrt{\frac{1}{N}\cdot\frac{1}{N-1}\sum_{b=1}^{N}(\bar{Z}_b - \bar{Z})^2}. \tag{20}$$

The naive $\sqrt{\widehat{V}/G}$ understates the error by the factor $\sqrt{\mathrm{IF}}$.

Burn-in and convergence. There is no theorem giving B (G p. 91); diagnostics are informal: trace plots, autocorrelations that die quickly, and agreement across chains started far apart (T p. 617). Stan reports $\widehat{R}$ and n_{eff} for this purpose (the heart-attack output in L03-09-hierarchical-models-shrinkage shows $n_{eff} = 766$ of 20,000 draws for η).

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9 · G — beyond this note (named)

Rao–Blackwellisation (estimate a marginal density by averaging the known conditional over the draws, G p. 105), Chib's marginal likelihood from Gibbs output (G §7.1.2), multiple switch points and hidden Markov regimes (Carlin et al. 1992; G p. 104). Not examinable here.

Links

- Prev: L03-06-monte-carlo-simulation; next: when a full conditional is not standard, L03-08-metropolis-hastings-hmc. Gibbs for hierarchical models: L03-09-hierarchical-models-shrinkage §5; for sparse regression: L03-10-sparsity-priors §6. Formulas: L03-11-formulas; code: L03-13-python §7.
- Earlier: the change point by profile likelihood, L01 §3; the Exp(1/2) prior on the same data, L01 §4; AR(1) equilibrium, L01 §7.
- Rules: rules/08 (Gibbs sampling, convergence diagnostics).
- Practice (statements in the books): **G 6.2** (p. 87, two-state chain), **G 7.1** (p. 105, verify (7.1)–(7.2)), **G 7.3** (p. 106, switch-point Gibbs on the coal data), **T 12.2** (p. 670, Gibbs for regression with AR errors), **T 12.3** (p. 670), **MS C.5** (p. 489, chained-data Gibbs).

G §7.2, §8.2.3, T §12.4, §12.7 · Metropolis–Hastings: propose anywhere, accept with the right probability

Claim. When a full conditional is not a standard distribution, propose a move from any density $q(x, y)$ and accept it with probability $\alpha(x, y) = \min\{1,\ f(y)q(y, x)/[f(x)q(x, y)]\}$; otherwise stay. This makes the chain **reversible** with respect to f, so f is invariant, and the unknown normalising constant cancels in the ratio. Gibbs is the special case where the proposal is the full conditional and $\alpha = 1$. The proposal's scale trades acceptance against step size: for a $N(0, 1)$ target and random-walk steps $N(0, s^2)$ the acceptance rate is exactly $\frac{2}{\pi}\arctan(2/s)$, **0.445 at** $s = 2.38$. The Bayesian logit is the standard use: a normal prior is not conjugate, a tailored random walk samples it, and the posterior means sit next to the MLE. Prev: L03-07-markov-chains-gibbs · Next: L03-09-hierarchical-models-shrinkage · Map: L03-00-map

Notation.

Symbol	Meaning	Range
f	target density (posterior), known up to a constant	—
$q(x, y)$	proposal density of y given the current x	—
$\alpha(x, y)$	acceptance probability	$[0, 1]$
s	random-walk proposal sd	> 0
β	logit coefficients	$\mathbb{R}^K$
P_i	$P(y_i = 1) = e^{x_i'\beta}/(1 + e^{x_i'\beta})$	$(0, 1)$
h_t	log-volatility (SV)	$\mathbb{R}$

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1 · U — accept or stay

Gibbs needs every full conditional in a form we can sample. Many models break this at one block: the logit coefficients, a GARCH parameter, a log-volatility. Metropolis–Hastings needs only the ability to **evaluate** the target up to a constant. It proposes a candidate from a convenient density and then corrects for the mismatch between proposal and target by accepting some moves and rejecting others. Unlike accept–reject (L03-06-monte-carlo-simulation §5), a rejection does not mean "try again": the chain **repeats its current value**, so values recur and draws are correlated (G p. 97).

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2 · C — reversibility and the acceptance probability (G pp. 96–98)

A kernel $p(x, y)$ is **reversible** with respect to f if

$$f(x)p(x, y) = f(y)p(y, x) \quad \text{for all } x, y. \tag{1}$$

Integrating (1) over x gives $\int f(x)p(x, y)dx = f(y)\int p(y, x)dx = f(y)$: **reversibility implies that** f **is invariant** (L03-07-markov-chains-gibbs (4)).

A proposal q is usually not reversible: for some pair, $f(x)q(x, y) > f(y)q(y, x)$, so moves $x \to y$ happen too often. Multiply by a factor α to restore balance (G (7.5)):

$$f(x)\,\alpha(x,y)\,q(x,y) = f(y)\,\alpha(y,x)\,q(y,x). \tag{2}$$

Moves $y \to x$ are too rare, so make them as frequent as possible: $\alpha(y,x) = 1$. Substitute into (2) and solve for $\alpha(x,y)$:

$$\alpha(x,y) = \frac{f(y)q(y,x)}{f(x)q(x,y)}. \tag{3}$$

Combining both cases gives G (7.6):

$$\alpha(x,y) = \min\Big\{\frac{f(y)\,q(y,x)}{f(x)\,q(x,y)},\ 1\Big\} \quad \text{if } f(x)q(x,y) > 0, \text{ and 0 otherwise.} \tag{4}$$

The normalising constant cancels. Only the ratio $f(y)/f(x)$ enters (4), so a posterior known as likelihood $\times$ prior is enough (G p. 98; T p. 622).

Algorithm 7.4. Given x: (1) draw $Y \sim q(x,\cdot)$; (2) draw $U \sim U(0,1)$; if $U \le \alpha(x,Y)$ return Y, otherwise return x. Theorem 7.2 (Tierney): a π-irreducible Metropolis kernel is Harris recurrent, so (5) of L03-07-markov-chains-gibbs applies.

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3 · C — two standard kernels (G p. 99; T §12.4.1–12.4.2)

Random walk. $y = x + u$ with u symmetric about 0, so $q(x,y) = q(y,x)$ and (4) becomes

$$\alpha(x,y) = \min\Big\{\frac{f(y)}{f(x)}, 1\Big\}. \tag{5}$$

This is the original **Metropolis** algorithm (T §12.4.1, symmetric "jumping distribution"): uphill moves are always accepted, downhill moves with probability equal to the density ratio. Hastings's generalisation (T §12.4.2) allows asymmetric q through the correction in (4).

Independence kernel. $q(x,y) = q(y)$, so

$$\alpha(x,y) = \min\Big\{\frac{f(y)/q(y)}{f(x)/q(x)}, 1\Big\}, \tag{6}$$

a random walk on the importance ratio f/q (compare L03-06-monte-carlo-simulation (9)). Greenberg recommends a **tailored** kernel: a fat-tailed density (a multivariate t with small ν) centred at the posterior mode with scale equal to the negative inverse Hessian there (p. 99).

Worked — Beta(3, 4) from $U(0,1)$ proposals (G Algorithm 7.5). With $q = 1$ on $(0,1)$ and $f(x) \propto x^2(1-x)^3$, (6) gives

$$\alpha(x^{(g-1)}, U_2) = \min\Big\{\frac{U_2^2(1-U_2)^3}{(x^{(g-1)})^2(1-x^{(g-1)})^3},\ 1\Big\}. \tag{7}$$

Greenberg: 5000 draws after 500 burn-in, acceptance 0.57, sample mean 0.4296 against $3/7 = 0.4286$. The stationary acceptance rate of an independence sampler with $q = 1$ is $\iint \min\{f(x), f(y)\}\,dx\,dy$

(average (6) over $x \sim f$, $y \sim U$), which for Beta(3, 4) is **0.5726**. A seeded Python run: acceptance 0.581, mean 0.4224.

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4 · C — the proposal scale: an exact acceptance curve

Take the target $N(0,1)$ and random-walk proposals $y = x + u$, $u \sim N(0, s^2)$. The stationary acceptance rate is $E[\min\{1, r\}]$ with $r = f(y)/f(x)$, $x \sim f$. Split it:

$$E[\min\{1,r\}] = P(r \geq 1) + E[r\,\mathbf{1}\{r < 1\}]. \tag{8}$$

The second term is $\iint f(y)q(y-x)\mathbf{1}\{f(y) < f(x)\}dx\,dy$; relabelling $x \leftrightarrow y$ and using the symmetry of q turns it into $P(r > 1)$. So the acceptance rate is $2P(f(y) > f(x)) = 2P(|x+u| < |x|)$. Write $u = sz$ with $z \sim N(0,1)$ independent of x: $|x + sz| < |x| \iff z(2x + sz) < 0$, the double wedge between the lines $z = 0$ and $z = -2x/s$. That wedge has angle $\arctan(2/s)$ on each side, and (x, z) is rotation-invariant, so its probability is $2\arctan(2/s)/(2\pi)$. Hence

$$\text{acceptance}(s) = \frac{2}{\pi}\arctan\frac{2}{s}. \tag{9}$$

s	0.1	0.5	1	2.38	8	20
acceptance (9)	0.968	0.844	0.705	**0.445**	0.156	0.063

The right panel of the L03-07-markov-chains-gibbs mixing figure plots (9) with seeded runs on it. Small s accepts almost everything but crawls (high autocorrelation, as with $\rho \to 1$ in Gibbs); large s proposes bold moves that are almost always rejected, and the chain sits still. Greenberg's warning (p. 98) in numbers: neither a high nor a low acceptance rate is a goal in itself. The value $s = 2.38$ (for one dimension) is the scale that minimises the autocorrelation time of this chain (Gelman, Roberts and Gilks; G-layer), and the tailored kernels in §6 use $2.38^2/K$ times the inverse Hessian.

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5 · C — Gibbs is MH with acceptance 1 (G pp. 100–101)

In a block update, propose Y_1 from the full conditional $f(x_1 \mid x_2)$. Then (4) reads

$$\alpha(x_1, Y_1 \mid x_2) = \frac{f(Y_1, x_2)\, f(x_1 \mid x_2)}{f(x_1, x_2)\, f(Y_1 \mid x_2)} = \frac{f(Y_1, x_2)\, f(x_1, x_2)/f(x_2)}{f(x_1, x_2)\, f(Y_1, x_2)/f(x_2)} = 1. \tag{10}$$

Every proposal is accepted. This licenses **MH-within-Gibbs** (G Algorithm 7.6): Gibbs steps for the blocks with standard conditionals, MH steps for the rest, in any order within a sweep. Tsay's **Griddy Gibbs** (§12.4.3) is a third option for a univariate nonstandard conditional: evaluate it on a grid, build an approximate inverse cdf, and draw by L03-06-monte-carlo-simulation (4).

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6 · Worked — the Bayesian logit (G §8.2.3; Bayesian-logistic-regression.html)

Model. $P(y_i = 1) = P_i = e^{x_i'\beta}/(1 + e^{x_i'\beta})$, so $\log \frac{P_i}{1-P_i} = x_i'\beta$: the log-odds are linear in the covariates (G p. 127). Likelihood and normal prior $\beta \sim N(\beta_0, B_0)$:

$$\pi(\beta \mid y) \propto \prod_{i=1}^{n} P_i^{y_i}(1-P_i)^{1-y_i} \cdot \exp\Big[-\frac{1}{2}(\beta-\beta_0)' B_0^{-1}(\beta-\beta_0)\Big], \qquad \log \pi(\beta \mid y) = \sum_i [y_i x_i'\beta - \log(1+e^{x_i'\beta})] - \frac{1}{2}(\beta-\beta_0)' B_0^{-1}(\beta-\beta_0) \tag{11}$$

(11) is not a standard form, and latent logistic data do not make it conditionally conjugate (G Ex. 8.8), so Greenberg uses MH with a random walk or a tailored independence kernel (p. 128).

Reading coefficients for a prior (G pp. 127–128). $\partial P_i/\partial x_{ik} = \beta_k P_i(1 - P_i)$, largest at $P_i = \frac{1}{2}$, so the largest effect of covariate k on the probability is $\beta_k/4$. The probit's is $0.4\beta_k$ (from $\phi(0) = 0.3989$), so logit coefficients are about **1.6 times** probit ones. Greenberg elicits $\beta_{01} = \log(1/9) = -2.1972$ from "a 10% chance of owning a computer" for the baseline household, and reports $P(\beta_2 > 0 \mid y) = 0.98$, $P(\beta_3 > 0 \mid y) = 0.88$ (Table 8.3).

Seeded check (L03-13 §8). $n = 2000$ simulated observations with $\beta = (-1, 0.8, 0.5)$; prior $N(0, 100I)$; random walk with covariance $2.38^2/3$ times the MLE covariance, 10,000 draws after 1000 burn-in. MLE $(-1.072,\ 0.845,\ 0.602)$; posterior means $(-1.074,\ 0.850,\ 0.605)$; acceptance 0.32.

The site page (5000 bank clients; loan on income and education). The GLM fit gives intercept -11.304 (s.e. 0.419), income 0.0596 (0.0024), education (2 or 3 vs 1) 4.055 (0.221). The Bayesian fit (package `UPG`, a Gibbs sampler based on Pólya–Gamma data augmentation, G-layer) gives posterior means -10.82 (0.39), 0.06 (0.00) and 3.82 (0.21), with every 95% interval excluding 0. The posterior means are 4–6% smaller in absolute value than the MLE ($-10.82/-11.30 = 0.957$, $3.82/4.06 = 0.942$): the package's default prior shrinks toward 0. The page's payoff is the **posterior of a probability**: draw by draw, $P(\text{loan} \mid \text{income}, \text{education}) = 1/(1 + e^{-x'\beta^{(g)}})$, plotted at incomes 145, 170 and 193 (the 90%, 95% and 99% income quantiles) for both education groups (L03-06-monte-carlo-simulation §8).

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7 · C — stochastic volatility by MCMC (T §12.7; petrobras page)

Tsay's univariate SV model (12.20)–(12.21):

$$r_t = x_t'\beta + a_t, \quad a_t = \sqrt{h_t}\,\epsilon_t, \qquad \ln h_t = \alpha_0 + \alpha_1 \ln h_{t-1} + v_t, \quad \epsilon_t \sim N(0,1),\ v_t \sim N(0, \sigma_v^2), \tag{12}$$

with $|\alpha_1| < 1$. The likelihood $f(R \mid \beta, \alpha) = \int f(R \mid \beta, H) f(H \mid \alpha) dH$ is an n-dimensional integral, but treating $H = (h_1, \ldots, h_n)$ as augmented parameters (L03-07-markov-chains-gibbs §7) gives three blocks: $\beta \mid H$ is a weighted regression (Result 12.1a after dividing by $\sqrt{h_t}$); $\alpha \mid H$ is an AR(1) regression for $\ln h_t$; and each h_t given its neighbours has (12.23)

$$f(h_t \mid \cdot) \propto h_t^{-1.5} \exp\Big[-\frac{(r_t - x_t'\beta)^2}{2h_t} - \frac{(\ln h_t - \mu_t)^2}{2\sigma^2}\Big], \tag{13}$$

with $\mu_t = [\alpha_0(1-\alpha_1) + \alpha_1(\ln h_{t+1} + \ln h_{t-1})]/(1 + \alpha_1^2)$ and $\sigma^2 = \sigma_v^2/(1 + \alpha_1^2)$. (13) is not standard: Jacquier, Polson and Rossi draw it by Metropolis, Tsay by Griddy Gibbs.

The **Petrobras page** (daily PBR log-returns since 2019) first compares iid models by maximum likelihood: Gaussian $\hat{\sigma} = 0.0328$; Student-t with $\hat{\nu} = 3$ and scale 0.0193, much heavier tails. It then fits four SV variants with Kastner's `stochvol` (Gaussian or t errors, with or without leverage $\text{cor}(\epsilon_t, u_t) = \rho$), priors $\mu \sim N(0, 100^2)$, $(\phi+1)/2 \sim \text{Beta}(5, 1.5)$, $\sigma^2 \sim G(1/2, 1/2)$, $(\rho+1)/2 \sim \text{Beta}(4, 4)$, $\nu \sim \text{Exp}(0.1)$, and plots the posterior median volatility paths. The model and its estimation belong to Class 8; here it is the example of MCMC with thousands of latent states.

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8 · G — Hamiltonian Monte Carlo and Stan (exemplos-1-2-3-R-STAN.txt)

The lecturer's Stan programs state a model and priors and let the software sample: example 1, a normal mean and sd with $\mu \sim N(1.7, 0.3^2)$ and $\sigma \sim$ half-Cauchy$(0, 1)$; example 2, a regression with $N(0, 1)$ priors on α, β and half-Cauchy$(0, 2.5)$ on σ; example 3, an SV model with Student-t errors on DAX returns, with $\phi \sim N(0.8, 0.3^2)$ on $(-1, 1)$, $\sigma \sim G(0.5, 0.5)$, $\nu \sim G(1, 0.05)$ and h_1 from the stationary distribution. Stan's sampler is **Hamiltonian Monte Carlo** (NUTS): it treats $-\log \pi(\theta \mid y)$ as a potential energy, adds a random momentum, and follows the gradient-driven trajectory for many steps before an MH accept/reject. Proposals land far from the current point yet are accepted with high probability, which beats random walks in high dimensions. The hierarchical Beta–Binomial of L03-09-hierarchical-models-shrinkage is fitted this way; `Bayesian-beta-regression.html` fits a Beta regression (Lula's vote share on abstention, population size and state dummies) with the `betaBayes` MCMC sampler alongside the MLE. Python equivalents (PyMC, ArviZ) are not installed in this project; the notes write every sampler in NumPy. Named only; not examinable.

Links

- Prev: L03-07-markov-chains-gibbs; next: Gibbs and HMC on hierarchical models, L03-09-hierarchical-models-shrinkage. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §8.
- Earlier: the logit by maximum likelihood (IRLS), L02 §4.
- Rules: rules/08 (Metropolis–Hastings).
- Practice (statements in the books): **G 7.5** (p. 106, random-walk MH for the Laplace; proposal scale), **G 7.8** (p. 107, MH-within-Gibbs), **G 8.8** (p. 132, logit posterior), **T 12.4** (p. 670, SV by MCMC).

G §4.5–4.6, the two hierarchical site pages · Hierarchical models: estimate each unit by borrowing strength from the others

> **Claim.** When units (hospitals, women, funds) are similar but not identical, treat their parameters as **exchangeable** draws from a common distribution whose parameters are themselves estimated. Each unit's posterior mean is then a weighted average of its own estimate and the group mean, with weight $\eta/(\eta + n_i)$ on the group: small units are pulled hardest. For 13 Manhattan hospitals the death rates range from **2.9% to 13.0%** separately; partially pooled, from **5.7% to 8.9%**, around $E(\mu \mid y) = 7.2\%$. An exact grid computation reproduces the lecturer's Stan output (posterior median of μ 0.0714 vs 0.0715). For 68 women's weight-gain curves the Gibbs sampler of the random-coefficient model separates within-woman noise ($\sigma \approx 1.10$ kg) from between-woman spread, which a pooled regression lumps into one residual sd of 3.61 kg. Prev: L03-08-metropolis-hastings-hmc · Next: L03-10-sparsity-priors · Map: L03-00-map

Notation.

Symbol	Meaning	Range
y_i, n_i	deaths and heart-attack cases at hospital i	$i = 1, \dots, 13$
p_i	death probability of hospital i	$(0, 1)$
μ	mean of the population of hospital rates	$(0, 1)$
η	precision of that population, $a + b$ of the Beta (a prior sample size)	> 0
w_i	weight on μ in $E(p_i \mid \cdot)$, $\eta/(\eta + n_i)$	$(0, 1)$
y_{ij}, x_{ij}	weight gain (kg) and week of visit j of woman i	$i \le 68$, $j \le n_i$
α_i, β_i	woman i's intercept and slope	$\mathbb{R}$
$\alpha, \beta, \tau_\alpha, \tau_\beta$	population means and precisions of the coefficients	—

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1 · U — three ways to estimate 13 rates (hierarchical-beta-binomial.html §1)

The page's data: 13 Manhattan hospitals, 2366 heart-attack cases, 163 deaths. Mount Sinai Roosevelt loses 6 of 46 patients (13.0%), "more than four times the rate of Harlem Hospital Center at 2.857%" (1 of 35). Three analyses:

- **Complete pooling:** one p for all: $163/2366 = 6.89\%$. Ignores real differences between hospitals.
- **No pooling:** 13 separate binomials. A hospital with 35 cases gets an estimate as noisy as its sample, and nothing learned about the other 12 hospitals is used.
- **Partial pooling (hierarchical):** the p_i are draws from a common Beta whose mean and spread are estimated. Each estimate leans on the group in proportion to how little data the hospital has.

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2 · C — exchangeability (G §4.5)

Random variables $z_1, \dots, z_n$ are **exchangeable** if their joint distribution is unchanged by any permutation of the indices (G p. 50); iid variables are exchangeable, but exchangeable variables need not be independent. Applied to parameters: if nothing but the data distinguishes hospital 3 from hospital 7, the prior should treat $(p_1, \dots, p_{13})$ symmetrically, and the natural way is iid draws **given** hyperparameters, with uncertainty about those hyperparameters making the p_i dependent a priori. Greenberg's heteroskedastic regression (4.8)–(4.9) is the same idea for variances: each observation's precision λ_i is a draw from $G(\nu/2, \nu/2)$, and marginally the errors are Student-t (L03-06-monte-carlo-simulation (6)).

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3 · C — the hierarchical model (G §4.6)

Greenberg's three levels (4.10)–(4.12):

$$y \sim f(y \mid \theta), \qquad \theta \sim \pi(\theta \mid \theta_0), \qquad \theta_0 \sim \pi(\theta_0 \mid \theta_{00}), \tag{1}$$

with θ_{00} fixed. Two remarks (G p. 53). First, θ_0 is not identified by the data: $f(y \mid \theta, \theta_0) = f(y \mid \theta)$, so the data inform θ_0 only through θ. Second, θ_0 can be integrated out, $\pi(\theta \mid \theta_{00}) = \int \pi(\theta \mid \theta_0)\pi(\theta_0 \mid \theta_{00}) d\theta_0$; the hierarchy is a way to **build** a prior with dependence across units and to compute with it, not a different model.

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4 · Worked — the Beta–Binomial for the hospitals (hierarchical-beta-binomial.html)

Model (the page's Stan file).

$$y_i \mid p_i \sim \mathrm{Bin}(n_i, p_i), \qquad p_i \mid \mu, \eta \sim \mathrm{Beta}(\mu\eta,\ (1{-}\mu)\eta), \qquad \mu \sim U(0,1), \qquad p(\eta) = \frac{n_1}{(n_1+\eta)^2},\ n_1 = 10. \tag{2}$$

The Beta has mean μ and variance $\mu(1-\mu)/(\eta+1)$, so η acts as a prior sample size: large η means the hospitals are nearly identical. The prior on η is proper ($\int_0^\infty n_1/(n_1+\eta)^2 d\eta = 1$) with median $n_1 = 10$ and a heavy right tail.

Integrate the p_i out. For one hospital, Beta–binomial conjugacy (L03-02-conjugate-sequential-updating (4)) gives the marginal likelihood

$$p(y_i \mid \mu, \eta) = \binom{n_i}{y_i} \frac{B(\mu\eta + y_i,\ (1-\mu)\eta + n_i - y_i)}{B(\mu\eta,\ (1-\mu)\eta)}, \tag{3}$$

so the posterior of the two hyperparameters is

$$p(\mu, \eta \mid y) \propto \frac{n_1}{(n_1+\eta)^2} \prod_{i=1}^{13} \frac{B(\mu\eta + y_i,\ (1-\mu)\eta + n_i - y_i)}{B(\mu\eta,\ (1-\mu)\eta)}. \tag{4}$$

(4) is two-dimensional: it can be evaluated **exactly on a grid** (here 1000×1400 points in $\mu \times \log\eta$), with no MCMC at all.

Each hospital, given the hyperparameters. Conjugacy again:

$$p_i \mid \mu, \eta, y \sim \text{Beta}(\mu\eta + y_i,\ (1-\mu)\eta + n_i - y_i), \qquad E(p_i \mid \mu, \eta, y) = \frac{\mu\eta + y_i}{\eta + n_i} = w_i\,\mu + (1-w_i)\frac{y_i}{n_i}, \quad w_i = \frac{\eta}{\eta + n_i}. \tag{5}$$

The second form splits $\mu\eta + y_i$ exactly as L03-02-conjugate-sequential-updating (6). Averaging (5) over (4) gives $E(p_i \mid y)$.

Stan vs the exact grid. The page ran NUTS with 4 chains × 5000 post-warm-up draws (20,000 in all).

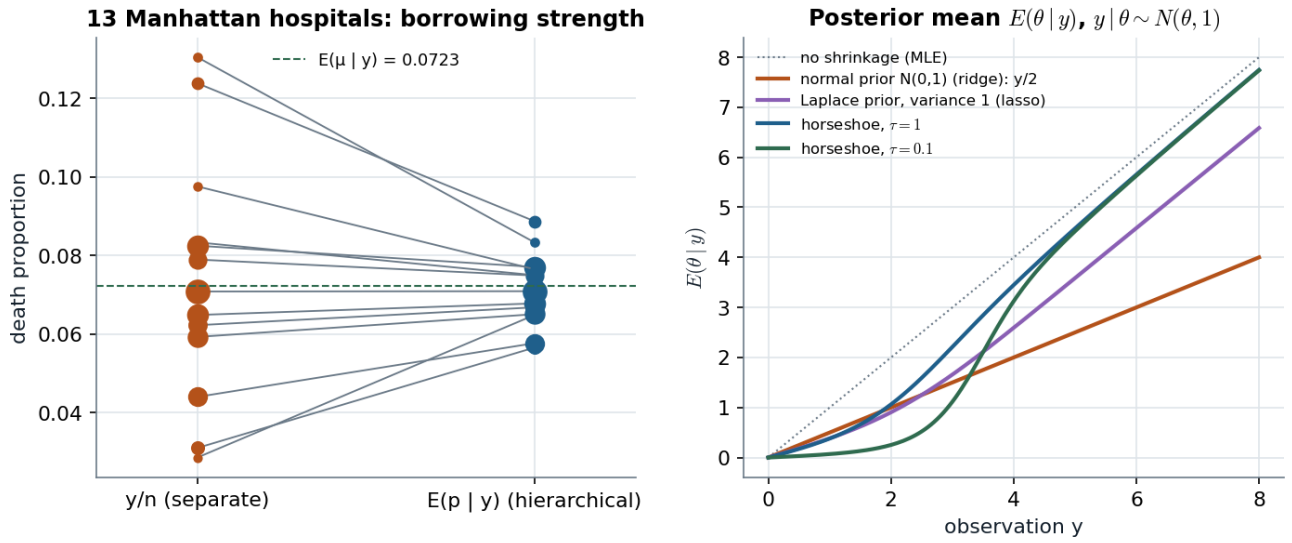
Quantity	Stan 5% / 50% / 95% (page §6)	Exact grid 5% / 50% / 95%
μ	0.0595 / 0.0715 / 0.0874	0.0599 / 0.0714 / 0.0874
η	49.2 / 225.7 / 2234	49.9 / 229.4 / 2759
sd of the Beta, $\sqrt{\mu(1-\mu)/(\eta+1)}$	0.0054 / 0.0170 / 0.0381	0.0048 / 0.0169 / 0.0376

The centre agrees to Monte Carlo error. The upper tail of η does not: Stan reports $n_{eff} = 766$ for η and a posterior sd of 1008 around a mean of 556, so its 95% quantile is noisy, while the grid's is exact. **Heavy-tailed hyperparameters are where MCMC output needs the most checking.**

Shrinkage in numbers. At the posterior median $\eta \approx 229$:

Hospital	y_i/n_i	n_i	$w_i = \eta/(\eta + n_i)$	$E(p_i \mid y)$ (exact)
Harlem	$1/35 = 0.0286$	35	0.87	0.0649
Bellevue	$4/129 = 0.0310$	129	0.64	0.0566
NYP Lower Manhattan	$4/41 = 0.0976$	41	0.85	0.0765
NYP Allen	$13/105 = 0.1238$	105	0.69	0.0887
Mount Sinai Roosevelt	$6/46 = 0.1304$	46	0.83	0.0833
NYP Columbia	$25/353 = 0.0708$	353	0.39	0.0709

The 13 separate rates have sd 0.0300; the 13 posterior means, 0.0090. Every posterior mean lies between its own y_i/n_i and $E(\mu \mid y) = 0.0723$.



Left: (5) averaged over the exact posterior (4) for the 13 hospitals (hierarchical-beta-binomial.html §8, recomputed). Right: shrinkage profiles of L03-10-sparsity-priors.

Read the left panel as the page's §8 plot. Large hospitals (big dots) barely move: NYP Columbia, with 353 cases and a rate already at the centre, stays at 0.071. The small ones move most. Lines cross: Harlem (1 death in 35) starts below Bellevue (4 in 129) and ends above it (0.0649 vs 0.0566), and Roosevelt starts above NYP Allen and ends below it (0.0833 vs 0.0887). Ranking hospitals by raw rates and by posterior means does not give the same league table, because a small hospital's extreme rate is mostly noise. The hierarchical model does not decide this by fiat: η, and hence how much to pool, is estimated from how dispersed the 13 rates are relative to binomial noise.

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5 · Worked — the hierarchical Gaussian linear model (hierarchical-gaussian-linear-model.html)

Data. Weight gain (kg) of $I = 68$ pregnant women at 5 to 7 prenatal visits (UFRJ maternity clinic; Gamerman and Lopes 2006, Ex. 5.6): 8, 33 and 27 women have 5, 6 and 7 visits, $n = 427$ pairs (x_{ij}, y_{ij}), x in weeks.

No pooling and full pooling. The page first fits 68 separate regressions (intercepts, slopes and residual sds vary widely) and then one pooled regression: $\hat{y} = -4.526 + 0.476x$, residual sd 3.612 (verified in L03-13-python §9 from the page's own data).

The model (page §5).

$$y_{ij} \sim N(\alpha_i + \beta_i x_{ij},\ \sigma^2), \qquad \alpha_i \sim N(\alpha, 1/\tau_\alpha), \quad \beta_i \sim N(\beta, 1/\tau_\beta), \tag{6}$$

$$\alpha \sim N(0, 1/P_\alpha),\ \beta \sim N(0, 1/P_\beta),\ \tau_\alpha, \tau_\beta \sim G(a, b),\ \sigma^2 \sim IG(a, b), \qquad P_\alpha = P_\beta = a = b = 1. \tag{7}$$

There are $p = 5 + 2I = 141$ unknowns; the joint posterior has no closed form, but every full conditional does ("the Gibbs sampler becomes the standard solution").

Full conditionals. Each follows from keeping the factors of (6)–(7) that contain the block (L03-07-markov-chains-gibbs §4).

- (α_i, β_i)*:* a normal regression of woman i's n_i points with prior precision $A^{-1} = \mathrm{diag}(\tau_\alpha, \tau_\beta)$ centred at (α, β); by L03-03-normal-measurement-physicists §7,

$$(\alpha_i, \beta_i) \mid \cdot \sim N_2(m_i, V_i), \qquad V_i = (A^{-1} + X_i'X_i/\sigma^2)^{-1}, \quad m_i = V_i(A^{-1}(\alpha, \beta)' + X_i'y_i/\sigma^2). \tag{8}$$

(The page's text prints V_i without the inverse and $X_i'y_i$ without the $1/\sigma^2$; its code has both.)

- α *(and* β*):* the 68 intercepts are 68 "observations" of α with precision τ_α:

$$\alpha \mid \cdot \sim N\Big(\frac{\tau_\alpha \sum_i \alpha_i}{\tau_\alpha I + P_\alpha},\ \frac{1}{\tau_\alpha I + P_\alpha}\Big). \tag{9}$$

- τ_α *(and* τ_β*):* $\tau_\alpha \mid \cdot \sim G(a + I/2,\ b + \sum_i (\alpha_i - \alpha)^2/2)$.
- σ^2*:* $\sigma^2 \mid \cdot \sim IG(a + n/2,\ b + \sum_i \sum_j (y_{ij} - \alpha_i - \beta_i x_{ij})^2/2)$.

(The page writes β_i's prior density with $1/\tau_\alpha$ in the joint posterior; it should be $1/\tau_\beta$.)

Results (NumPy port, 3000 draws after 500 burn-in, posterior medians). $\alpha = -3.66$, $\beta = 0.465$ kg per week, between-woman sds $1/\sqrt{\tau_\alpha} = 3.39$ and $1/\sqrt{\tau_\beta} = 0.217$, within-woman sd $\sigma = 1.10$. The pooled residual sd 3.61 mixed two things the hierarchy separates: measurement-to-measurement noise around each woman's line (1.10) and real differences between women's lines. The page's §5.6 plots each woman's posterior median coefficients against her OLS ones: points move toward the population line, most for women whose own fit is least certain.

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6 · C — shrinkage as the common thread

Every shrinkage formula in this unit has the form

$$\text{estimate}_i = w_i \times (\text{prior or group centre}) + (1 - w_i) \times (\text{unit's own estimate}), \qquad w_i = \frac{\text{prior precision}}{\text{prior precision} + \text{data precision}_i}. \tag{10}$$

Where	Prior / group centre	w_i
Beta–Bernoulli (L03-02-conjugate-sequential-updating (6))	$\alpha_0/(\alpha_0 + \beta_0)$	$(\alpha_0 + \beta_0)/(\alpha_0 + \beta_0 + n)$
Normal measurement (L03-03-normal-measurement-physicists (8))	μ_0	$\sigma^2/(\sigma^2 + \tau_0^2)$
Hospitals, (5)	μ (estimated)	$\eta/(\eta + n_i)$
Random coefficients, (8)	(α, β) (estimated)	matrix weight $V_i A^{-1}$
Ridge (L03-10-sparsity-priors §2)	0	$\lambda/(d_j^2 + \lambda)$ per principal direction

What the hierarchy adds is that the centre and the weight are **learned from the ensemble** instead of fixed by the analyst. The sparsity priors of L03-10-sparsity-priors go one step further and make the weight different for each coefficient.

Links

- Prev: L03-08-metropolis-hastings-hmc (Stan's NUTS fitted the hospital model); next: priors that shrink noise and leave signals alone, L03-10-sparsity-priors. Formulas: L03-11-formulas; pitfalls: L03-12-pitfalls; code: L03-13-python §9.
- Later: hierarchical models, Tobit and probit in Class 7 (G §8.1–8.2 in the Class 7 readings).
- Rules: rules/07, rules/08; random effects and panels, rules/03.
- Practice (statements in the books): **G 8.1** (p. 132, Gibbs regression on birth-weight data), **G 4.7** (p. 59, an exchangeable but dependent urn; background).

Sparse-priors deck, sparse-regression simulation, AME §3.6, I §6.2 · Sparsity priors: shrink the noise, leave the signals alone

Claim. With many coefficients and few of them non-zero, the prior should pull small estimates hard toward zero and leave large ones almost untouched. A normal prior cannot: its posterior mean is y times a constant (ridge, which is the posterior **mode** under a Gaussian prior, as lasso is under a Laplace prior). **Scale mixtures of normals** give each coefficient its own variance. The **horseshoe** (half-Cauchy local scales) puts a Beta(1/2, 1/2) prior on the shrinkage factor $\kappa = 1/(1+\lambda^2)$, with mass at both ends: $\kappa \approx 1$ kills noise, $\kappa \approx 0$ keeps signals. With $y \mid \theta \sim N(\theta, 1)$ and $\tau = 1$, an observation $y = 6$ is estimated at **5.66** under the horseshoe, **4.59** under a Laplace prior of the same variance and **3** under the normal prior. In the lecturer's simulation ($n = 200$, 3 signals among 50 coefficients) the horseshoe's null estimates are about **8 times** closer to zero than ridge's, with the signals equally well estimated. Prev: L03-09-hierarchical-models-shrinkage · Next: L03-11-formulas · Map: L03-00-map

Notation.

Symbol	Meaning	Range
β_j / θ_i	regression coefficient / normal mean	$\mathbb{R}$
Ψ_j, λ_j^2	coefficient-specific (local) variance	> 0
τ	global scale	> 0
$\kappa_i = 1/(1+\lambda_i^2)$	shrinkage factor (with $\sigma = \tau = 1$)	$(0, 1)$
$C^+(0, a)$	half-Cauchy with scale a	—
$\mathrm{Ga}(c, d)$	gamma, shape c, rate d (Griffin–Brown)	—
λ (ridge/lasso)	penalty weight	≥ 0

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1 · U — the problem with one variance for everyone

A normal prior $\beta_j \sim N(0, v)$ shrinks every OLS coefficient by the same factor. If v is small enough to silence 97 null coefficients, it also drags the three real ones toward zero; if it is large enough to spare the signals, it lets the noise through. Griffin and Brown (deck p. 5): a zero-mean normal prior "performs poorly if there are large differences in the size of regression coefficients". The fix is to let each coefficient carry its own variance, drawn from a distribution with **a lot of mass near zero and heavy tails**.

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2 · C — ridge and lasso as posterior modes (AME §3.6; I pp. 250–251)

Let $y_i \mid x_i, \beta \sim N(\beta' x_i, \sigma^2)$ with σ^2 known. The log posterior is the log-likelihood plus the log prior:

$$\log \pi(\beta \mid y) = -\frac{1}{2\sigma^2} \sum_i (y_i - \beta' x_i)^2 + \log \pi(\beta) + c. \tag{1}$$

Gaussian prior $\beta_j \overset{iid}{\sim} N(0,\tau^2)$: $\log\pi(\beta) = -\sum_j \beta_j^2/(2\tau^2) + c$. Multiply (1) by $-2\sigma^2$: maximising the posterior is minimising

$$\sum_i (y_i - \beta' x_i)^2 + \frac{\sigma^2}{\tau^2}\sum_j \beta_j^2, \tag{2}$$

the ridge criterion (I (6.5), p. 240) with $\lambda = \sigma^2/\tau^2$. The posterior is normal, so mode = mean.

Laplace prior $\pi(\beta_j) = \frac{1}{2b}e^{-|\beta_j|/b}$: $\log\pi(\beta) = -\sum_j |\beta_j|/b + c$, and the same step gives

$$\sum_i (y_i - \beta' x_i)^2 + \frac{2\sigma^2}{b}\sum_j |\beta_j|, \tag{3}$$

the lasso criterion (I (6.7), p. 244) with $\lambda = 2\sigma^2/b$. AME (p. 42): "under the Bayesian approach, the tuning parameter λ is defined by the prior distribution"; a tighter prior means a larger λ, and choosing λ by cross-validation "is equivalent to choosing the prior that gives the best predictive power". Two cautions: the lasso's zeros are a property of the **mode**; the posterior **mean** under a Laplace prior is never exactly zero. And the ISLP Fig. 6.11 picture makes the shapes plain: the Laplace density has a sharp peak at 0, the normal a rounded one.

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3 · C — scale mixtures of normals (Griffin–Brown §3, deck pp. 5–7)

$$\pi(\beta_j) = \int N(\beta_j \mid 0, \Psi_j)\, dG(\Psi_j) \qquad \Longleftrightarrow \qquad \beta_j \mid \Psi_j \sim N(0,\Psi_j), \quad \Psi_j \sim G. \tag{4}$$

Every such prior has heavier-than-normal tails (unless G is a point mass). Special cases:

Mixing distribution G	Prior on β_j
point mass at v	normal (ridge)
$z_j\delta_{\sigma_\beta^2} + (1-z_j)\delta_0$, $P(z_j=1)=w$	spike-and-slab: $w\,N(0,\sigma_\beta^2) + (1-w)\,\delta_0$
exponential	double exponential (Bayesian lasso, Park and Casella 2008)
$\mathrm{Ga}(\lambda,\ 1/(2\gamma^2))$	normal–gamma
$\Psi_j = \lambda_j^2\tau^2$, $\lambda_j \sim C^+(0,1)$	horseshoe

(4) is the composition idea of L03-06-monte-carlo-simulation §4: the Student-t was the case $\Psi \sim$ inverse-gamma. It also makes Gibbs sampling easy: given the Ψ_j, the coefficients are normal.

Normal–gamma moments. With $\Psi \sim \mathrm{Ga}(\lambda, 1/(2\gamma^2))$, $E\Psi = 2\lambda\gamma^2$ and $V\Psi = \lambda(2\gamma^2)^2$. By (4),

$$V(\beta) = E[V(\beta\mid\Psi)] = E\Psi = 2\lambda\gamma^2, \qquad E\beta^4 = E[3\Psi^2] = 3\big[\lambda(2\gamma^2)^2 + \lambda^2(2\gamma^2)^2\big], \tag{5}$$

so the kurtosis is $E\beta^4/V(\beta)^2 = 3(\lambda+\lambda^2)/\lambda^2 = 3 + 3/\lambda$: excess kurtosis $3/\lambda$, Griffin and Brown's statement. As λ falls, the prior "place[s] a lot of mass close to zero but at the same time [has] heavy tails" (their Fig. 1: variance 2 with $\lambda = 0.1,\ 0.333,\ 1$). A simulation check with $\lambda = 0.333$ and unit variance: sample excess kurtosis 8.9 against $3/0.333 = 9.0$. $\lambda = 1$ is the Bayesian lasso.

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4 · C — the horseshoe (Carvalho, Polson and Scott 2010, deck pp. 1–2)

For $y \mid \theta \sim N_p(\theta, \sigma^2 I)$:

$$\theta_i \mid \lambda_i \sim N(0, \lambda_i^2), \qquad \lambda_i \mid \tau \sim C^+(0, \tau), \qquad \tau \mid \sigma \sim C^+(0, \sigma), \qquad p(\sigma^2) \propto 1/\sigma^2. \tag{6}$$

"Each θ_i is mixed over its own λ_i, and each λ_i has a half-Cauchy prior with common scale τ."

Posterior mean as shrinkage. Fix $\sigma = \tau = 1$. Given λ_i, the normal–normal update (L03-03-normal-measurement-physicists (8)) with prior variance λ_i^2 and data variance 1 gives $E(\theta_i \mid y, \lambda_i) = (1 - \kappa_i) y_i$ with $\kappa_i = 1/(1 + \lambda_i^2)$. Averaging over $\lambda_i \mid y$:

$$E(\theta_i \mid y) = \int_0^1 (1 - \kappa_i)\, y_i\, p(\kappa_i \mid y)\, d\kappa_i = \{1 - E(\kappa_i \mid y)\}\, y_i. \tag{7}$$

$E(\kappa_i \mid y)$ "can be interpreted as the amount of shrinkage toward zero, a posteriori".

Why "horseshoe". With $\lambda \sim C^+(0, 1)$, $p(\lambda) = \frac{2}{\pi(1+\lambda^2)}$. Change variables to $\kappa = 1/(1 + \lambda^2)$: $\lambda = \sqrt{(1-\kappa)/\kappa}$ and $|d\lambda/d\kappa| = \frac{1}{2}\kappa^{-3/2}(1-\kappa)^{-1/2}$. Substitute, using $1 + \lambda^2 = 1/\kappa$:

$$p(\kappa) = \frac{2}{\pi}\,\kappa \cdot \frac{1}{2}\kappa^{-3/2}(1-\kappa)^{-1/2} = \frac{1}{\pi}\kappa^{-1/2}(1-\kappa)^{-1/2}, \tag{8}$$

the Beta(1/2, 1/2) density, U-shaped like a horseshoe. "The left side of the horseshoe, $\kappa_i \approx 0$, yields virtually no shrinkage, and describes signals. The right side, $\kappa_i \approx 1$, yields near-total shrinkage and describes noise."

The posterior of κ. Since $y_i \mid \kappa_i \sim N(0, 1 + \lambda_i^2) = N(0, 1/\kappa_i)$, the likelihood is $\kappa_i^{1/2} e^{-\kappa_i y_i^2/2}$; times (8):

$$p(\kappa_i \mid y) \propto (1 - \kappa_i)^{-1/2}\, e^{-\kappa_i y_i^2/2}. \tag{9}$$

Small $|y_i|$: (9) keeps the pole at $\kappa = 1$, so $E(\kappa \mid y)$ is near 1 and θ is shrunk hard. Large $|y_i|$: the exponential kills κ away from 0 and $E(\kappa \mid y) \to 0$: the signal is left alone.

Other priors through κ (CPS Table 1, Fig. 2). Double exponential: $p(\kappa) \propto \kappa^{-2} e^{-1/(2\kappa)}$, which vanishes at $\kappa = 0$, so even large signals keep some shrinkage. Cauchy: $\kappa^{-1/2}(1-\kappa)^{-3/2} e^{-\kappa/(2(1-\kappa))}$. Strawderman–Berger: $\kappa^{-1/2}$. Normal–Jeffreys: $\kappa^{-1}(1-\kappa)^{-1}$ (improper at both ends). Only the horseshoe is unbounded at both ends and integrable.

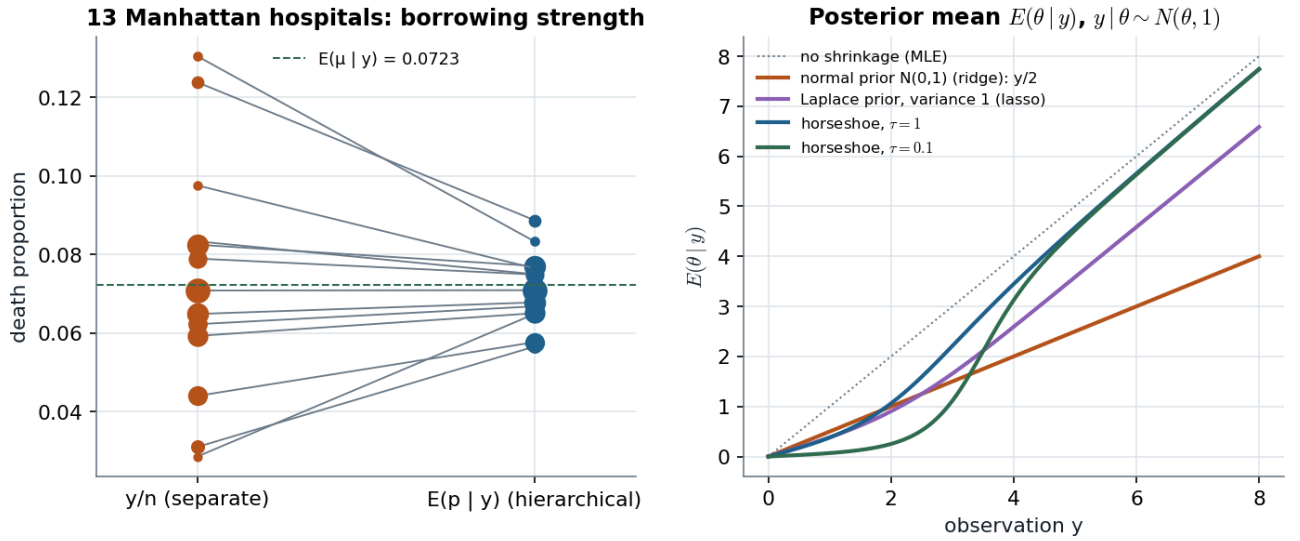
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5 · Worked — shrinkage profiles

$y \mid \theta \sim N(\theta, 1)$, three priors with unit variance where defined: normal $N(0, 1)$; Laplace with $b = 1/\sqrt{2}$ (variance $2b^2 = 1$); horseshoe with $\tau = 1$ and $\tau = 0.1$. Posterior means by quadrature of (7)–(9):

y	Normal: $y/2$	Laplace	Horseshoe $\tau = 1$
2	1.000	0.904	1.063
6	3.000	4.586	5.656
large	$y/2$	$\approx y - \sqrt{2}$	$\to y$

For large y the Laplace prior subtracts a constant $1/b = \sqrt{2}$ (the slope of its log density), so big signals stay biased by 1.41; the horseshoe's bias vanishes.



Right: $E(\theta \mid y)$ under the four priors (quadrature of (7)–(9) and of the Laplace posterior). Left: L03-09-hierarchical-models-shrinkage §4.

The comparative static in τ: shrinking the global scale to 0.1 makes the horseshoe treat every $|y| < 2.5$ as noise (posterior mean below 10% of y at $y = 1$) while still letting $y = 6$ through almost unshrunk. That is the "global–local" design: τ sets how sparse the vector is believed to be, the λ_i let individual coefficients escape.

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6 · Worked — the sparse-regression simulation (sparseregression-simulation-R.txt)

Design (the R script). $n = 200$, $k_{max} = 100$ standard-normal covariates, $\sigma = 0.1$, $\beta = (1, 2, 3, 0, \dots, 0)$, $y = X\beta + \varepsilon$; fit with the first $k \in \{10, 20, 40, 50, 100\}$ columns under ridge, lasso and horseshoe priors (R package `bayesreg`, 1000 burn-in + 1000 draws), and box-plot the posterior draws of $\beta_1, \beta_2, \beta_3$ and of the nulls. The PDF's boxplots show the three signals estimated to within about ± 0.01 by all three priors, and the null coefficients spread over roughly ± 0.04 for ridge and lasso, noticeably tighter for the horseshoe, more so as k grows.

NumPy port (L03-13-python §10). `bayesreg` is R-only, so the three samplers are written out (no intercept; the simulated covariates have mean zero). Each is a Gibbs sampler on the scale-mixture representation (4) with $\beta_j \mid \cdot \sim N(0, \sigma^2 D_j)$:

$$\beta \mid \cdot \sim N_k(A^{-1}X'y,\ \sigma^2 A^{-1}), \quad A = X'X + D^{-1}; \qquad \sigma^2 \mid \cdot \sim IG\Big(\frac{n+k}{2},\ \frac{\mathrm{RSS} + \beta' D^{-1}\beta}{2}\Big). \tag{10}$$

The priors differ only in D and its update:

- **ridge:** $D = \tau^2 I$, $\tau \sim C^+(0,1)$, drawn through the auxiliary-variable form $\tau^2 \mid \xi \sim IG(\frac{1}{2}, \frac{1}{\xi})$, $\xi \sim IG(\frac{1}{2}, 1)$ (Makalic and Schmidt 2016);
- **Bayesian lasso (Park–Casella):** $D = \mathrm{diag}(\tau_j^2)$, $1/\tau_j^2 \mid \cdot \sim \mathrm{InvGauss}(\sqrt{\lambda^2\sigma^2/\beta_j^2},\ \lambda^2)$, $\lambda^2 \mid \cdot \sim G(1 + k,\ 0.1 + \sum_j \tau_j^2/2)$;

- **horseshoe:** $D = \tau^2 \mathrm{diag}(\lambda_j^2)$ with $\lambda_j^2 \mid \cdot \sim IG(1,\ \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau^2\sigma^2})$, $\tau^2 \mid \cdot \sim IG(\frac{k+1}{2},\ \frac{1}{\xi} + \sum_j \frac{\beta_j^2}{2\sigma^2\lambda_j^2})$, $\nu_j \mid \cdot \sim IG(1, 1 + 1/\lambda_j^2)$, $\xi \mid \cdot \sim IG(1, 1 + 1/\tau^2)$.

Results (1500 draws after 500 burn-in, seeded). All three priors put the posterior means of $\beta_1, \beta_2, \beta_3$ within 0.03 of $(1, 2, 3)$. Median absolute posterior mean over the null coefficients:

	ridge	lasso	horseshoe
$k = 10$	0.0055	0.0052	0.0031
$k = 50$	0.0046	0.0045	**0.00055**

With 47 nulls the horseshoe's global scale τ learns that the vector is sparse and shrinks the noise eight times harder than ridge, without touching the signals. (A coding trap met on the way: drawing one gamma variate for the whole vector of λ_j^2 instead of one per coordinate makes the local scales move together, and τ then drifts to zero.)

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7 · Worked — reading the two papers' tables (deck pp. 3, 8–9)

Vanguard mutual funds (CPS Table 4; Carvalho and Scott 2009). $n = 86$ weekly returns on $p = 59$ funds; out-of-sample prediction with five estimates of the 59×59 covariance matrix. Risk relative to Bayesian model averaging (BMA = 1.00):

	MLE	Lasso AND	Lasso OR	Horseshoe	BMA
squared-error loss	10.63	1.25	2.12	**1.07**	1.00
absolute-error loss	3.51	1.22	1.47	**1.04**	1.00

With p close to n the MLE is useless (ten times the risk). The horseshoe, a single continuous prior, comes within 4–7% of full model averaging over sparse graphical models and beats both lasso variants.

NIR spectroscopy (Griffin–Brown §5.3, Table 3). Predict fat content of 215 meat samples from 100 absorbance channels; training/monitoring/test split 129/43/43; "all data" uses $n = 172$, "small" a random 60 (so $p > n$). Test RMSE:

	All data	Small ($n = 60 < p$)
normal–gamma	**1.94**	**2.59**
lasso	3.54	3.09

The posterior median of the shape λ is 0.020 (95% interval 0.016–0.026): far below the lasso's implicit $\lambda = 1$, i.e. a much spikier, heavier-tailed prior (excess kurtosis $3/0.02 = 150$). Their Fig. 7 shows why: the normal–gamma posterior means are zero almost everywhere with a few large coefficients around channels 30–40, while the lasso's oscillate across all 100 channels.

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8 · G — spike-and-slab and model selection (named)

The spike-and-slab prior of §3 puts positive probability on exact zeros, so its posterior gives $P(z_j = 1 \mid y)$, the probability that variable j is in the model: variable selection as model comparison (L03-05-model-comparison-bayes-factors) over 2^k submodels, sampled by Gibbs on the z_j or by reversible jump (G §8.3, p. 130). Continuous shrinkage priors (normal–gamma, horseshoe) trade exact zeros for much easier computation. Large Bayesian VARs (Class 9) use the same priors.

Links

- Prev: L03-09-hierarchical-models-shrinkage (one common weight per group); next: the unit's formula sheet, L03-11-formulas. Pitfalls: L03-12-pitfalls; code: L03-13-python §10.
- Earlier: best subset, AIC/BIC and overfitting, L01 §6; bias–variance, L02 §1.
- Rules: rules/07; ridge/lasso as frequentist shrinkage, rules/06.
- Reading: AME §3.6 (pp. 42–43); I §6.2 (pp. 240–251); the slides for the horseshoe and normal–gamma (no book on disk).

Classes 3+4 · Formulas — the exam specification, every result, every number

Prev: L03-10-sparsity-priors · Next: L03-12-pitfalls · Map: L03-00-map

Equations are cited by the **section note's own numbering**: "L03-03 (8)" is equation (8) of L03-03-normal-measurement-physicists. Every number in §2 is asserted in L03-13-python (script `L03_python_checks.py`).

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0 · The exam specification, mapped

0.1 · The site's description (`Aula/COURSE.md`)

"Introduction to Bayesian learning: Prior, posterior, and predictive distributions; Sequential Bayesian updating and conjugate analysis, Bayes factor, posterior model probability, model selection; Monte Carlo & Markov Chain Monte Carlo methods; Sparsity in linear and log-linear models."

#	Site item (verbatim)	Where it is settled
i	"Prior, posterior, and predictive distributions"	L03-01-bayes-rule-discrete §2–§3; L03-02-conjugate-sequential-updating §4; L03-03-normal-measurement-physicists §2–§3
ii	"Sequential Bayesian updating and conjugate analysis"	L03-01-bayes-rule-discrete §4; L03-02-conjugate-sequential-updating §2–§6
iii	"Bayes factor, posterior model probability, model selection"	L03-05-model-comparison-bayes-factors
iv	"Monte Carlo & Markov Chain Monte Carlo methods"	L03-06-monte-carlo-simulation; L03-07-markov-chains-gibbs; L03-08-metropolis-hastings-hmc
v	"Sparsity in linear and log-linear models"	L03-10-sparsity-priors (linear); the logit of L03-08-metropolis-hastings-hmc §6 is the log-linear (log-odds) case

0.2 · The goals in `Map/class-plan.md`, item by item

#	Goal	Note and section
1	prior × likelihood ∝ posterior	L03-01 §2 ((3), (6))
2	the predictive distribution	L03-01 (4); L03-02 §4; L03-03 §2
3	sequential updating (Beta–Bernoulli, Pareto–Uniform, discrete priors)	L03-01 §4, §7; L03-02 §3, §6
4	conjugate analysis (normal with four priors, the physicists)	L03-02 §2, §5; L03-03 §3–§5

#	Goal	Note and section
5	Bayes estimators and loss functions	L03-04 §2–§9
6	Bayes factors, posterior model probabilities, model selection	L03-05 §2–§8
7	Monte Carlo and MCMC (Gibbs, MH, HMC/Stan)	L03-06; L03-07 §4–§8; L03-08 §2–§5, §8 (HMC named)
8	hierarchical Beta–Binomial and Gaussian models	L03-09 §4–§5
9	sparsity priors (horseshoe, normal–gamma)	L03-10 §3–§7

Final exam: no code (`Map/class-plan.md`). Expect derivations of posteriors, predictives and full conditionals, the reading of MCMC output, and Bayes-factor arithmetic.

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1 · Definitions and results

1.1 · Bayes' rule and the predictive

Result	Statement	Derived in
Bayes' rule (G (2.1))	$p(\theta \mid x) = p(x \mid \theta)p(\theta)/p(x)$	L03-01 (3)
Predictive (total probability)	$p(x) = \sum_k p(x \mid \theta_k)p(\theta_k)$ or $\int p(x \mid \theta)p(\theta)d\theta$	L03-01 (4)
Proportional form (G (2.3))	$p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$	L03-01 (6)
Sequential updating (G (3.1))	$p(\theta \mid y_1, y_2) \propto f(y_2 \mid y_1, \theta)\, p(\theta \mid y_1)$	L03-01 (9)
Posterior odds	posterior ratio = likelihood ratio × prior ratio	L03-01 §6

1.2 · Conjugate families

Data	Prior	Posterior	Posterior mean	Note
Bernoulli, s_n in n	$\text{Beta}(\alpha_0, \beta_0)$	$\text{Beta}(\alpha_0 + s_n, \beta_0 + n - s_n)$	$w\frac{\alpha_0}{\alpha_0+\beta_0} + (1-w)\bar{x}$, $w = \frac{\alpha_0+\beta_0}{\alpha_0+\beta_0+n}$	L03-02 (4)–(6)
Poisson	$\text{Gamma}(\alpha_0, \beta_0)$	$\text{Gamma}(\alpha_0 + s_n, \beta_0 + n)$	$(\alpha_0 + s_n)/(\beta_0 + n)$	L03-02 §5
Exponential(θ)	$\text{Gamma}(\alpha_0, \beta_0)$	$\text{Gamma}(\alpha_0 + n, \beta_0 + \sum x_i)$	$(\alpha_0 + n)/(\beta_0 + \sum x_i)$	L03-02 §5; L03-04 (12)–(13)
$U(0, \theta)$	$\text{Pareto}(\theta_0, \alpha_0)$	$\text{Pareto}(\max\{x_{(n)}, \theta_0\}, \alpha_0 + n)$	$\alpha_1\theta_1/(\alpha_1 - 1)$; median $\theta_1 2^{1/\alpha_1}$	L03-02 (14)–(15)
$N(\theta, \sigma^2)$, σ known	$N(\mu_0, \tau_0^2)$	$N(w\mu_0 + (1-w)x,\ w\tau_0^2)$, $w = \frac{\sigma^2}{\sigma^2+\tau_0^2}$	precisions add: $\tau_1^{-2} = \sigma^{-2} + \tau_0^{-2}$	L03-03 (5)–(8)

Data	Prior	Posterior	Posterior mean	Note
Normal regression	$N(\beta_0, \sigma^2 B_0) \times IG$	$\bar{\beta} = B_1(X'y + B_0^{-1}\beta_0)$, $B_1 = (X'X + B_0^{-1})^{-1}$	multivariate t marginal	L03-03 §7

Result	Statement	Note
Uniform-prior Beta	mode $= s_n/n =$ MLE; mean $(1+s_n)/(2+n)$; variance (8)	L03-02 (7)–(8)
Bernoulli predictive	$P(x_{n+1} = 1 \mid x) = (\alpha_0 + s_n)/(\alpha_0 + \beta_0 + n) =$ posterior mean	L03-02 (11)
Normal predictive	$x \sim N(\mu_0, \sigma^2 + \tau_0^2)$	L03-03 (2)
Large samples	posterior $\approx N(\hat{\theta}, v/n)$ if $p(\hat{\theta}) \neq 0$	L03-02 (16)
t prior influence	$\psi(\theta) = (\nu+1)(\theta-\mu_0)/(\nu s^2 + (\theta-\mu_0)^2)$: bounded, $\to 0$	L03-03 (12)

1.3 · Bayes estimators

Loss	Bayes estimator	Note
$(\theta - a)^2$	posterior mean; minimum expected loss $= V(\theta \mid x)$	L03-04 (4)–(5)
$\lvert\theta - a\rvert$	posterior median ($d\rho/da = 2F - 1$)	L03-04 (7)–(8)
0–1	posterior mode (MAP)	L03-04 (9)–(10)
bilinear (a, b)	quantile $a/(a+b)$	L03-04 (11)
Bayes risk	$r(\pi, d) = \int \rho(d(x) \mid x) g(x) dx$, minimised by the Bayes estimator	L03-04 (15)–(16)
Exp MLE bias	$E(n/\sum x_i) = n\theta/(n-1)$	L03-04 (14)
Credible / HPD interval	$\int_{t_1}^{t_2} \pi(\theta \mid x) d\theta = 1 - \epsilon$; HPD = shortest, equal densities at the ends	L03-04 §10

1.4 · Model comparison

Result	Statement	Note
Marginal likelihood (G (3.4))	$m_i(y) = \int f_i(y \mid \theta_i) \pi_i(\theta_i) d\theta_i$	L03-05 (3)
Posterior odds	$R_{12} = \frac{p_1}{p_2} B_{12}$, $B_{12} = m_1/m_2$; $P(M_1 \mid y) = 1/(1+(p_2/p_1)/B_{12})$	L03-05 (4)

Result	Statement	Note
Michaela–Lila	$m_1 = B(s_1+s_2+1, 2m-s_1-s_2+1)$, $m_2 = B(s_1+1, m-s_1+1)B(s_2+1, m-s_2+1)$	L03-05 (5)–(6)
Large-sample	$\log B_{12} \approx \log LR - \frac{d_1-d_2}{2}\log n + O(1)$	L03-05 (8)
Improper priors	B_{12} scales by the arbitrary c_1/c_2	L03-05 (9)
Model averaging	$f(y_f \mid y) = \sum_i P(M_i \mid y) f_i(y_f \mid y)$	L03-05 (10)

1.5 · Monte Carlo

Result	Statement	Note
MC average, n.s.e.	$\bar{g}_M = \frac{1}{M}\sum g(X^{(i)})$, n.s.e. $= \sqrt{\widehat{V}/M}$	L03-06 (1)–(2)
Hit-or-miss π	se $= 4\sqrt{p(1-p)/M}$, $p = \pi/4$	L03-06 (3)
Inverse cdf	$F^{-1}(U) \sim F$; truncation $F^{-1}(F(c_1) + U[F(c_2) - F(c_1)])$	L03-06 (4)
t as scale mixture	$u \mid \lambda \sim N(0, \sigma^2/\lambda)$, $\lambda \sim G(\nu/2, \nu/2) \Rightarrow u \sim t_\nu(0, \sigma^2)$	L03-06 (5)–(6)
Accept–reject	accept if $u \le f/(cg)$; acceptance $1/c$	L03-06 (7)
Importance sampling	$\frac{1}{G}\sum g(X)f(X)/h(X)$; self-normalised (10); variance needs heavy-tailed h	L03-06 (9)–(11)
SIR	resample proposals with weights $\propto f/g$	L03-06 §7
ZIP MLE	$\hat{\lambda}/(1-e^{-\hat{\lambda}}) = \sum y/(n-n_0)$, $\hat{\pi} = 1 - \bar{y}/\hat{\lambda}$	L03-06 (14)

1.6 · MCMC

Result	Statement	Note
Invariant distribution	$\pi P = \pi$; $\pi(y) = \int \pi(x)p(x,y)dx$	L03-07 (2), (4)
Ergodic average (Thm 6.3)	$\frac{1}{n}\sum f(X_m) \to E_\pi f$	L03-07 (5)
Gibbs invariance	$\int f(y_1 \mid x_2) f(y_2 \mid y_1) f(x)dx = f(y)$	L03-07 (7)
Normal mean/precision Gibbs	$h \mid \mu \sim G(\frac{\alpha_0+n}{2}, \frac{\delta_0+\sum(y_i-\mu)^2}{2})$; $\mu \mid h \sim N(\frac{h_0\mu_0+hn\bar{y}}{h_0+hn}, (h_0+hn)^{-1})$	L03-07 (9)–(10)
Switch point	$\lambda_1 \mid k \sim G(\alpha_0+S_k, \beta_0+k)$, $\lambda_2 \mid k \sim G(\alpha_0+S_n-S_k, \beta_0+n-k)$, $P(k \mid \lambda) \propto e^{k(\lambda_2-\lambda_1)}(\lambda_1/\lambda_2)^{S_k}$; exact $P(k \mid y)$ (15)	L03-07 (13)–(15)

Result	Statement	Note
Gibbs on $N_2(0,\rho)$	x_1 chain is AR(1) with ρ^2; $\rho_h=\rho^{2h}$; IF $=(1+\rho^2)/(1-\rho^2)$	L03-07 (17)–(19)
Batch-means n.s.e.	$\sqrt{\frac{1}{N}\frac{1}{N-1}\sum(\bar{Z}_b-\bar{Z})^2}$	L03-07 (20)
Reversibility	$f(x)p(x,y)=f(y)p(y,x)\Rightarrow f$ invariant	L03-08 (1)
MH acceptance (G (7.6))	$\alpha=\min\{\frac{f(y)q(y,x)}{f(x)q(x,y)},1\}$; RW: $f(y)/f(x)$; independence: $\frac{f(y)/q(y)}{f(x)/q(x)}$	L03-08 (4)–(6)
RW acceptance on $N(0,1)$	$\frac{2}{\pi}\arctan(2/s)$	L03-08 (9)
Gibbs = MH	$\alpha=1$ when the proposal is the full conditional	L03-08 (10)

1.7 · Hierarchical models and sparsity

Result	Statement	Note
Hierarchy (G (4.10)–(4.12))	$y\sim f(y\mid\theta)$, $\theta\sim\pi(\theta\mid\theta_0)$, $\theta_0\sim\pi(\theta_0\mid\theta_{00})$; θ_0 not identified	L03-09 (1)
Beta–Binomial marginal	$p(y_i\mid\mu,\eta)=\binom{n_i}{y_i}B(\mu\eta+y_i,(1-\mu)\eta+n_i-y_i)/B(\mu\eta,(1-\mu)\eta)$	L03-09 (3)
Partial pooling	$E(p_i\mid\mu,\eta,y)=w_i\mu+(1-w_i)y_i/n_i$, $w_i=\eta/(\eta+n_i)$	L03-09 (5)
Random coefficients	$(\alpha_i,\beta_i)\mid\cdot\sim N_2(m_i,V_i)$, $V_i=(A^{-1}+X_i'X_i/\sigma^2)^{-1}$	L03-09 (8)
Ridge / lasso as MAP	$\lambda=\sigma^2/\tau^2$ (Gaussian prior); $\lambda=2\sigma^2/b$ (Laplace)	L03-10 (2)–(3)
Scale mixture	$\beta\mid\Psi\sim N(0,\Psi)$, $\Psi\sim G$	L03-10 (4)
Normal–gamma	variance $2\lambda\gamma^2$, excess kurtosis $3/\lambda$	L03-10 (5)
Horseshoe	$E(\theta\mid y)=(1-E(\kappa\mid y))y$; $\kappa\sim$ Beta(1/2, 1/2); $p(\kappa\mid y)\propto(1-\kappa)^{-1/2}e^{-\kappa y^2/2}$	L03-10 (7)–(9)

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2 · Every number

Quantity	Value	Note
Diego: $P(X=1)$; $P(\text{ill}\mid X=1)$	0.785; **0.8471**	L03-01 (7)–(8)
Diego: $P(\text{healthy}\mid X=0)$	0.8372 (slide: 0.8373)	L03-01 §3
Diego: $P(Y=1\mid X=1)$; after X,Y; Y alone	0.8448; **0.9928**; 0.9830	L03-01 (10)–(11)
Three coins: $P(HHH)$; posteriors	0.618; **0.0674, 0.3932, 0.5394**	L03-01 (12)

Quantity	Value	Note
Three coins after 4 heads; coin 3 after 10	0.0364, 0.3818, 0.5819; 0.741	L03-01 §5
Stone: chickenpox vs smallpox	0.988 vs 0.011 (ratio ≈ 89)	L03-01 (13)
"Four candles"	0.885 vs 0.115	L03-01 §6
G Fig. 2.2 posterior means	$5/14 = 0.357$; $17/54 = 0.315$	L03-02 §2
Pareto example (seeded)	$\theta_1 = 1.8923$, $\alpha_1 = 50.1$, mean 1.9308, median 1.9186, 90% (1.8942, 2.0089)	L03-02 §6
Physicists A, B	$N(890, 17.89^2)$, $N(840, 35.78^2)$; $w = 0.8$, 0.2	L03-03 §4
Predictive sds A, B	44.72, 89.44	L03-03 §2
C, D posterior means (medians)	891.12 (892.41); 876.08 (884.30)	L03-03 §5
Normal vs t prior at $x = 1100$	940.0 vs 1022.0	L03-03 §6
Beta(2,10): mode, median, mean	0.100, 0.148, 0.167; min quadratic loss 0.0107	L03-04 §7
Beta(2,10) 95%: equal-tail, HPD	(0.0228, 0.4128), (0.0063, 0.3675)	L03-04 §10
Bayes risk, $n = 10$, uniform prior	$1/72 = 0.0139$ vs MLE $1/60 = 0.0167$	L03-04 (18)
Table 3.2	$-2.506, \ldots, 0.297$ ($m = 10$); $-30.775, \ldots, 0.756$ ($m = 100$)	L03-05 §4
$m = 10$, (2, 8): $\log_{10} m_1$, $\log_{10} m_2$	-6.5888, -5.3892; BF -1.200; BIC -1.024	L03-05 §4, §6
$\hat{\pi}$ se at $M = 1000$, 10^4	0.0519, 0.0164	L03-06 (3)
Accept–reject Beta(3,3)	$c = 1.875$, acceptance 0.5333	L03-06 §5
Normal from Laplace	$c = 1.3155$, acceptance 0.7602 (G prints 0.6577)	L03-06 (8)
Importance sampling target	0.8302 (infinite-variance estimator)	L03-06 §6
ZIP	$n = 45$; MLE (4.223, 0.526); MM (7.159, 0.721)	L03-06 §7.2
Correlation, uniform prior	2.5/50/97.5%: 0.105, 0.567, 0.784	L03-06 §7.1
P^{10}; π	(0.339, 0.661; 0.330, 0.670); (1/3, 2/3)	L03-07 §2
Switch point	mode 1891, $P = 0.240$; $E(\lambda_1 \mid y) = 3.106$, $E(\lambda_2 \mid y) = 0.930$	L03-07 §6
Inefficiency factors	1.67, 9.5, 999.5 for $\rho = 0.5, 0.9, 0.999$	L03-07 (19)
Beta(3,4) independence MH	acceptance 0.5726 exact (G: 0.57); mean $3/7 = 0.4286$	L03-08 §3
RW acceptance at $s = 2.38$	0.445	L03-08 (9)
Site logit: GLM vs Bayes	$-11.304/0.0596/4.055$ vs $-10.82/0.06/3.82$	L03-08 §6
Hospitals: μ median; $E(\mu \mid y)$	0.0714 (Stan 0.0715); 0.0723	L03-09 §4
Pooled weight-gain OLS	$-4.526 + 0.476x$, sd 3.612; hierarchical $\sigma = 1.10$	L03-09 §5

Quantity	Value	Note
Shrinkage at $y = 6$	normal 3.00, Laplace 4.59, horseshoe 5.66	L03-10 §5
Vanguard risk ratios (SE)	MLE 10.63, horseshoe 1.07, BMA 1.00	L03-10 §7
NIR RMSE	normal–gamma 1.94 / 2.59 vs lasso 3.54 / 3.09	L03-10 §7

Classes 3+4 · Pitfalls — the confusions, each with its minimal counterexample, and an original True/False self-test

Prev: L03-11-formulas · Next: L03-13-python · Map: L03-00-map

Every number below is asserted in L03-13-python. Equations are cited by the section note's own numbering.

Part I — the confusions this material produces

1 · $P(\textbf{data} \mid \textbf{state})$ vs $P(\textbf{state} \mid \textbf{data})$

The confusion. A test with 95% sensitivity means a positive patient is 95% likely to be ill.

Counterexample (Diego, L03-01 (8)). $P(X = 1 \mid \text{ill}) = 0.95$, but $P(\text{ill} \mid X = 1) = 0.847$ with a 70% prior, and it would be far lower with a rare disease (Stone: smallpox 0.011 despite the better likelihood).

> **The likelihood is a forward probability; the posterior needs the prior and the predictive.** → L03-01-bayes-rule-discrete §1, §6.

2 · Sequential vs batch updating

The confusion. Updating one test at a time double-counts or loses information.

Counterexample (L03-01 §4). Sequential: $0.70 \to 0.8471 \to 0.992762$. Batch on $(X = 1, Y = 1)$: 0.992762. Equal whenever the data are conditionally independent given θ (L03-01 (9)). → L03-01-bayes-rule-discrete §4; L03-02-conjugate-sequential-updating §3.

3 · Posterior mode, mean and MLE under a uniform prior

The confusion. A flat prior makes the Bayes estimate equal the MLE.

Counterexample (L03-04 §7). Uniform prior, $s = 1$, $n = 10$: the **mode** is 0.100 = MLE, but the **mean** (the quadratic-loss Bayes estimator) is $2/12 = 0.167$ and the median 0.148. "Flat" fixes the mode, not the estimator. → L03-02-conjugate-sequential-updating (7); L03-04-bayes-estimators-loss.

4 · Plug-in prediction vs the predictive distribution

The confusion. $P(\text{next success}) = \hat{\theta}_{MLE}$.

Counterexample (L03-02 (11)). After 0 successes in n trials, the MLE predicts 0 forever; the predictive under a uniform prior is $1/(n + 2)$. The predictive averages over θ's uncertainty. → L03-02-conjugate-sequential-updating §4.

5 · A normal prior's pull vs a heavy-tailed prior's pull

The confusion. Two priors with the same mean and variance give the same posterior.

Counterexample (L03-03 §6). Normal $N(900, 20^2)$ and $t_5(900, 240)$ (variance 400 too). At $x = 850$: 890.0 vs 891.1. At $x = 1100$: 940.0 vs 1022.0. The normal pull is proportional to the conflict; the t pull is bounded and fades. → L03-03-normal-measurement-physicists (11)–(12).

6 · "The Bayes estimator is biased, the MLE is not"

The confusion. The deck's slide 13 contrast taken as a general rule.

Counterexample (L03-04 (14)). Exponential data: $E(n/\sum x_i) = n\theta/(n-1)$, so the MLE is biased too. Bayes estimators are optimal for Bayes risk ($1/72 < 1/60$ in B Ex. 4.4.1), not for unbiasedness. → L03-04-bayes-estimators-loss §8–§9.

7 · Credible vs confidence interval

The confusion. "95% probability that θ is in my confidence interval."

Counterexample (L03-04 §10). A credible interval conditions on the data and is a statement about θ; a confidence interval is a statement about the procedure over repeated samples (G p. 31). For the skewed Beta(2, 10) posterior, the equal-tail and HPD intervals differ: (0.023, 0.413) vs (0.006, 0.368). → L03-04-bayes-estimators-loss §10.

8 · Maximised likelihood vs marginal likelihood

The confusion. The bigger model always wins because it fits at least as well.

Counterexample (L03-05 §4). Michaela and Lila both at 0.5 with $m = 100$: the two-probability model's maximised likelihood equals the one-probability model's, yet $\log_{10} B_{12} = +0.756$ for the simpler model. The marginal likelihood averages over the prior and charges $\frac{1}{2}\log n$ per extra parameter. → L03-05-model-comparison-bayes-factors §6.

9 · Improper priors in Bayes factors

The confusion. "Non-informative" priors are safe everywhere.

Counterexample (L03-05 (9)). With $c\,\pi(\theta)$ improper, B_{12} scales by c_1/c_2, an arbitrary number. Fine for one model's posterior, meaningless for comparing models. → L03-05-model-comparison-bayes-factors §7.

10 · Accept–reject vs Metropolis–Hastings

The confusion. Both reject proposals, so both give independent draws.

Counterexample (L03-06 §5, L03-08 §1). Accept–reject retries until acceptance: accepted values are iid from f. MH keeps the current value on rejection: the chain repeats values and its draws are correlated. → L03-06-monte-carlo-simulation (7); L03-08-metropolis-hastings-hmc §2.

11 · A printed Monte Carlo standard error vs a valid one

The confusion. Any simulation's standard error formula can be trusted.

Counterexample (L03-06 §6). Greenberg's importance sampler for the truncated exponential has $\int g^2 f^2/h = \infty$ because $h = \text{Beta}(2, 3)$ vanishes at 0. Seeded run: 0.800 with "se" 0.009 against the truth 0.830. For MCMC, the naive se ignores autocorrelation by a factor $\sqrt{\text{IF}}$. → L03-06-monte-carlo-simulation (11); L03-07-markov-chains-gibbs (19)–(20).

12 · High acceptance rate vs good mixing

The confusion. An MH sampler accepting 95% of proposals works well.

Counterexample (L03-08 (9)). On $N(0,1)$ with $s = 0.1$, acceptance is 0.968 but every step is tiny and the chain crawls; $s = 2.38$ accepts 0.445 and mixes far better. → L03-08-metropolis-hastings-hmc §4.

13 · Many Gibbs draws vs many independent draws

The confusion. 5000 draws give a Monte Carlo error of $\sigma/\sqrt{5000}$.

Counterexample (L03-07 (18)–(19)). For the bivariate normal with $\rho = 0.999$, IF = 999.5: 5000 draws carry the information of about 5. Greenberg's means of 1.084 and -0.623 from two starts show it. → L03-07-markov-chains-gibbs §8.

14 · Lasso zeros vs posterior zeros

The confusion. The Bayesian lasso sets coefficients exactly to zero.

Counterexample (L03-10 §2). The lasso estimate is the posterior **mode** under a Laplace prior; the posterior **mean** is never exactly zero, and at large $|y|$ it stays $1/b$ below y (4.59 at $y = 6$ vs the horseshoe's 5.66). → L03-10-sparsity-priors §2, §5.

15 · Complete pooling, no pooling, partial pooling

The confusion. Either all hospitals share one rate or each has its own estimate.

Counterexample (L03-09 §4). Hierarchical posterior means run from 0.057 to 0.089, between the separate estimates (0.029–0.130) and the pooled 0.069, with weights $\eta/(\eta + n_i)$ estimated from the data. → L03-09-hierarchical-models-shrinkage.

Part II — True/False self-test (original)

Scoring (rules/00): V/F score = $\max\{0,\ \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count as wrong. Difficulty: **[E]** easy · **[M]** medium · **[P]** hard. Numbers are illustrative unless a note is cited. Answers follow each block.

Block A — Bayes' rule

Setting (items 1–5). A Brazilian bank's fraud filter flags 98% of fraudulent card transactions and 3% of legitimate ones. One transaction in a hundred is fraudulent.

1. **[E]** A flagged transaction is more likely fraudulent than not.
2. **[M]** The probability that a random transaction is flagged is 0.0395.
3. **[M]** If a second, independent filter with the same error rates also flags it, the probability of fraud exceeds 0.9.
4. **[P]** If the fraud rate doubled to 2%, the probability of fraud given one flag would double.
5. **[E]** With the two filters, the posterior depends on which filter is consulted first.

Answers.

1. **False.** By L03-01 (3)–(4): $0.98 \times 0.01/(0.98 \times 0.01 + 0.03 \times 0.99) = 0.0098/0.0395 = 0.248$. The filter's 98% is $P(\text{flag} \mid \text{fraud})$, not $P(\text{fraud} \mid \text{flag})$ (Part I, 1).
2. **True.** The predictive (L03-01 (4)): $0.0098 + 0.0297 = 0.0395$.
3. **True.** Each flag multiplies the odds by the likelihood ratio $0.98/0.03 = 32.67$. Prior odds 1/99; after two flags $32.67^2/99 = 10.78$, so $P = 10.78/11.78 = 0.915$ (L03-01 (9), sequential = batch).

4. **False.** The odds double, not the probability: odds $= (0.02/0.98) \times 32.67 = 0.667$, so $P = 0.400$, up from 0.248. The posterior ratio is likelihood ratio × prior ratio (L03-01 §6).
5. **False.** With conditionally independent data the joint likelihood is a product, and products commute (L03-01 (9); Part I, 2).

Block B — conjugate updating and Bayes estimators

Setting (items 6–10). A startup models the conversion probability θ of its landing page with a Beta(2, 8) prior. Of 30 visitors, 6 sign up.

6. **[E]** The posterior is Beta(8, 32), and its mean is a weighted average of the prior mean and the MLE with weight 0.25 on the prior.
7. **[M]** Under absolute-error loss the Bayes estimate is 0.20.
8. **[M]** The probability that the next visitor signs up, given the data, is 0.20.
9. **[P]** Had the startup recorded visitors until the 6th sign-up (which came at visitor 30), its posterior would differ.
10. **[M]** As the number of visitors grows with the conversion share fixed, the prior's weight in the posterior mean falls like $1/n$.

Answers.

6. **True.** L03-02 (4): $(2 + 6,\ 8 + 24)$. By (6), $w = (2 + 8)/(2 + 8 + 30) = 0.25$; both centres equal 0.2 here (2/10 and 6/30), so the mean is $8/40 = 0.20$.
7. **False.** Absolute loss gives the median (L03-04 (8)). Beta(8, 32) is right-skewed, so the median, 0.195, is below the mean 0.200. The mode is $7/38 = 0.184$.
8. **True.** The predictive equals the posterior mean, L03-02 (11): 8/40.
9. **False.** Inverse sampling multiplies the likelihood by a different constant only; the kernel $\theta^6(1-\theta)^{24}$ and hence the posterior are unchanged (the likelihood principle, L01-01 §10). Frequentist bias and p-values would change.
10. **True.** $w = (\alpha_0 + \beta_0)/(\alpha_0 + \beta_0 + n) \approx 10/n$ (L03-02 (6), §7).

Block C — the normal model and Bayes factors

Setting (items 11–15). An analyst's prior for next year's inflation is $\theta \sim N(4, 0.5^2)$ (%). A survey gives one reading $x \mid \theta \sim N(\theta, 1)$, and $x = 6$.

11. **[E]** The posterior mean is 4.4%.
12. **[M]** Because the reading conflicts with the prior, the posterior sd exceeds the prior sd of 0.5.
13. **[P]** A Student-t_5 prior with the same centre and variance would put the posterior mean closer to 6 than the normal prior does.
14. **[P]** Comparing $M_1 : \theta = 4$ with $M_2 : \theta \sim N(4, 0.5^2)$, the reading favours the sharp hypothesis M_1.
15. **[M]** Making M_2's prior very diffuse, $N(4, 100^2)$, moves the Bayes factor toward M_1, even though $x = 6$ is unchanged.

Answers.

11. **True.** $w = 1/(1 + 0.25) = 0.8$; L03-03 (8): $0.8 \times 4 + 0.2 \times 6 = 4.4$.
12. **False.** In the normal–normal model $V(\theta \mid x) = w\tau_0^2 = 0.8 \times 0.25 = 0.2$, sd 0.447, whatever x is (L03-03 (8)). Conflict does not widen a conjugate normal posterior; it does widen a t-prior posterior (L03-03 §6).
13. **False.** The conflict is $2/\sqrt{1.25} = 1.79$ predictive sd, too mild for the t prior's bounded influence to show. Near its centre a t prior with the same variance is more peaked, so it pulls slightly harder:

quadrature gives 4.387 against 4.400. The t posterior overtakes the normal one only for larger conflicts (L03-03 Fig., right panel; Part I, 5).

14. **False.** $m_1 = \phi(2) = 0.0540$; m_2 is the predictive density $N(6 \mid 4, 1.25) = 0.0720$; $B_{12} = 0.75$, weak evidence for M_2 (L03-05 (3)–(4)).
15. **True.** m_2 becomes $N(6 \mid 4, 10001) \approx 0.0040$, so $B_{12} \approx 13.5$: a diffuse prior spreads M_2's predictive mass over values never seen (L03-05 §7, the Jeffreys–Lindley effect).

Block D — Monte Carlo and MCMC

Setting (items 16–21). A team computes posterior summaries by simulation.

16. [**E**] To halve the Monte Carlo error of an average of independent draws, they need four times as many draws.
17. [**M**] An accept–reject sampler with $c = 2.5$ needs on average 2.5 proposals per accepted draw.
18. [**M**] A random-walk Metropolis sampler that accepts 90% of its proposals is mixing well.
19. [**P**] In a two-block Gibbs sampler on a bivariate normal posterior with correlation 0.95, each chain's lag-1 autocorrelation is about 0.90, and about 20 draws carry the information of one independent draw.
20. [**M**] The Metropolis–Hastings acceptance probability requires the posterior's normalising constant.
21. [**E**] Gibbs sampling is Metropolis–Hastings with every proposal accepted.

Answers.

16. **True.** The n.s.e. is $\sqrt{V/M}$ (L03-06 (2)).
17. **True.** Each proposal is accepted with probability $1/c$ (L03-06 (7)), so the number of proposals is geometric with mean c.
18. **False.** High acceptance usually means tiny steps: on $N(0, 1)$, $s = 0.1$ accepts 96.8% and crawls (L03-08 (9); Part I, 12).
19. **True.** $\rho_1 = \rho^2 = 0.9025$ and IF $= (1 + 0.9025)/(1 - 0.9025) = 19.5$ (L03-07 (18)–(19)).
20. **False.** It enters only through $f(y)/f(x)$, where the constant cancels (L03-08 (4)).
21. **True.** With the full conditional as proposal, $\alpha = 1$ (L03-08 (10)).

Block E — hierarchical and sparsity priors

Setting (items 22–25). A regulator compares death rates across hospitals with a Beta–Binomial hierarchy; a data scientist chooses priors for a regression with 200 candidate predictors.

22. [**M**] In the hierarchical model, a hospital with more cases is pulled more strongly toward the group mean.
23. [**P**] The population precision η does not enter the binomial likelihood of the counts given the p_i; the data inform it only through how dispersed the p_i appear.
24. [**M**] With a Gaussian prior, the ridge estimate is both the posterior mode and the posterior mean.
25. [**M**] Under a horseshoe prior, a very large observation is shrunk by roughly a constant amount, as under the lasso.

Answers.

22. **False.** The weight on the group mean is $\eta/(\eta + n_i)$ (L03-09 (5)): more cases, less pull. Harlem (35 cases) moves from 0.029 to 0.065; NYP Columbia (353) stays at 0.071.
23. **True.** $f(y \mid p, \eta) = f(y \mid p)$: η is not identified through the likelihood (G p. 53; L03-09 §3), and is learned from the spread of the p_i relative to binomial noise.
24. **True.** The posterior is normal, so mode = mean (L03-10 (2)); under a Laplace prior they differ.

25. **False.** The Laplace prior keeps a constant bias $1/b$; the horseshoe's $E(\kappa \mid y) \to 0$, so large observations are left almost unshrunk (5.66 at $y = 6$ vs 4.59; L03-10 §5).

Links

- All section notes: L03-00-map. Formulas: L03-11-formulas. Code with the asserts: L03-13-python.
- Style: rules/00.

Classes 3+4 · Computation in Python — every routine, with the assert that must pass

> Every computation the unit uses, in Python (numpy, scipy, statsmodels; no R, and no PyMC/ArviZ, which are not installed), each checked with an `assert` against the number printed in the deck, the site page, the book or the section note. The code below is the script `Notas/L03/L03_python_checks.py`, split into its sections. Run `python Notas/L03/L03_python_checks.py` (about one minute). It prints one `ok` line per check and ends with `ALL L03 CHECKS PASSED`; the output shown under each section is from the run of 2026-10-29. Prev: L03-12-pitfalls · Map: L03-00-map · Formulas: L03-11-formulas

What is reused, not rewritten.

Already in the project	What it gives	How this script uses it
`Notas/L01/companions/check_companions.py`	the coal series, loaded from `.agent/notes/coal-mining-data.md` (`cc.Y`, `cc.YEARS`)	**imported** for the switch point (§7)
`Aula/class-03-04/materials/hierarchical-gaussian-linear-model.html`	the 68 × 7 weight-gain and week matrices in the lecturer's R code	**parsed at run time** (§9), never copied
`figures/make_figures.py`	the eight figures, each with its own asserts	**cited**, not imported (importing would redraw every figure and rerun the samplers)
`companions/check_companions.py`	the two companions' read-outs vs their JavaScript	separate script; same formulas as §3 and §5

Conventions. `near(x, printed, tol)` passes when `x` matches a printed value to `tol` (default half a unit in the third decimal). Seeds are fixed; R's random streams cannot be reproduced in Python, so wherever the lecturer's page simulates, the script uses the page's design with a NumPy seed and checks formulas, not the page's draws. Monte Carlo tolerances are about four standard errors. Every sampler (Gibbs, MH, the horseshoe) is written out in NumPy.

---

0 · Setup and helpers

```
"""Classes 3+4 (L03) reference implementation, with assert self-checks against the
 ↪  decks' numbers.

Run:  python Notas/L03/L03_python_checks.py      (stops at the first failed assert;
 ↪  about 1 min)

Sections §1-§10 are the code blocks of L03-13-python.md, in the same order; equation
 ↪  numbers
"(k)" refer to the section note named in each header. Reused, not copied: the coal
 ↪  series and
its loader come from Notas/L01/companions/check_companions.py; the 68-women
 ↪  weight-gain data are
parsed at run time from the lecturer's page
 ↪  Aula/class-03-04/materials/hierarchical-gaussian-
```

```
linear-model.html. Every Monte Carlo draw is seeded; MC checks allow about four
 ↪  standard errors.
Python only: numpy, scipy, statsmodels (PyMC/ArviZ are not installed, so every
 ↪  sampler is
written out in NumPy).
"""
import pathlib
import re
import sys

import numpy as np
import statsmodels.api as sm
from scipy import integrate, optimize, stats
from scipy.special import betaln, gammaln, lambertw, logsumexp

HERE = pathlib.Path(__file__).resolve().parent
ROOT = HERE.parents[1]
MAT = ROOT / "Aula" / "class-03-04" / "materials"
sys.path.insert(0, str(ROOT / "Notas" / "L01" / "companions"))
import check_companions as cc  # noqa: E402  coal series loaded from
 ↪  .agent/notes/coal-mining-data.md

sys.stdout.reconfigure(encoding="utf-8")


def near(x, printed, tol=5e-4):
    """True when x matches a printed value to `tol` (default: half a unit in the 3rd
     ↪  decimal)."""
    return abs(float(x) - float(printed)) <= tol


def ok(tag, **vals):
    print(f"ok  {tag:<28}" + "  ".join(f"{k}={v:.6g}" if isinstance(v, float) else
     ↪  f"{k}={v}"
                                       for k, v in vals.items()))
```

---

1 · Bayes' rule, discrete

Bayes' rule on a finite space: Diego's two tests (slides 12–24), the three coins, Stone's base rates, the discrete prior of the site page. Section note: L03-01-bayes-rule-discrete.

```
def posterior(prior, like):
    """Bayes' rule on a finite parameter space: returns (posterior, predictive p(x)).

    prior: prior probabilities (sum to 1); like: p(x | theta_k) for the observed x.
    """
    prior, like = np.asarray(prior, float), np.asarray(like, float)
    joint = prior * like
    return joint / joint.sum(), joint.sum()
```

```
# Diego (ingredients deck, slides 5-25): theta = 1 ill. P(X=1|0)=0.40, P(X=0|1)=0.05; prior 0.70.
pX1 = np.array([0.40, 0.95])              # P(X=1 | theta=0), P(X=1 | theta=1)
pY1 = np.array([0.04, 0.99])              # second test: P(Y=1|0)=0.04, P(Y=0|1)=0.01
prior = np.array([0.30, 0.70])
post_x1, pred_x1 = posterior(prior, pX1)
post_x0, pred_x0 = posterior(prior, 1 - pX1)
post_xy, pred_y = posterior(post_x1, pY1)
post_y, _ = posterior(prior, pY1)
assert near(pred_x1, 0.785, 1e-12) and near(post_x1[1], 0.8471338, 1e-7)          # slides 12, 16
assert near(pred_x0, 0.215, 1e-12) and near(post_x0[0], 0.8372093, 1e-7)          # slide 19 prints 0.8373093
assert near(pred_y, 0.8447771, 1e-7) and near(post_xy[1], 0.992762, 1e-6)          # slides 21-22
assert near(post_y[1], 0.982979, 1e-6)                                            # slide 24
batch, _ = posterior(prior, pX1 * pY1)                                            # both tests at once
assert np.allclose(batch, post_xy)
assert near(pX1[1] / pX1[0], 2.375, 1e-12) and near((1 - pX1[0]) / (1 - pX1[1]), 12, 1e-12)  # ML ratios
ok("§1 Diego", pX1=pred_x1, post=post_x1[1], postX0=post_x0[0], postXY=post_xy[1], postY=post_y[1])

# Three coins (question0.pdf): P(head) = 0.5, 0.9, 1; uniform prior; n heads in a row.
p_head = np.array([0.5, 0.9, 1.0])
coins = {n: posterior(np.full(3, 1 / 3), p_head**n) for n in range(0, 11)}
post3, pred3 = coins[3]
assert near(pred3, 0.618, 1e-12) and np.allclose(post3, [0.0674, 0.3932, 0.5394], atol=5e-5)
post4_seq, _ = posterior(post3, p_head)                                           # update with the 4th head
assert np.allclose(post4_seq, coins[4][0])
ok("§1 three coins", P3H=pred3, c1=post3[0], c2=post3[1], c3=post3[2],
   c1_4=coins[4][0][0],
   c3_4=coins[4][0][2], c3_10=coins[10][0][2])

# Stone (2013), Figs 1.4 and 1.7: posterior = likelihood x prior / p(x), p(x) as printed.
assert near(0.8 * 0.1 / 0.081, 0.988) and near(0.9 * 0.001 / 0.081, 0.011)
assert near(0.6 * 0.9 / 0.61, 0.885) and near(0.7 * 0.1 / 0.61, 0.115)
two_disease, _ = posterior([0.1, 0.001], [0.8, 0.9])                              # renormalised on 2 diseases
ok("§1 Stone", chickenpox=0.8 * 0.1 / 0.081, smallpox=0.9 * 0.001 / 0.081,
   two_norm_c=two_disease[0])
```

```
# Discrete prior on theta (bernoulli-trials-discrete-prior.html): {0.3, 0.5, 0.8}
 ↪  w.p. {0.4, 0.5, 0.1}.
def discrete_theta_posterior(s, n, thetas=(0.3, 0.5, 0.8), prior=(0.4, 0.5, 0.1)):
    """Posterior over a finite set of Bernoulli probabilities after s successes in n
     ↪  trials."""
    t = np.asarray(thetas)
    return posterior(prior, t**s * (1 - t) ** (n - s))[0]


d40 = discrete_theta_posterior(40, 50)                                          #
 ↪  80% heads in 50 trials
assert d40[2] > 0.999
ok("§1 discrete prior", s40n50_theta08=d40[2],
 ↪  s5n10_theta05=discrete_theta_posterior(5, 10)[1])
```

Output:

```
ok  §1 Diego                    pX1=0.785  post=0.847134  postX0=0.837209
 ↪  postXY=0.992762  postY=0.982979
ok  §1 three coins              P3H=0.618  c1=0.0674218  c2=0.393204  c3=0.539374
 ↪  c1_4=0.0363668  c3_4=0.581869  c3_10=0.74093
ok  §1 Stone                    chickenpox=0.987654  smallpox=0.0111111
 ↪  two_norm_c=0.988875
ok  §1 discrete prior           s40n50_theta08=0.999674  s5n10_theta05=0.737439
```

---

2 · Conjugate updating

Beta–Bernoulli (G Fig. 2.2), sequential = batch, the predictive as an integral, Gamma–Poisson (L01-04's numbers), Pareto–Uniform (the page's (α_1+1) slip). Section note: L03-02-conjugate-sequential-updating.

```
# Beta-Bernoulli, Greenberg Fig. 2.2: Beta(2,2) prior; n = 10, s = 3 and n = 50, s =
 ↪  15.
def beta_update(a0, b0, s, n):
    """Beta(a0,b0) prior x Bernoulli likelihood with s successes in n trials ->
     ↪  Beta(a1,b1)."""
    return a0 + s, b0 + n - s


for (n, s, mean) in ((10, 3, 5 / 14), (50, 15, 17 / 54)):
    a1, b1 = beta_update(2, 2, s, n)
    w = (2 + 2) / (2 + 2 + n)                                                   #
 ↪  prior weight in (2.5)
    assert near(a1 / (a1 + b1), mean, 1e-12) and near(w * 0.5 + (1 - w) * s / n,
     ↪  mean, 1e-12)
a_seq = beta_update(*beta_update(2, 2, 3, 10), 12, 40)                          #
 ↪  10 trials then 40 more
assert a_seq == beta_update(2, 2, 15, 50)                                       #
 ↪  sequential = batch
# Predictive P(y_{n+1}=1 | y) = posterior mean (G p. 33); check against the integral.
a1, b1 = beta_update(2, 2, 3, 10)
```

```
pred = integrate.quad(lambda t: t * stats.beta.pdf(t, a1, b1), 0, 1)[0]
assert near(pred, a1 / (a1 + b1), 1e-10)
# Uniform prior: mode = MLE s/n; mean (1+s)/(2+n) (sequential-bayes page).
a1, b1 = beta_update(1, 1, 1, 10)
assert near((a1 - 1) / (a1 + b1 - 2), 0.1, 1e-12) and near(a1 / (a1 + b1), 1 / 6,
 ↪  1e-12)
ok("§2 beta-bernoulli", mean10=5 / 14, mean50=17 / 54, pred=pred)

# Gamma-Poisson: L01-04 coal 1851-1891 (n = 41, s = 127), Exp(1/2) = G(1, 1/2) prior.
a1, b1 = 1 + 127, 0.5 + 41
assert near(a1 / b1, 3.084) and near((a1 - 1) / b1, 3.060) and near(np.sqrt(a1) / b1,
 ↪  0.273)
gmed = stats.gamma.ppf(0.5, a1, scale=1 / b1)
ok("§2 gamma-poisson (L01-04)", mean=a1 / b1, median=gmed, mode=(a1 - 1) / b1)

# Pareto-Uniform (sequential-bayes page, section 2): U(0, theta) data, Pareto(theta0,
 ↪  alpha0) prior.
def pareto_update(theta0, alpha0, x):
    """Pareto(theta0, alpha0) prior x U(0,theta) likelihood -> Pareto(max(theta0,
     ↪  x_(n)), alpha0 + n)."""
    return max(theta0, float(np.max(x))), alpha0 + len(x)


def pareto_mean(t, a):
    return a * t / (a - 1)


rng = np.random.default_rng(165431)                                             #
 ↪  seed of the R page; draws differ
xu = rng.uniform(0, 2, 50)
t1, al1 = pareto_update(1.5, 0.1, xu)
q = stats.pareto.ppf([0.05, 0.5, 0.95], al1, scale=t1)
assert near(q[1], t1 * 2 ** (1 / al1), 1e-10)                                   #
 ↪  median theta1 2^(1/alpha1)
mc = stats.pareto.rvs(al1, scale=t1, size=400_000, random_state=1).mean()
assert abs(mc - pareto_mean(t1, al1)) < 4 * t1 * np.sqrt(al1 / ((al1 - 1) ** 2 * (al1
 ↪  - 2))) / np.sqrt(4e5)
assert pareto_mean(t1, al1) > t1 > t1 * al1 / (al1 + 1)                         #
 ↪  the page's (alpha1+1) form is below the support
ok("§2 pareto-uniform", xmax=float(xu.max()), theta1=t1, alpha1=al1,
 ↪  mean=pareto_mean(t1, al1),
   q05=q[0], q50=q[1], q95=q[2])
```

Output:

```
ok  §2 beta-bernoulli           mean10=0.357143  mean50=0.314815  pred=0.357143
ok  §2 gamma-poisson (L01-04)   mean=3.08434  median=3.07631  mode=3.06024
ok  §2 pareto-uniform           xmax=1.89226  theta1=1.89226  alpha1=50.1
 ↪  mean=1.9308  q05=1.8942  q50=1.91862  q95=2.00886
```

---

3 · Normal measurement, four priors

Physicists A–D: conjugate formulas for A and B, quadrature for C and D against the page's SIR table, bounded influence of the t prior. Section note: L03-03-normal-measurement-physicists.

```
SIG, X0 = 40.0, 850.0                          # x | theta ~ N(theta, 40^2), x = 850
 ↪  (slides 28, 39)
PRI = {"A": (900.0, 20.0), "B": (800.0, 80.0)}


def normal_posterior(x, sig, mu0, tau0):
    """Conjugate normal update (3)-(5) of L03-03: returns (mean, sd, weight on the
     ↪  prior mean)."""
    w = sig**2 / (sig**2 + tau0**2)
    return w * mu0 + (1 - w) * x, np.sqrt(w * tau0**2), w


mA, sA, wA = normal_posterior(X0, SIG, *PRI["A"])
mB, sB, wB = normal_posterior(X0, SIG, *PRI["B"])
assert near(mA, 890, 1e-9) and near(sA, 17.9, 0.05) and near(mB, 840, 1e-9) and
 ↪  near(sB, 35.7, 0.1)
assert near(np.hypot(SIG, 20), 44.72, 0.005) and near(np.hypot(SIG, 80), 89.44,
 ↪  0.005)  # predictive sd
TH = np.linspace(400, 1300, 180001)


def t_prior(th, mu=900.0, s2=240.0, nu=5):
    return stats.t.pdf((th - mu) / np.sqrt(s2), nu) / np.sqrt(s2)


def prior_D(th):
    return (stats.norm.pdf(th, 900, 20) + stats.norm.pdf(th, 800, 80) + t_prior(th))
     ↪  / 3


def grid_summary(prior_vals, lo=600.0, hi=1000.0, x=X0, sig=SIG):
    """Posterior mean, median, P(theta < x), 5% quantile on [lo, hi] by quadrature on
     ↪  TH."""
    keep = (TH >= lo) & (TH <= hi)
    th, dens = TH[keep], (prior_vals * stats.norm.pdf(x, TH, sig))[keep]
    cdf = integrate.cumulative_trapezoid(dens, th, initial=0)
    dens, cdf = dens / cdf[-1], cdf / cdf[-1]
    return (integrate.trapezoid(th * dens, th), np.interp(0.5, cdf, th), np.interp(x,
     ↪  th, cdf),
            np.interp(0.05, cdf, th))


SIR = {"A": (890.06, 890.10, 0.01, 860.52), "B": (840.02, 840.09, 0.61, 781.18),   #
 ↪  fisicos-ABCD.html
       "C": (891.16, 892.45, 0.02, 860.30), "D": (876.15, 884.35, 0.19, 806.53)}
```

```
EXACT = {"A": grid_summary(stats.norm.pdf(TH, 900, 20)), "B":
 ↪  grid_summary(stats.norm.pdf(TH, 800, 80)),
       "C": grid_summary(t_prior(TH)), "D": grid_summary(prior_D(TH))}
for k, (mean, med, p, q05) in EXACT.items():
    m0, md0, p0, q0 = SIR[k]
    assert abs(mean - m0) < 0.3 and abs(med - md0) < 0.6 and abs(p - p0) < 0.006 and
     ↪  abs(q05 - q0) < 1.0, (k, EXACT[k])
assert near(5 / 3 * 240, 400, 1e-12)                                          #
 ↪  V(prior C) = V(prior A), slide 45
ok("§3 physicists", A=EXACT["A"][0], B=EXACT["B"][0], C=EXACT["C"][0],
 ↪  Cmed=EXACT["C"][1],
   D=EXACT["D"][0], Dmed=EXACT["D"][1], PDltx=EXACT["D"][2], sdA=sA, sdB=sB)


def post_mean_t(x):
    return grid_summary(t_prior(TH), lo=400, hi=1300, x=x)[0]


for x in (900.0, 1200.0):                          # far conflict: the t prior lets the
 ↪  data win
    assert abs(post_mean_t(x) - normal_posterior(x, SIG, 900, 20)[0]) < (0.5 if x ==
     ↪  900 else 1e9)
assert post_mean_t(1100.0) > normal_posterior(1100.0, SIG, 900, 20)[0] + 20
ok("§3 bounded influence", tmean_x1100=post_mean_t(1100.0),
 ↪  nmean_x1100=normal_posterior(1100.0, SIG, 900, 20)[0])
```

Output:

```
ok  §3 physicists               A=890  B=839.999  C=891.122  Cmed=892.409  D=876.078
 ↪  Dmed=884.299  PDltx=0.18696  sdA=17.8885  sdB=35.7771
ok  §3 bounded influence        tmean_x1100=1021.97  nmean_x1100=940
```

---

4 · Bayes estimators and loss

Expected-loss curves and their minima, the Exp–Gamma example, Bolfarine's Bayes risk 1/(6(n+2)), equal-tail vs HPD intervals. Section note: L03-04-bayes-estimators-loss.

```
POST = stats.beta(2, 10)                           # uniform prior, n = 10, s = 1
 ↪  (L01-01's sample)
mean, med, mode = POST.mean(), POST.median(), (2 - 1) / (2 + 10 - 2)
a_grid = np.linspace(0.0005, 0.6, 2400)
tq = np.linspace(0, 1, 20001)
fq = POST.pdf(tq)
quad_loss = [integrate.trapezoid((a - tq) ** 2 * fq, tq) for a in a_grid]
abs_loss = [integrate.trapezoid(np.abs(a - tq) * fq, tq) for a in a_grid]
eps = 0.01
zo_loss = [1 - (POST.cdf(a + eps) - POST.cdf(a - eps)) for a in a_grid]
assert abs(a_grid[np.argmin(quad_loss)] - mean) < 5e-4 and
 ↪  abs(a_grid[np.argmin(abs_loss)] - med) < 5e-4
assert abs(a_grid[np.argmin(zo_loss)] - mode) < 2e-3
```

```
assert near(min(quad_loss), POST.var(), 1e-6)                                  #
 ↪ minimum expected loss = V(theta|x)
qa, qb = 1.0, 3.0                                                              #
 ↪ bilinear: a|.| above, b|.| below
lin_loss = [integrate.trapezoid(np.where(tq > a, qa * (tq - a), qb * (a - tq)) * fq,
 ↪ tq) for a in a_grid]
assert abs(a_grid[np.argmin(lin_loss)] - POST.ppf(qa / (qa + qb))) < 5e-4
ok("§4 loss functions", mean=mean, median=med, mode=mode, minquad=min(quad_loss),
 ↪ q25=POST.ppf(0.25))

# Exp-Gamma (notas-bayesianas slides 4-5, 13-15): theta ~ G(a0,b0), x_i ~ Exp(theta).
rng = np.random.default_rng(7)
xe = rng.exponential(1 / 2.0, 30)
a0, b0 = 2.0, 1.0
bayes, mle = (a0 + len(xe)) / (b0 + xe.sum()), len(xe) / xe.sum()
assert abs(bayes - mle) < 0.2 and near((0 + len(xe)) / (0 + xe.sum()), mle, 1e-12)  #
 ↪ vague prior -> MLE
# B Thm 4.4.1 / Ex. 4.4.1: uniform prior, Bernoulli, Bayes risk of (S+1)/(n+2) is
 ↪ 1/(6(n+2)).
n = 10
risk = lambda th: (n * th * (1 - th) + (1 - 2 * th) ** 2) / (n + 2) ** 2          #
 ↪ MSE of (S+1)/(n+2)
assert near(integrate.quad(risk, 0, 1)[0], 1 / (6 * (n + 2)), 1e-12)
assert integrate.quad(lambda th: th * (1 - th) / n, 0, 1)[0] > 1 / (6 * (n + 2))   #
 ↪ MLE's Bayes risk is larger
ok("§4 exp-gamma, Bayes risk", bayes=bayes, mle=mle, risk_bayes=1 / 72, risk_mle=1 /
 ↪ 60)

# Bayesian intervals (B Ex. 5.5.1; G §3.2.2): equal-tail vs HPD for Beta(2,10).
et = POST.ppf([0.025, 0.975])
res = optimize.minimize_scalar(lambda lo: POST.ppf(POST.cdf(lo) + 0.95) - lo,
 ↪ bounds=(0, POST.ppf(0.05)),
                               method="bounded")
hpd = (res.x, POST.ppf(POST.cdf(res.x) + 0.95))
assert hpd[1] - hpd[0] < et[1] - et[0] and near(POST.pdf(hpd[0]), POST.pdf(hpd[1]),
 ↪ 1e-3)
ok("§4 intervals", et_lo=et[0], et_hi=et[1], hpd_lo=hpd[0], hpd_hi=hpd[1])
```

Output:

```
ok  §4 loss functions           mean=0.166667  median=0.147963  mode=0.1
 ↪ minquad=0.0106838  q25=0.08761
ok  §4 exp-gamma, Bayes risk    bayes=2.16359  mle=2.17546  risk_bayes=0.0138889
 ↪ risk_mle=0.0166667
ok  §4 intervals                et_lo=0.0228312  et_hi=0.41278  hpd_lo=0.00630176
 ↪ hpd_hi=0.367513
```

---

5 · Bayes factors

Greenberg's Table 3.2 from Beta functions (and from his Γ form), the BIC-type approximation, coins as models. Section note: L03-05-model-comparison-bayes-factors.

```
def log10_bf_coins(m, s1, s2, alpha=1.0):
    """log10 B12 for M1: theta1 = theta2 vs M2: theta1 != theta2, Beta(alpha, alpha)
     ↪ priors (G p. 37-38)."""
    lm1 = betaln(alpha + s1 + s2, alpha + 2 * m - s1 - s2) - betaln(alpha, alpha)
    lm2 = sum(betaln(alpha + s, alpha + m - s) - betaln(alpha, alpha) for s in (s1,
 ↪ s2))
    return (lm1 - lm2) / np.log(10)


TABLE32 = {0.1: (-2.506, -30.775), 0.2: (-1.200, -15.793), 0.3: (-0.348, -6.316),
           0.4: (0.138, -0.975), 0.5: (0.297, 0.756)}
for p, (b10, b100) in TABLE32.items():
    for m, printed in ((10, b10), (100, b100)):
        got = log10_bf_coins(m, round(p * m), round((1 - p) * m))
        assert near(got, printed), (p, m, got, printed)
# Greenberg's own form (Gamma functions, (3.5)) equals the Beta-function form.
m, s1, s2 = 10, 2, 8
g_m1 = gammaln(s1 + s2 + 1) + gammaln(2 * m - s1 - s2 + 1) - gammaln(2 * m + 2)
g_m2 = sum(gammaln(s + 1) + gammaln(m - s + 1) - gammaln(m + 2) for s in (s1, s2))
assert near((g_m1 - g_m2) / np.log(10), log10_bf_coins(m, s1, s2), 1e-12)


def log10_bf_bic(m, s1, s2):
    """Large-sample approximation: log LR - (d1 - d2)/2 log n with n = 2m tosses (G
     ↪ p. 37)."""
    ll = lambda s, k: (s * np.log(s / k) if s else 0) + ((k - s) * np.log(1 - s / k)
 ↪ if k - s else 0)
    return (ll(s1 + s2, 2 * m) - ll(s1, m) - ll(s2, m) + 0.5 * np.log(2 * m)) /
     ↪ np.log(10)


ok("§5 table 3.2", m10_p01=log10_bf_coins(10, 1, 9), m100_p01=log10_bf_coins(100, 10,
 ↪ 90),
   m10_p05=log10_bf_coins(10, 5, 5), bic_m100_p05=log10_bf_bic(100, 50, 50),
 ↪ exact_m100_p05=log10_bf_coins(100, 50, 50))
# Posterior model probability with even prior odds; improper prior constants c1/c2
 ↪ rescale B12 (G §4.2).
b = 10 ** log10_bf_coins(10, 2, 8)
assert near(1 / (1 + 1 / b), b / (1 + b), 1e-15)
# Three coins as three models: the marginal likelihood of "3 heads" under each coin
 ↪ is the likelihood itself.
assert near(0.729 / 0.125, 5.832, 1e-12)
ok("§5 coins as models", BF_c3_c1=1 / 0.125, BF_c2_c1=0.729 / 0.125, PM1_2v8=b / (1
 ↪ + b))
```

Output:

```
ok  §5 table 3.2                m10_p01=-2.50603  m100_p01=-30.7748  m10_p05=0.296769
 ↪  bic_m100_p05=1.15051  exact_m100_p05=0.756273
ok  §5 coins as models          BF_c3_c1=8  BF_c2_c1=5.832  PM1_2v8=0.0594014
```

---

6 · Monte Carlo

π by hit-or-miss, accept–reject (Beta(3,3); N(0,1) from a Laplace), importance sampling and its infinite variance, the t as a scale mixture, SIR, the ZIP MLE (n = 45), the correlation posterior. Section note: L03-06-monte-carlo-simulation.

```
# pi by hit-or-miss (slides 53-56): theta = P(x^2+y^2<1) = pi/4, se of 4*theta_hat.
se = lambda M: 4 * np.sqrt(np.pi / 4 * (1 - np.pi / 4) / M)
assert near(se(1000), 0.0519, 5e-5) and near(se(10000), 0.0164, 5e-5)
rng = np.random.default_rng(12345)
u = rng.uniform(size=(10000, 2))
pi_hat = 4 * np.mean((u**2).sum(1) < 1)
assert abs(pi_hat - np.pi) < 4 * se(10000)
ok("§6 pi", se1000=se(1000), se10000=se(10000), pi_hat=pi_hat, z_deck=(3.196 - np.pi)
 ↪  / se(1000))

# Accept-reject, Beta(3,3) from U(0,1) (G Algorithm 5.3): c = 1.875, acceptance 1/c.
c = stats.beta.pdf(0.5, 3, 3)
assert near(c, 1.875, 1e-12) and near(1 / c, 0.5333) and near(stats.beta.pdf(0.15, 3,
 ↪  3), 0.4877, 5e-5)
rng = np.random.default_rng(1)
u1, u2 = rng.uniform(size=(2, 200_000))
acc = u2 <= stats.beta.pdf(u1, 3, 3) / c
assert abs(acc.mean() - 1 / c) < 4 * np.sqrt(0.5333 * 0.4667 / 2e5)
assert abs(u1[acc].mean() - 0.5) < 0.003 and abs(u1[acc].var() - 1 / 28) < 0.0008  #
 ↪  Beta(3,3): var 1/28
# N(0,1) from a Laplace proposal (G Algorithm 5.4): with the normalised target c =
 ↪  sqrt(2e/pi).
c_norm = np.sqrt(2 * np.e / np.pi)
assert near(c_norm, 1.3155, 5e-5) and near(np.sqrt(np.e / (2 * np.pi)), 0.6577, 5e-5)
rng = np.random.default_rng(2)
xL, uL = rng.exponential(size=200_000), rng.uniform(size=200_000)
accL = uL <= np.exp(xL - xL**2 / 2 - 0.5)                                        #
 ↪  Greenberg's test (5.2)
assert abs(accL.mean() - 1 / c_norm) < 0.004 and abs(accL.mean() - 0.6577) > 0.09   #
 ↪  0.760, not 0.6577
assert abs(np.mean(xL[accL] ** 2) - 1) < 0.01
ok("§6 accept-reject", c_beta=c, acc_beta=acc.mean(), c_norm=c_norm,
 ↪  acc_norm=accL.mean())

# Importance sampling (G Algorithm 5.5): E[(1+x^2)^-1], x ~ Exp(1) truncated to
 ↪  [0,1]; h = Beta(2,3).
exact = integrate.quad(lambda x: np.exp(-x) / (1 + x**2), 0, 1)[0] / (1 - np.exp(-1))
rng = np.random.default_rng(3)
xb = rng.beta(2, 3, 10_000)
```

```
wts = np.exp(-xb) / (1 - np.exp(-1)) / stats.beta.pdf(xb, 2, 3)
g_is = np.mean(wts / (1 + xb**2))
se_is = np.std(wts / (1 + xb**2), ddof=1) / 100
# h(x) = 12x(1-x)^2 vanishes at 0 where f g does not: E_h[(f g / h)^2] = int f^2 g^2
 ↪  / h diverges like log,
# so the estimator has infinite variance and its printed se (here and in G p. 71) is
 ↪  not reliable.
var_dens = lambda x: (np.exp(-x) / (1 - np.exp(-1)) / (1 + x**2)) ** 2 /
 ↪  stats.beta.pdf(x, 2, 3)
growth = integrate.quad(lambda t: var_dens(np.exp(t)) * np.exp(t), np.log(1e-8),
 ↪  np.log(1e-4))[0]                                          # ~ f(0)^2/12 *
 ↪  log(1e4)
assert near(exact, 0.8302) and abs(g_is - exact) < 0.05
assert abs(growth - (1 / (1 - np.exp(-1))) ** 2 / 12 * np.log(1e4)) < 0.01
ok("§6 importance sampling", exact=exact, estimate=g_is, se_printed=se_is,
 ↪  var_growth=growth)

# Composition: t_nu as a normal scale mixture (G §4.5, §5.2).
rng = np.random.default_rng(4)
lam = rng.gamma(5 / 2, 2 / 5, 400_000)                                    #
 ↪  G(nu/2, nu/2), rate nu/2
tdraw = rng.normal(0, 1 / np.sqrt(lam))
assert abs(np.var(tdraw) - 5 / 3) < 0.03 and abs(np.mean(np.abs(tdraw) > 2.571) -
 ↪  0.05) < 0.002
ok("§6 composition t5", var=float(np.var(tdraw)), tail=float(np.mean(np.abs(tdraw) >
 ↪  2.571)))

# SIR (MS §B.6.2; fisicos-ABCD.html): physicist C's posterior from U(600,1000)
 ↪  proposals.
rng = np.random.default_rng(125532)
th = rng.uniform(600, 1000, 1_000_000)
w = stats.norm.pdf(X0, th, SIG) * t_prior(th)
draws = rng.choice(th, size=1_000_000, p=w / w.sum())
assert abs(draws.mean() - EXACT["C"][0]) < 0.15
ess = w.sum() ** 2 / (w**2).sum()
ok("§6 SIR physicist C", mean=float(draws.mean()), exact=EXACT["C"][0],
 ↪  ess=float(ess))

# Zero-inflated Poisson (ZIP-fishing.html): the frequency table gives 45 groups, not
 ↪  42.
counts = {0: 24, 1: 6, 2: 2, 3: 4, 4: 2, 5: 3, 7: 1, 8: 1, 14: 1, 16: 1}
yz = np.repeat(list(counts), list(counts.values()))
n, n0, ybar, s2 = len(yz), int((yz == 0).sum()), yz.mean(), yz.var(ddof=1)
zbar = yz.sum() / (n - n0)
lam_mle = float(np.real(lambertw(-zbar * np.exp(-zbar)))) + zbar
pi_mle = 1 - ybar / lam_mle
lam_mm, pi_mm = (s2 + ybar**2) / ybar - 1, (s2 - ybar) / (s2 + ybar**2 - ybar)
assert n == 45 and near(lam_mle, 4.223) and near(pi_mle, 0.526) and near(lam_mm,
 ↪  7.159) and near(pi_mm, 0.721)
```

```
assert near(lam_mle / (1 - np.exp(-lam_mle)), zbar, 1e-10)                                  # the score equation


def zip_loglik(lam, pi):
    """ZIP log-likelihood: n0 log(pi + (1-pi)e^-lam) + (n-n0)[log(1-pi) - lam] + sum(y) log(lam) + const."""
    return n0 * np.log(pi + (1 - pi) * np.exp(-lam)) + (n - n0) * (np.log(1 - pi) - lam) + yz.sum() * np.log(lam)


L, P = np.meshgrid(np.linspace(1, 8, 701), np.linspace(0.1, 0.9, 801), indexing="ij")
lw = zip_loglik(L, P)
wz = np.exp(lw - lw.max())
wz /= wz.sum()
zip_post = (float((wz * L).sum()), float((wz * P).sum()))
ok("§6 ZIP", n=n, lam_mle=lam_mle, pi_mle=pi_mle, lam_mm=lam_mm, pi_mm=pi_mm, Elam=zip_post[0], Epi=zip_post[1])

# Correlation coefficient (correlationcoefficient.html): 14 standardised height/weight pairs.
hw = np.array([178, 70, 175, 80, 184, 82, 200, 89, 183, 80, 173, 65, 185, 70, 162, 70, 190, 85,
               174, 80, 180, 87, 165, 74, 176, 72, 174, 63, 197, 120], float).reshape(15, 2)[:14]
z = (hw - hw.mean(0)) / hw.std(0, ddof=1)
S11, S12 = (z**2).sum(), (z[:, 0] * z[:, 1]).sum()
rho = np.linspace(-0.9999, 0.9999, 200001)
loglik_rho = -len(z) / 2 * np.log(1 - rho**2) - (S11 - 2 * rho * S12) / (2 * (1 - rho**2))


def rho_quantiles(logprior):
    lp = loglik_rho + logprior
    d = np.exp(lp - lp.max())
    cdf = integrate.cumulative_trapezoid(d, rho, initial=0)
    return np.interp([0.025, 0.5, 0.975], cdf / cdf[-1], rho)


q_unif = rho_quantiles(0 * rho)
q_page = rho_quantiles(-0.5 * np.log(1 + rho**2))                                           # what the page codes
q_arcs = rho_quantiles(-0.5 * np.log(1 - rho**2))                                           # the stated arc-sine prior
assert np.allclose(q_unif, [0.102, 0.569, 0.784], atol=0.015)
assert np.allclose(q_page, [0.081, 0.558, 0.783], atol=0.015)
ok("§6 correlation", r=float(np.corrcoef(z.T)[0, 1]), unif=str(np.round(q_unif, 3)),
   page=str(np.round(q_page, 3)), arcsine=str(np.round(q_arcs, 3)))
```

Output:

```
ok  §6 pi                        se1000=0.0519304  se10000=0.0164218  pi_hat=3.1452
 ↪  z_deck=1.0477
ok  §6 accept-reject             c_beta=1.875  acc_beta=0.53385  c_norm=1.31549
 ↪  acc_norm=0.760505
ok  §6 importance sampling       exact=0.830217  estimate=0.800335
 ↪  se_printed=0.00887156  var_growth=1.92086
ok  §6 composition t5            var=1.65953  tail=0.050205
ok  §6 SIR physicist C           mean=891.133  exact=891.122  ess=144275
ok  §6 ZIP                       n=45  lam_mle=4.2229  pi_mle=0.526392  lam_mm=7.15909
 ↪  pi_mm=0.720635  Elam=4.26584  Epi=0.524483
ok  §6 correlation               r=0.614559  unif=[0.105 0.567 0.784]  page=[0.089
 ↪  0.556 0.781]  arcsine=[0.13  0.589 0.794]
```

---

7 · Markov chains and Gibbs

The two-state chain (6.2), the normal mean–precision Gibbs sampler, the coal switch point (exact p(k|y) vs Gibbs), Gibbs mixing on $N_2(0, \rho)$, batch means. Section note: L03-07-markov-chains-gibbs.

```
P2 = np.array([[0.750, 0.250], [0.125, 0.875]])                                  #
 ↪  G (6.2)
evals, evecs = np.linalg.eig(P2.T)
pi_inv = np.real(evecs[:, np.argmax(np.real(evals))])
pi_inv /= pi_inv.sum()
assert np.allclose(pi_inv, [1 / 3, 2 / 3]) and np.allclose(pi_inv @ P2, pi_inv)
assert np.allclose(np.linalg.matrix_power(P2, 10), [[0.339, 0.661], [0.330, 0.670]],
 ↪  atol=5e-4)
assert np.allclose(np.linalg.matrix_power(P2, 20),  3 * 2, atol=5e-4)
assert near(sorted(np.real(evals))[0], 0.625, 1e-12)                             #
 ↪  second root 1 - 0.25 - 0.125
ok("§7 two-state chain", pi1=pi_inv[0], P10_11=np.linalg.matrix_power(P2, 10)[0, 0],
 ↪  root2=0.625)
# AR(1) has invariant density N(0, s^2/(1-phi^2)) (G pp. 85-86): one step of the
 ↪  kernel preserves it.
phi, s_ar = 0.9, 1.0
v_inv = s_ar**2 / (1 - phi**2)
assert near(phi**2 * v_inv + s_ar**2, v_inv, 1e-12)


def gibbs_normal(y, mu0, h0, a0, d0, G, B, seed):
    """Algorithm 7.3: y_i ~ N(mu, 1/h), mu ~ N(mu0, 1/h0), h ~ G(a0/2, d0/2); returns
     ↪  (mu, h) draws."""
    rng = np.random.default_rng(seed)
    n, yb, mu = len(y), y.mean(), y.mean()
    out = np.empty((G, 2))
    for g in range(B + G):
        h = rng.gamma((a0 + n) / 2, 2 / (d0 + ((y - mu) ** 2).sum()))
        mu = rng.normal((h0 * mu0 + h * n * yb) / (h0 + h * n), 1 / np.sqrt(h0 + h *
 ↪  n))
        if g >= B:
            out[g - B] = mu, h
```

```
    return out


rng = np.random.default_rng(100)
yn = rng.normal(1.6, 0.2, 100)                                                  #
 ↪  exemplo1 data design (Stan file)
gn = gibbs_normal(yn, 1.7, 1 / 0.3**2, 2.0, 2 * 0.04, 5000, 1000, seed=101)
assert abs(gn[:, 0].mean() - yn.mean()) < 0.01 and abs(1 / np.sqrt(gn[:, 1].mean())
 ↪  - yn.std()) < 0.02
ok("§7 normal mean-precision", mu=float(gn[:, 0].mean()), ybar=float(yn.mean()),
 ↪  sigma=float(1 / np.sqrt(gn[:, 1].mean())))

# Switch point (G (7.3)-(7.4)) on the coal data, with the L01-04 prior G(1, 1/2) for
 ↪  both rates.
Y, YEARS = cc.Y, cc.YEARS
N, S = len(Y), np.cumsum(Y)
A0, B0 = 1.0, 0.5
k = np.arange(1, N + 1)


def switch_exact_logpost():
    """log p(k | y), k = 1..n, rates integrated out analytically (gamma integrals);
     ↪  uniform prior on k."""
    Sk, Rk = S, S[-1] - S
    return (gammaln(A0 + Sk) - (A0 + Sk) * np.log(B0 + k) + gammaln(A0 + Rk) - (A0 +
     ↪  Rk) * np.log(B0 + N - k))


lpk = switch_exact_logpost()
pk = np.exp(lpk - logsumexp(lpk))


def gibbs_switch(G, B, seed):
    """Three-block Gibbs for (lambda1, lambda2, k): two gamma draws and one discrete
     ↪  draw per sweep."""
    rng = np.random.default_rng(seed)
    kk, out = N // 2, np.empty((G, 3))
    for g in range(B + G):
        l1 = rng.gamma(A0 + S[kk - 1], 1 / (B0 + kk))
        l2 = rng.gamma(A0 + S[-1] - S[kk - 1], 1 / (B0 + N - kk))
        lp = k * (l2 - l1) + S * np.log(l1 / l2)
        kk = rng.choice(k, p=np.exp(lp - logsumexp(lp)))
        if g >= B:
            out[g - B] = l1, l2, kk
    return out


gs = gibbs_switch(20000, 1000, seed=31415)
freq = np.bincount(gs[:, 2].astype(int), minlength=N + 1)[1:] / len(gs)
tv = 0.5 * np.abs(freq - pk).sum()
```

```
kmode = int(k[np.argmax(pk)])
assert YEARS[kmode - 1] == 1891 and tv < 0.03
E_l1 = float((pk * (A0 + S) / (B0 + k)).sum())
E_l2 = float((pk * (A0 + S[-1] - S) / (B0 + N - k)).sum())
assert abs(gs[:, 0].mean() - E_l1) < 0.02 and abs(gs[:, 1].mean() - E_l2) < 0.01
win = (YEARS >= 1886) & (YEARS <= 1896)
ok("§7 switch point", mode_year=int(YEARS[kmode - 1]), p_mode=float(pk.max()),
 ↪  p_1886_96=float(pk[win].sum()),
   E_l1=E_l1, E_l2=E_l2, gibbs_l1=float(gs[:, 0].mean()), gibbs_l2=float(gs[:,
 ↪  1].mean()), TV=float(tv))


def acf(x, lags):
    x = x - x.mean()
    return np.array([np.dot(x[:-h], x[h:]) / np.dot(x, x) if h else 1.0 for h in
     ↪  lags])


def batch_nse(x, nb=100):
    """Numerical standard error by batch means (G §7.3): sd of batch means /
     ↪  sqrt(number of batches)."""
    bm = x[: len(x) // nb * nb].reshape(nb, -1).mean(1)
    return bm.std(ddof=1) / np.sqrt(nb)


def gibbs_binormal(rho, G, x2_0, seed):
    """Two-block Gibbs for N2(0, [[1, rho], [rho, 1]]) (G p. 94): x1 | x2 ~ N(rho x2,
     ↪  1 - rho^2)."""
    rng = np.random.default_rng(seed)
    x2, out, sd = x2_0, np.empty(G), np.sqrt(1 - rho**2)
    for g in range(G):
        x1 = rng.normal(rho * x2, sd)
        x2 = rng.normal(rho * x1, sd)
        out[g] = x1
    return out


for rho_ in (0.5, 0.9):
    ch = gibbs_binormal(rho_, 200_000, -2.0, seed=5)
    assert abs(acf(ch, [1])[0] - rho_**2) < 0.01
    iff = (1 + rho_**2) / (1 - rho_**2)
    assert abs(batch_nse(ch) ** 2 * len(ch) / ch.var() - iff) / iff < 0.25
assert 0.5**6 < 0.05 < 0.5**4 and 0.999 ** 40 > 0.9                              # G
 ↪  p. 103: lag 3 vs lag 20
ok("§7 Gibbs mixing", lag1_rho05=0.25, IF_rho05=(1 + 0.25) / (1 - 0.25),
 ↪  IF_rho0999=(1 + 0.998001) / (1 - 0.998001))
```

Output:

```
ok  §7 two-state chain          pi1=0.333333  P10_11=0.339397  root2=0.625
ok  §7 normal mean-precision    mu=1.60785  ybar=1.60769  sigma=0.195068
```

```
ok  §7 switch point             mode_year=1891  p_mode=0.240214  p_1886_96=0.981084
 ↪  E_l1=3.10636  E_l2=0.930185  gibbs_l1=3.10729  gibbs_l2=0.930675  TV=0.00993003
ok  §7 Gibbs mixing             lag1_rho05=0.25  IF_rho05=1.66667  IF_rho0999=999.5
```

---

8 · Metropolis-Hastings

Independence MH for Beta(3,4), the exact random-walk acceptance $(2/\pi)\arctan(2/s)$, a tailored random-walk sampler for the logit. Section note: L03-08-metropolis-hastings-hmc.

```
def mh_independence_beta34(G, B, seed):
    """Algorithm 7.5: target Beta(3,4) kernel, U(0,1) independence proposal; returns
     ↪  draws, acceptance."""
    rng = np.random.default_rng(seed)
    x, out, nacc = 0.5, np.empty(G), 0
    for g in range(B + G):
        y, u = rng.uniform(), rng.uniform()
        if u <= (y**2 * (1 - y) ** 3) / (x**2 * (1 - x) ** 3):
            x, nacc = y, nacc + (g >= B)
        if g >= B:
            out[g - B] = x
    return out, nacc / G


mh, acc_rate = mh_independence_beta34(5000, 500, seed=6)
f34 = stats.beta(3, 4).pdf
tg = np.linspace(0, 1, 2001)
acc_exact = integrate.trapezoid(integrate.trapezoid(np.minimum.outer(f34(tg),
 ↪  f34(tg)), tg), tg)  # E min(1, f(y)/f(x))
assert near(acc_exact, 0.57, 0.01) and abs(acc_rate - acc_exact) < 0.03 and
 ↪  abs(mh.mean() - 3 / 7) < 0.015
ok("§8 independence MH", acc_exact=acc_exact, acc_sim=acc_rate,
 ↪  mean=float(mh.mean()), true=3 / 7)


def rw_accept_exact(s):
    """Stationary acceptance of random-walk MH, N(0,1) target, N(0,s^2) increments:
     ↪  (2/pi) arctan(2/s)."""
    return 2 / np.pi * np.arctan(2 / s)


def rw_mh_normal(s, G, seed):
    rng = np.random.default_rng(seed)
    x, out, nacc = 0.0, np.empty(G), 0
    for g in range(G):
        y = x + s * rng.normal()
        if np.log(rng.uniform()) <= (x**2 - y**2) / 2:
            x, nacc = y, nacc + 1
        out[g] = x
    return out, nacc / G
```

```
for s_ in (0.5, 2.38, 8.0):
    ch, a = rw_mh_normal(s_, 200_000, seed=8)
    xg, ug = np.linspace(-9, 9, 1801), np.linspace(-9 * s_, 9 * s_, 3601)        # E
 ↪  min(1, f(x+u)/f(x))
    XG, UG = np.meshgrid(xg, ug, indexing="ij")
    integrand = np.minimum(1, np.exp((XG**2 - (XG + UG) ** 2) / 2)) *
 ↪  stats.norm.pdf(XG) * stats.norm.pdf(UG, 0, s_)
    num = integrate.trapezoid(integrate.trapezoid(integrand, ug, axis=1), xg)
    assert near(num, rw_accept_exact(s_), 2e-4) and abs(a - rw_accept_exact(s_)) <
     ↪  0.006
assert near(rw_accept_exact(2.38), 0.4449, 5e-5)
ok("§8 random-walk MH", acc_s05=rw_accept_exact(0.5), acc_s238=rw_accept_exact(2.38),
 ↪  acc_s8=rw_accept_exact(8.0))

# Bayesian logit by random-walk MH with a tailored scale (G §8.2.3), on seeded data:
 ↪  posterior ~ MLE.
rng = np.random.default_rng(9)
nl = 2000
Xl = np.column_stack([np.ones(nl), rng.normal(0, 1, nl), rng.binomial(1, 0.5, nl)])
yl = rng.binomial(1, 1 / (1 + np.exp(-(Xl @ np.array([-1.0, 0.8, 0.5])))))
glm = sm.Logit(yl, Xl).fit(disp=0)


def log_post_logit(b, B0inv=np.eye(3) / 100):
    eta = Xl @ b
    return yl @ eta - np.logaddexp(0, eta).sum() - 0.5 * b @ B0inv @ b             #
     ↪  N(0, 100 I) prior


Lch = np.linalg.cholesky(glm.cov_params() * 2.38**2 / 3)
b, lp, draws_l, nacc = glm.params.copy(), log_post_logit(glm.params),
 ↪  np.empty((10000, 3)), 0
for g in range(11000):
    prop = b + Lch @ rng.normal(size=3)
    lpp = log_post_logit(prop)
    if np.log(rng.uniform()) <= lpp - lp:
        b, lp, nacc = prop, lpp, nacc + 1
    if g >= 1000:
        draws_l[g - 1000] = b
assert np.allclose(draws_l.mean(0), glm.params, atol=0.03) and 0.2 < nacc / 11000 <
 ↪  0.5
ok("§8 logit RW-MH", mle=str(np.round(glm.params, 3)),
 ↪  post=str(np.round(draws_l.mean(0), 3)), acc=nacc / 11000)
# The site page (Bayesian-logistic-regression.html): GLM vs UPG posterior means,
 ↪  slope ratio.
assert near(-10.82 / -11.304484, 0.957, 5e-4) and near(3.82 / 4.055082, 0.942, 5e-4)
```

Output:

```
ok  §8 independence MH          acc_exact=0.572583  acc_sim=0.5812  mean=0.422404
 ↪  true=0.428571
ok  §8 random-walk MH           acc_s05=0.844042  acc_s238=0.444906  acc_s8=0.155958
ok  §8 logit RW-MH              mle=[-1.072  0.845  0.602]  post=[-1.074  0.85
 ↪  0.605]  acc=0.318455
```

---

9 · Hierarchical models

The hospital Beta–Binomial on an exact grid vs Stan; the 68-women data parsed from the lecturer's page, pooled OLS, the random-coefficient Gibbs sampler. Section note: L03-09-hierarchical-models-shrinkage.

```
YH = np.array([4, 1, 18, 7, 24, 16, 6, 19, 15, 13, 25, 11, 4])
NH = np.array([129, 35, 228, 84, 291, 270, 46, 293, 241, 105, 353, 250, 41])
assert YH.sum() == 163 and NH.sum() == 2366 and near(6 / 46, 0.13043, 5e-6) and
 ↪  near(1 / 35, 0.02857, 5e-6)
MU = np.linspace(0.0005, 0.25, 1000)
LE = np.linspace(np.log(0.5), np.log(1e7), 1400)                                #
 ↪  grid on log(eta)
MM, EE = np.meshgrid(MU, np.exp(LE), indexing="ij")
a_, b_ = MM * EE, (1 - MM) * EE
loglik_h = sum(betaln(a_ + yi, b_ + ni - yi) - betaln(a_, b_) for yi, ni in zip(YH,
 ↪  NH))
logprior_h = np.log(10) - 2 * np.log(10 + EE) + np.log(EE)                      #
 ↪  p(eta) d eta, in log eta
lpost = loglik_h + logprior_h
wh = np.exp(lpost - lpost.max())
wh /= wh.sum()


def grid_quantile(vals, weights, qs):
    o = np.argsort(vals.ravel())
    cw = np.cumsum(weights.ravel()[o])
    return np.interp(qs, cw, vals.ravel()[o])


q_mu = grid_quantile(MM, wh, [0.05, 0.5, 0.95])
q_eta = grid_quantile(EE, wh, [0.05, 0.5, 0.95])
sd_beta = np.sqrt(a_ * b_ / (EE**2 * (EE + 1)))
q_sd = grid_quantile(sd_beta, wh, [0.05, 0.5, 0.95])
Ep = np.array([(wh * (a_ + yi) / (EE + ni)).sum() for yi, ni in zip(YH, NH)])
assert np.allclose(q_mu, [0.0595, 0.0715, 0.0874], atol=0.0015)                 #
 ↪  Stan, page section 6
assert abs(q_eta[1] / 225.7 - 1) < 0.15 and abs(q_sd[1] - 0.0170) < 0.0015
Emu = float((wh * MM).sum())
assert np.all(np.abs(Ep - Emu) <= np.abs(YH / NH - Emu) + 1e-12) and Ep.std() < (YH
 ↪  / NH).std() / 2  # shrinkage
ok("§9 beta-binomial grid", mu=str(np.round(q_mu, 4)), eta=str(np.round(q_eta, 1)),
 ↪  sd=str(np.round(q_sd, 4)),
   Ep_roosevelt=float(Ep[6]), Ep_harlem=float(Ep[1]), pooled=YH.sum() / NH.sum(),
 ↪  Emu=Emu,
```

```
   sd_phat=float((YH / NH).std()), sd_Ep=float(Ep.std()))

# Hierarchical Gaussian linear model (hierarchical-gaussian-linear-model.html): parse the page's data.
page = (MAT / "hierarchical-gaussian-linear-model.html").read_text(encoding="utf-8")


def r_matrix(name):
    """Read `name <- matrix(c(...),7,68)` from the page's R code (column-major, NA allowed)."""
    body = re.search(rf"{name} &lt;- matrix\(c\((.*?)\),7,68\)", page, re.S).group(1)
    vals = [np.nan if v.strip() == "NA" else float(v) for v in body.replace("\n", "").split(",")]
    return np.array(vals).reshape(68, 7)                                  # row i = woman i


yw, xw = r_matrix("y"), r_matrix("x")
nobs = (~np.isnan(yw)).sum(1)
assert nobs.sum() == 427 and np.bincount(nobs)[5:].tolist() == [8, 33, 27]
rows = [(np.column_stack([np.ones(m), xw[i, :m]]), yw[i, :m]) for i, m in enumerate(nobs)]
Xp, yp = np.vstack([r[0] for r in rows]), np.concatenate([r[1] for r in rows])
ols = sm.OLS(yp, Xp).fit()
assert near(ols.params[0], -4.526386, 5e-6) and near(ols.params[1], 0.476334, 5e-6) and near(np.sqrt(ols.scale), 3.611520, 5e-6)


def gibbs_hier_gauss(G, B, seed, Pa=1.0, Pb=1.0, a=1.0, b=1.0):
    """Random-coefficient Gibbs: (alpha_i, beta_i) | . ~ N2; alpha, beta | . ~ N; tau's ~ G; sigma2 ~ IG."""
    rng = np.random.default_rng(seed)
    I, n = len(rows), len(yp)
    XtX = [X.T @ X for X, _ in rows]
    Xty = [X.T @ yv for X, yv in rows]
    ind = np.array([[np.polyfit(X[:, 1], yv, 1)[::-1] for X, yv in rows]])[0]
    al, be = ind[:, 0].mean(), ind[:, 1].mean()
    ta, tb, s2 = 1 / ind[:, 0].var(ddof=1), 1 / ind[:, 1].var(ddof=1), ols.scale
    out = np.empty((G, 5))
    coef = np.empty((I, 2))
    for g in range(B + G):
        iA = np.diag([ta, tb])
        for i in range(I):
            V = np.linalg.inv(iA + XtX[i] / s2)
            coef[i] = V @ (iA @ [al, be] + Xty[i] / s2) + np.linalg.cholesky(V) @ rng.normal(size=2)
        al = rng.normal(ta * coef[:, 0].sum() / (ta * I + Pa), 1 / np.sqrt(ta * I + Pa))
        be = rng.normal(tb * coef[:, 1].sum() / (tb * I + Pb), 1 / np.sqrt(tb * I + Pb))
```

```
        ta = rng.gamma(a + I / 2, 1 / (b + ((coef[:, 0] - al) ** 2).sum() / 2))
        tb = rng.gamma(a + I / 2, 1 / (b + ((coef[:, 1] - be) ** 2).sum() / 2))
        rss = sum(((yv - X @ cf) ** 2).sum() for (X, yv), cf in zip(rows, coef))
        s2 = 1 / rng.gamma(a + n / 2, 1 / (b + rss / 2))
        if g >= B:
            out[g - B] = al, be, 1 / np.sqrt(ta), 1 / np.sqrt(tb), np.sqrt(s2)
    return out


hg = gibbs_hier_gauss(3000, 500, seed=2025)
med_hg = np.median(hg, 0)
assert abs(med_hg[1] - ols.params[1]) < 0.03 and med_hg[4] < np.sqrt(ols.scale)    #
 ↪  within-woman sd < pooled sd
ok("§9 hierarchical gaussian", alpha=med_hg[0], beta=med_hg[1], sd_alpha=med_hg[2],
 ↪  sd_beta=med_hg[3], sigma=med_hg[4])
```

Output:

```
ok  §9 beta-binomial grid       mu=[0.0599 0.0714 0.0874]  eta=[  49.9  229.4 2759.4]
 ↪  sd=[0.0048 0.0169 0.0376]  Ep_roosevelt=0.0832845  Ep_harlem=0.0648683
 ↪  pooled=0.0688926  Emu=0.0723238  sd_phat=0.0299541  sd_Ep=0.00897594
ok  §9 hierarchical gaussian    alpha=-3.65992  beta=0.465101  sd_alpha=3.38924
 ↪  sd_beta=0.217395  sigma=1.10311
```

---

10 · Sparsity priors

Shrinkage profiles by quadrature, $\kappa \sim \mathrm{Beta}(1/2,1/2)$, normal–gamma kurtosis, Gibbs samplers for ridge, Bayesian lasso and horseshoe on the site's simulation design. Section note: L03-10-sparsity-priors.

```
def shrink_normal(y, v=1.0):
    return y * v / (1 + v)                                                    #
     ↪  y ~ N(theta,1), theta ~ N(0,v)


def shrink_laplace(y, b=1 / np.sqrt(2)):
    """E(theta | y) for y ~ N(theta, 1), theta ~ Laplace(0, b) (variance 2b^2), by
     ↪  quadrature."""
    f = lambda t, p: t**p * stats.norm.pdf(y - t) * np.exp(-abs(t) / b)
    return integrate.quad(f, -30, 30, args=(1,), points=[0, y])[0] /
     ↪  integrate.quad(f, -30, 30, args=(0,), points=[0, y])[0]


def shrink_horseshoe(y, tau=1.0):
    """E(theta | y) = y (1 - E(kappa | y)), kappa = 1/(1 + tau^2 lambda^2), lambda ~
     ↪  C+(0,1), sigma = 1."""
    # prior p(kappa) propto kappa^{-1/2}(1-kappa)^{-1/2} / (1 - kappa + tau^2 kappa);
     ↪  p(y | kappa) propto
    # kappa^{1/2} exp(-kappa y^2/2); the (1-kappa)^{-1/2} factor goes to quad's
     ↪  algebraic weight.
    g = lambda kp, p: kp**p * np.exp(-kp * y**2 / 2) / (1 - kp + tau**2 * kp)
    num = integrate.quad(lambda kp: g(kp, 1), 0, 1, weight="alg", wvar=(0, -0.5))[0]
```

```
    den = integrate.quad(lambda kp: g(kp, 0), 0, 1, weight="alg", wvar=(0, -0.5))[0]
    return y * (1 - num / den)


for yy in (0.5, 2.0, 6.0):                                                  #
 ↪  tau = 1: Be(1/2,1/2) on kappa
    lam_ = stats.halfcauchy.rvs(size=400_000, random_state=11)
    kap = 1 / (1 + lam_**2)
    wk = np.sqrt(kap) * np.exp(-kap * yy**2 / 2)
    assert abs(shrink_horseshoe(yy) - yy * (1 - (wk * kap).sum() / wk.sum())) < 0.02
assert abs(shrink_horseshoe(8.0) - 8.0) < 0.35 and abs(shrink_horseshoe(0.5)) < 0.2
assert abs(shrink_laplace(8.0) - (8.0 - np.sqrt(2))) < 0.05                  #
 ↪  Laplace: constant pull 1/b = sqrt(2)
kap = 1 / (1 + stats.halfcauchy.rvs(size=400_000, random_state=12) ** 2)
assert abs(np.mean(kap < 0.1) - stats.beta.cdf(0.1, 0.5, 0.5)) < 0.003       #
 ↪  kappa ~ Be(1/2, 1/2)
ok("§10 shrinkage profiles", hs_y2=shrink_horseshoe(2.0),
 ↪  hs_y6=shrink_horseshoe(6.0), lap_y2=shrink_laplace(2.0),
   lap_y6=shrink_laplace(6.0), nor_y2=shrink_normal(2.0))
# Normal-gamma (Griffin & Brown 2010): beta | Psi ~ N(0, Psi), Psi ~ Ga(lambda, 1/(2
 ↪  gamma^2)).
lam_ng, gam = 0.333, 1 / np.sqrt(2 * 0.333)                                  #
 ↪  variance 2 lambda gamma^2 = 1
psi = stats.gamma.rvs(lam_ng, scale=2 * gam**2, size=2_000_000, random_state=13)
bng = np.random.default_rng(14).normal(0, np.sqrt(psi))
assert abs(bng.var() - 2 * lam_ng * gam**2) < 0.02 and abs(stats.kurtosis(bng) - 3 /
 ↪  lam_ng) / (3 / lam_ng) < 0.15
ok("§10 normal-gamma", var=float(bng.var()), ex_kurt=float(stats.kurtosis(bng)),
 ↪  theory=3 / lam_ng)


def gibbs_sparse(X, y, prior, G, B, seed):
    """Gibbs for y = X beta + e: ridge (beta_j ~ N(0, s2 tau2)), Bayesian lasso
     ↪  (Park-Casella, exponential
    mixing with a G(1, 0.1) prior on lambda^2) or horseshoe (Makalic-Schmidt
 ↪  auxiliary variables)."""
    rng = np.random.default_rng(seed)
    n, p = X.shape
    XtX, Xty = X.T @ X, X.T @ y
    beta, s2, tau2, lam2, nu, xi, l2 = np.zeros(p), 1.0, 1.0, np.ones(p), np.ones(p),
 ↪  1.0, 1.0
    out = np.empty((G, p))
    ig = lambda shape, scale: scale / rng.gamma(shape, size=np.shape(scale))   # one
 ↪  gamma per coordinate
    for g in range(B + G):
        dinv = {"ridge": np.full(p, 1 / tau2), "lasso": 1 / lam2, "horseshoe": 1 /
 ↪  (tau2 * lam2)}[prior]
        Lc = np.linalg.cholesky(XtX + np.diag(dinv))
        mean = np.linalg.solve(Lc.T, np.linalg.solve(Lc, Xty))
        beta = mean + np.sqrt(s2) * np.linalg.solve(Lc.T, rng.normal(size=p))
```

```
        rss = ((y - X @ beta) ** 2).sum()
        s2 = ig((n + p) / 2, (rss + (beta**2 * dinv).sum()) / 2)
        if prior == "ridge":
            tau2 = ig((p + 1) / 2, 1 / xi + (beta**2).sum() / (2 * s2))
            xi = ig(1, 1 + 1 / tau2)
        elif prior == "lasso":
            lam2 = 1 / rng.wald(np.sqrt(l2 * s2 / beta**2), l2)
            l2 = rng.gamma(1 + p, 1 / (0.1 + lam2.sum() / 2))
        else:
            lam2 = ig(1, 1 / nu + beta**2 / (2 * tau2 * s2))
            tau2 = ig((p + 1) / 2, 1 / xi + (beta**2 / lam2).sum() / (2 * s2))
            nu, xi = ig(1, 1 + 1 / lam2), ig(1, 1 + 1 / tau2)
        if g >= B:
            out[g - B] = beta
    return out


rng = np.random.default_rng(123456)                                             #
 ↪ sparseregression-simulation-R.txt design
nS, kmax = 200, 100
X1 = rng.normal(size=(nS, kmax))
beta_true = np.r_[1, 2, 3, np.zeros(kmax - 3)]
yS = X1 @ beta_true + rng.normal(0, 0.1, nS)
spread = {}
for kk in (10, 50):
    X = X1[:, :kk]
    for pr in ("ridge", "lasso", "horseshoe"):
        dr = gibbs_sparse(X, yS, pr, 1500, 500, seed=kk)
        pm = dr.mean(0)
        assert np.allclose(pm[:3], [1, 2, 3], atol=0.03), (kk, pr, pm[:3])
        spread[(kk, pr)] = float(np.median(np.abs(pm[3:])))
assert spread[(50, "horseshoe")] < spread[(50, "ridge")]
ok("§10 sparse regression", **{f"k{kk}_{pr}": v for (kk, pr), v in spread.items()})
```

Output:

```
ok  §10 shrinkage profiles      hs_y2=1.06253  hs_y6=5.65575  lap_y2=0.904472
 ↪ lap_y6=4.58579  nor_y2=1
ok  §10 normal-gamma            var=0.996666  ex_kurt=8.91747  theory=9.00901
ok  §10 sparse regression       k10_ridge=0.00548398  k10_lasso=0.00518957
 ↪ k10_horseshoe=0.00313458  k50_ridge=0.00455048  k50_lasso=0.00454065
 ↪ k50_horseshoe=0.000545924
```

---

11 · Numbers of the L03-12 self-test

The numbers of the True/False self-test. Section note: L03-12-pitfalls.

```
post_f, pred_f = posterior([0.99, 0.01], [0.03, 0.98])                          #
 ↪ Block A: fraud filter
odds2 = 0.01 / 0.99 * (0.98 / 0.03) ** 2
odds_2pct = 0.02 / 0.98 * 0.98 / 0.03
```

```
assert near(post_f[1], 0.248) and near(pred_f, 0.0395, 1e-12) and near(odds2 / (1 +
 ↪  odds2), 0.915)
assert near(odds_2pct / (1 + odds_2pct), 0.400)
b8 = stats.beta(8, 32)                                                          #
 ↪  Block B: Beta(2,8) + 6 of 30
assert near(b8.mean(), 0.200, 1e-12) and near(b8.median(), 0.195) and near(7 / 38,
 ↪  0.184)
s2t = 0.25 * 3 / 5                                                              #
 ↪  Block C: N(4, 0.5^2), x = 6
tt = np.linspace(-10, 20, 300001)
dt = stats.t.pdf((tt - 4) / np.sqrt(s2t), 5) / np.sqrt(s2t) * stats.norm.pdf(6, tt,
 ↪  1)
t_mean = integrate.trapezoid(tt * dt, tt) / integrate.trapezoid(dt, tt)
assert near(normal_posterior(6, 1, 4, 0.5)[0], 4.4, 1e-12) and near(t_mean, 4.387)
bf_c = stats.norm.pdf(6, 4, 1) / stats.norm.pdf(6, 4, np.sqrt(1.25))
bf_d = stats.norm.pdf(6, 4, 1) / stats.norm.pdf(6, 4, np.sqrt(1 + 100**2))
assert near(bf_c, 0.75, 5e-3) and near(bf_d, 13.5, 0.05)
assert near((1 + 0.95**2) / (1 - 0.95**2), 19.5, 0.02)                          #
 ↪  Block D
ok("§11 self-test", fraud=post_f[1], two_flags=odds2 / (1 + odds2),
 ↪  median=b8.median(), t_mean=t_mean,
   bf=bf_c, bf_diffuse=bf_d)
```

Output:

```
ok  §11 self-test               fraud=0.248101  two_flags=0.915102  median=0.194969
 ↪  t_mean=4.3874  bf=0.749441  bf_diffuse=13.5369
```

The run ends with:

```
ALL L03 CHECKS PASSED
```

Links

- Map: L03-00-map; formulas: L03-11-formulas; pitfalls: L03-12-pitfalls.
- Companion checks: `Notas/L03/companions/check_companions.py`; figures: `Notas/L03/figures/make_figu`r
- Class 1 reference code: L01 §11.
- Rules: rules/07, rules/08.